Functional Analysis

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This text consists of a series of articles on the mathematical area of functional analysis and of a final article concerning with the applications of the abstract theory to linear partial differential equations, which arise as models for problems in mathematical physics and engineering science.

Functional analysis was mainly developed in the first half of the 20^{th} century, and valuable contributions have been done by D. Hilbert, S. Banach, F. Riesz, J. von Neumann, F. E. Browder, L. Schwartz, J.-L. Lions and others.

The topics within functional analysis are mainly divided into two parts: the linear and the non-linear theory. In our treatise we shall exclusively deal with the linear one, hence the fundamental concept of "linear operator" will play a central role in our considerations. Especially what historically is meant by the concept of unbounded operator is of great importance, since the theory of this specific class of mappings can be successfully applied to the study of partial differential operators and their related boundary and initial-value problems.

It is assumed that the reader is familiar with elementary concepts in linear algebra, general topology and measure theory. In fact, functional analysis can be seen as the synthesis of the both first subjects, especially with respect to infinite-dimensional linear spaces. Further preliminaries will be introduced as soon as there is the need to use them.

It is not the idea of this text to give all the proofs of the stated theorems, except in the case where these are elementary. Instead, the main purpose is a transparent and overall presentation of the topics, especially for readers, which have learned functional analysis in the past and want to have a refresh of the main results now.

I wrote the treatise in honour of Prof. Dr. P. Werner (1932-2018), who conveyed my fascination of these beautiful mathematical topics by his lessons hold in 1984 and 1985 at Mathematical Institute A of University Stuttgart.

Version 1.0 from 2026-07-19; Email: *khoeflinger@t-online.de*

Table of Contents

XI. Coercive Sesquilinear Forms

XII. Applications to Differential Operators

I.
Topological and Normed Linear Spaces

Our aim in this first section is to review those fundamental classes of topological linear spaces, in which the analysis of so-called *linear operators* is usually done. These classes include, besides some generic classes like locally convex spaces and completely metrizable spaces, especially the class of normed linear spaces and related subclasses, where we want to emphasize Banach and Hilbert spaces. Finally, there are a lot of normed linear spaces appearing in higher analysis, whose elements are either arbitrary functions or measures.

§1. Generalities on Topological Linear Spaces

Basically, the scalar field $\mathbb{K}$ of the linear spaces under consideration in this text will either be the field $\mathbb{R}$ of real numbers or the field $\mathbb{C}$ of complex numbers. Notice that these fields are furnished with the so-called standard topology, whose subbase is given by the totality of open intervals in the real line and by the totality of open disks in the complex plane, respectively. In general, if results on topological or normed linear spaces prove to be independent of the choice of the scalar field, we shall use the symbol $\mathbb{K}$ instead of $\mathbb{R}$ or $\mathbb{C}$.

By a *topological linear space* is meant an ordinary vector space V over a topological field $\mathbb{K}$ (usually that of real or complex numbers), together with a topology $\mathcal{T}_V$ such that addition $(x, y) \mapsto x + y$ and outer multiplication $(\lambda, x) \mapsto \lambda x$ are continuous. Hence a multiplicity of topological concepts, like the *closure* of a set, the *neighbourhood* of a point, the *convergence* of a sequence, the *continuity* of a mapping and others are meaningful in the context of topological linear spaces.

We continue with the introduction of a basic property that our topological linear spaces under consideration have to own: all singletons $\{x\}$ with $x \in V$ are closed sets, which is equivalent to say that the topology $\mathcal{T}_V$ fulfills HAUSDORFF's separation axiom. In other words, each pair of points $x, y \in X$ can be separated by open sets $\mathcal{O}_x, \mathcal{O}_y$, that is, there are $\mathcal{O}_x \subset V$ and $\mathcal{O}_y \subset V$ with $x \in \mathcal{O}_x$, $y \in \mathcal{O}_y$ and $\mathcal{O}_x \cap \mathcal{O}_y = \emptyset$. The validness of this separation axiom implies not only that every singleton $\{x\}$ is closed, but also that the limit of a convergent sequence of elements in V is unique.

Among a lot of properties, a topological linear space may have, there are two special ones that are of importance for our purposes:

1. A topological linear space V is said to be *metrizable*, if there is a function $d: V \times V \to [0, \infty)$ fulfilling the properties of a *metric*[1] such that the metric topology associated with d resembles the given topology $\mathcal{T}_V$.
2. A sequence $\{x_n\}_{n\in\mathbb{N}}$ with values in a topological linear space V is said to be Cauchy if for each neighbourhood $\mathcal{U}$ of $0 \in V$ there is an integer $N_{\mathcal{U}}$ such that $x_n - x_m \in \mathcal{U}$ for all $m, n > N_{\mathcal{U}}$. Then V is called *sequentially complete*, if every Cauchy sequence converges to a unique element $x \in V$. In the setting of metric spaces, however, the definition of Cauchy sequence may be simplified by means of the metric: $\{x_n\}_{n\in\mathbb{N}}$ is Cauchy, if for each $\epsilon > 0$ there is an integer N_ϵ such that $x_m - x_n \in B_\epsilon(0)$ for all $m, n > N_\epsilon$, where $B_\epsilon(0) := \{x \in V : d(x, 0) < \epsilon\}$ is the open ball with radius ϵ and center $0 \in V$.

For a topological linear space being metrizable and sequentially complete at the same time, both concepts of completeness are compatible to each other, that is, such a space is complete as a metric space if and only if it is sequentially complete as a topological linear space. This equivalence is founded by the fact that a metric space fulfills the *first countability axiom.*

In the sequel we also use the notion of *continuous embedding* of a topological linear space X into a topological linear space Y, which is nothing but a continuous and *injective* linear mapping $\iota: X \to Y$[2].
If ι is surjective and ι^{-1} is continuous as well, then both mappings are called *linear homeomorphisms* between X and Y. By $X \cong Y$ we mean that X and Y are linear-homeomorphic to each another.

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[1] Given a non-empty set M, a mapping $d: M \times M \to \mathbb{R}$ is called a *metric* on M iff the following four conditions hold for all $x, y, z \in M$:

(M-1) $d(x, y) \geq 0$,
(M-2) $d(x, y) = 0$ if and only if $x = y$,
(M-3) $d(x, y) = d(y, x)$;
(M-4) $d(x, z) \leq d(x, y) + d(y, z)$ (*triangle inequality*).

Such a function d gives rise to a topology in M, called the *metric topology* associated with d, and a subbase is given by all open balls $B_r(x) := \{y \in M : d(x, y) < r\}$ with center $x \in M$ and radius $r > 0$.

[2] The linearity of ι means that $\iota(\alpha x + \beta y) = \alpha\iota(x) + \beta\iota(y)$ holds for all $\alpha, \beta \in \mathbb{K}$ and $x, y \in X$.

§2. Seminorms and Locally Convex Spaces

The class of *locally convex spaces* can be seen as the most important one within all kinds of topological linear spaces. The topology $\mathcal{T}_V$ of a locally convex space V distinguishes by the fact that its origin $\mathsf{0}$ (and thus any other element in V) owns a neighbourhood base consisting of *convex*, *balanced* and *absorbing* sets.

It turns out that each such pair $(V, \mathcal{T}_V)$ gives rise to a family of *seminorms* on the space V. By the concept of seminorm we mean a mapping $|\cdot| : V \to \mathbb{R}$ having the properties

(SN-1) $|\lambda x| = |\lambda| \cdot |x|$ for $\lambda \in \mathbb{K}$ *(homogenity)*,

(SN-2) $|x+y| \;\leq\; |x| + |x|$ for $x, y \in V$ *(subadditivity)*,

which especially implies $|x| \leq 0$ for $x \in V$ and $|\mathsf{0}| = 0$.

Moreover, if $\mathcal{U} \subset V$ forms a convex, balanced and absorbing neighbourhood of the origin $\mathsf{0}$, then the *Minkowski functional*

$$\begin{cases} |\cdot|_{\mathcal{U}} : V \to \mathbb{R}_0^+, \\ x \mapsto |x|_{\mathcal{U}} \;:=\; \inf\{\lambda \geq 0 : x \in \lambda\mathcal{U}\} \end{cases}$$

fulfills all the properties of a seminorm. On the other hand, a family $\mathfrak{F}$ of seminorms defined on a linear space V leads in a natural way to a locally convex topology $\mathcal{T}_V$: it is the coarsest topology on V such that all seminorms in $\mathfrak{F}$ remain continuous.

In summary, there is a one-to-one correspondence between linear spaces endowed with locally convex topologies and those endowed with families of seminorms. Notice that a locally convex topology is Hausdorff iff the family of seminorms is *separating*, i. e. for each element $x \neq 0$ there is at least one seminorm $|\cdot|_\alpha$ such that $|x|_\alpha \neq 0$.

There are a two subclasses of locally convex spaces, we want to review in the following:

1. A topological linear space V is called *F-space*, if it is also a *complete* metric space with translation-invariant metric d, i. e.

$$d(x,y) \;=\; d(x+z, y+z) \;\text{ for all } x, y, z \in V,$$

and the metric topology coincides with the given topology on V.

2. If the topology of the F-space V is locally convex, then V is said to be a *Fréchet space*[3]. Alternatively, a locally convex space is

[3] A metrizable locally convex space that is not necessarily complete, is called a *pre-Fréchet space.*

Fréchet if and only if it is complete and its family of seminorms is mostly countable. In this case we may define a translation-invariant metric on a Fréchet space V according to

$$d(x,y) \ := \ \sum_{n=1}^{\infty} p_n \cdot \frac{|x-y|_n}{1+|x-y|_n} \qquad (x,y \in V)$$

where $\{p_n\}_{n=1}^{\infty}$ is any sequence of positive real numbers, whose infinite sum converges.

§3. Normed Linear Spaces

Based on the preceded results, a normed linear space is nothing but a pre-Fréchet space, whose family of separating seminorms consists of only one element $\|\cdot\| \ := \ |\cdot|_0$, which of course must be a separating one, that is, $\|x\| = 0$ implies $x = \mathsf{0}$.

It is possible to introduce the concepts of norm and normed linear space in a direct way, i. e. without using the theory on families of seminorms and locally convex spaces. Following this approach, a *normed space* X is a $\mathbb{K}$-linear space that differs from other abstract linear spaces by the presence of a particular mapping $\|\cdot\| : X \to \mathbb{R}$, denoted the *norm* on X, which fulfills for all $x, y \in X$ and $\lambda \in \mathbb{K}$

(N-1) $\quad \|\lambda\, x\| \ = \ |\lambda| \cdot \|x\|$,

(N-2) $\quad \|x+y\| \ \leq \ \|x\| + \|y\|$ *(triangle inequality)*,

(N-3) $\quad \|x\| \geq 0$ and $\|x\| = 0 \Leftrightarrow x = \mathsf{0}$.

Linear spaces can be furnished with different norms. Then two arbitrary norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ defined on a linear space X are called *equivalent* to each another, if there are constants $c_1, c_2 > 0$ such that $c_1 \|x\|_\alpha \leq \|x\|_\beta \leq c_2 \|x\|_\alpha$ for all $x \in X$.

Each normed linear space X can be equipped with a very natural topology $\mathcal{T}$, called the *norm topology* on X. It can be defined either by the assertion that $\mathcal{T} \ := \ \mathcal{T}_{\|\cdot\|}$ is the smallest topology on X such that the norm mapping remains continuous, or it is defined to be the metric topology $\mathcal{T}_d$, when X is considered as a metric space with metric function $d(x,y) \ := \ \|x-y\|$ for all $x, y \in X$. In fact, both definitions lead to the same collection of open subsets in X, so $\mathcal{T} \ := \ \mathcal{T}_{\|\cdot\|} = \mathcal{T}_d$, which justifies to speak of *the* topology of a normed linear space. Clearly, equivalent norms on X own one and the same topology. For example, if X is a finite-dimensional linear space, then all norms on X are equivalent and each of them generates the so-called *standard topology* on X.

if X and Y are normed linear spaces and ι is a *norm-preserving* linear homeomorphism between X and Y, that is, $\|\iota x\|_Y = \|x\|_X$ for all $x \in X$, we denote this mapping a *norm-isomorphism* between X and Y, and in this case both spaces are called *norm-isomorphic* to each other (we again use the notion $X \cong Y$ in this case).

A normed linear space X is called *separable* if there is a mostly countable subset lying dense in X.

We finally introduce the concept of convergence for sequences in a *normed* linear space X. Any sequence $\{x_n\}_{n\in\mathbb{N}}$ in X is called *strongly convergent* or simply *norm convergent* to $x \in X$, if it converges to x with respect to the norm topology, that is, if $\lim_{n\to\infty} \|x_n - x\| = 0$.

§4. Strictly and Uniformly Convex Spaces

In this paragraph we consider certain normed linear spaces, in which the best approximation of arbitrary elements by other elements contained in a convex subset is always possible.

A *uniformly convex space* is a *real* normed linear space X having the following property: for every $\epsilon > 0$ there is $\delta > 0$ such that for any elements x, y of the closed unit ball $B_1(0)$ of X the validness of the inequality $\|(x + y)/2\| > 1 - \delta$ implies $\|x - y\| < \epsilon$. In other words, if the center $(x + y)/2$ of the line segment $[x, y]$ between elements x, y belonging to the *unit sphere* $S_1(0) := \partial B_1(0)$ tends to $S_1(0)$, then the x and y tend together. This property is possibly lost when switching to another but equivalent norm on X.

A normed linear space X is called *strictly convex*, if the conditions $\|x\| \le 1$, $\|y\| \le 1$ and $x \ne y$ imply $\|x + y\|/2 < 1$. This means, that the *inner* line segment (x, y) between $x, y \in S_1(0)$ lies completely in the interior of $B_1(0)$, that is $\|\,\alpha x + (1 - \alpha)\, y\,\| < 1$ for $0 < \alpha < 1$. Each linear subspace of a uniformly convex Banach space owns this feature. In the light of the forthcoming theorem I.1, it is an important fact that uniformly convex spaces are also strictly convex.

Recall that a subset $\mathcal{C}$ of a linear space is called *convex*, if the line segment $[x, y]$ of any pair of points $x, y \in \mathcal{C}$ belongs to $\mathcal{C}$ again. Then many approximation problems may be formulated as follows: it is to find an element y_0 in a convex set $\mathcal{C} \subset X$ such that y_0 has minimal distance to a given element $y \notin \mathcal{C}$. The following statement, known as the *approximation theorem*, provides sufficient conditions regarding existence and uniqueness of a solution for that problem:

THEOREM I.1. Let $(X, \|\cdot\|)$ be a real normed linear space, let $y \in X$ and $\mathcal{C} \subset X$ be a *closed* and *convex* subset. Then we have:

(a) *Uniqueness*: if X is *strictly convex*, then there is mostly one element $y_0 \in \mathcal{C}$ such that $\|y_0 - y\| = \inf_{x \in \mathcal{C}} \|y - x\|$.

(b) *Existence*: if X is a *uniformly convex Banach space* then there is a unique element $y_0 \in V$ such that $\|y_0 - y\| = \inf_{x \in \mathcal{C}} \|y - x\|$.

§5. Banach and Hilbert Spaces

A normed linear space X is said to be *sequentially complete* (or briefly *complete*), if for any Cauchy sequence $\{x_n\}_{n \in \mathbb{N}}$ in X there is $x \in X$ such that $\lim_{n \to \infty} \|x_n - x\| = 0$. Equivalently, the space X is complete if and only if for every absolutely convergent series $\sum_{n=1}^{\infty} \|x_n\| < \infty$ there is $x \in X$ such that $\|x - x_n\| \to 0$ for $n \to \infty$. Each complete normed linear space is denoted a *Banach space.* Like in the non-complete case, a Banach space X with norm $\|\cdot\|$ is nothing but a Fréchet space, whose separating family of seminorms contains only this norm.

A complex (real) *Hilbert space* differs from an arbitrary Banach space by the presence of an *inner product*, which is nothing but a symmetric and positive sesquilinear (bilinear) form on such a space. This form defines the norm and the metric topology of the Hilbert space, but also determines the inner geometry to be close to that of an Euclidean space.

Let us make these assertions more precise. Given a real or complex linear space H, an *inner product* in H is defined to be a mapping $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{K}$, which is

(I-1) linear in the first argument,

(I-2) either linear or antilinear in the second argument (conditional upon, whether $\mathbb{K}$ is the real or the complex number field),

together with the additional properties

(I-3) $\langle y, x\rangle = \langle x, y\rangle$ ($\mathbb{K} = \mathbb{R}$) rsp. $\langle y, x\rangle = \overline{\langle x, y\rangle}$ ($\mathbb{K} = \mathbb{C}$),

(I-4) $\langle x, x\rangle \geq 0$ for all $x \neq 0$,

(I-5) $\langle x, x\rangle = 0$ implies $x = 0$.

Property (I-3) is denoted the *Hermitean symmetry* of $\langle \cdot, \cdot \rangle$, while (I-4) and (I-5) together mean *positive definiteness.* The presence of the inner product gives rise to define a norm in H according to

$$\|x\| := \langle x, x\rangle^{1/2} \qquad (x \in H), \tag{5.1}$$

so each linear space furnished with an inner product is normed in a natural manner. The inner product and the norm are coupled by the *Cauchy-Schwarz inequality*

$$|\langle x, y\rangle| \leq \|x\| \cdot \|y\| \qquad (x, y \in H),$$

which implies that the mapping $\langle\,\cdot\,,\,\cdot\,\rangle : H \times H \to \mathbb{K}$ is continuous. An inner product space H is called a *Hilbert space* or shortly *Hilbertian*, if it is complete with respect to the norm defined by (5.1).

The pleasant geometric properties of inner product spaces may be attributed to the validness of the *parallelogram equation*

$$2\,\|x\|^2 + 2\,\|y\|^2 = \|x+y\|^2 + \|x-y\|^2 \qquad \text{for all } x, y \in H. \tag{5.2}$$

In general, the norm $\|\cdot\|$ on a linear space X can be induced by an inner product if and only if (5.2) holds. In this case and if X is a complex linear space, the inner product can be recovered from the norm by the *polarization formula*

$$\langle x, y\rangle = \frac{1}{4}\left(\,\|x+y\|^2 - \|x-y\|^2 + i\,\|x+iy\|^2 - i\,\|x-iy\|^2\,\right).$$

If X is a real space, however, the polarization formula simplifies to

$$\langle x, y\rangle = \frac{1}{4}\left(\,\|x+y\|^2 - \|x-y\|^2\,\right).$$

In the following we want to point out that Hilbert spaces differ from general Banach spaces by a couple of desirable properties:

1. *Orthogonality.* Two elements $x, y \in H$ are called *orthogonal* to each other, if $\langle x, y\rangle = 0$ (notation: $x \perp y$). Any subset $\mathfrak{O} \subset H$ is said to be *orthogonal* or an *orthogonal system*, if all elements in $\mathfrak{O}$ are orthogonal to each other. Finally, the *orthogonal space* $\mathcal{A}^{\perp}$ of a subset $\mathcal{A} \subset H$ contains exactly the elements $y \in H$, which are orthogonal to all $x \in \mathcal{A}$, and forms a closed linear subspace of H.

2. *Projection Theorem.* Let $\mathcal{A} \subset H$ be a *closed* linear subspace. Then for each $x \in H$ there is exactly one $x_0 \in \mathcal{A}$ such that

$$\|x - x_0\| = \inf\{\,\|x-y\| : y \in \mathcal{A}\,\}.$$

The element x_0 is called the *projection* of x onto $\mathcal{A}$ and minimizes the distance between $\mathcal{A}$ and x. The associated linear mapping $\pi_{\mathcal{A}} : x \in H \mapsto x_0 \in \mathcal{A}$ is also denoted the *projection* of H onto $\mathcal{A}$. In general, by a *projection* is meant any linear mapping $\pi : H \to H$

satisfying $\pi \circ \pi = \pi$ and $\langle \pi x, y\rangle = \langle x, \pi y\rangle$ for all $x, y \in H$.

3. *Decomposition Theorem.* Using the notation of orthogonal space $\mathcal{A}^{\perp}$, the projection of an element x onto $\mathcal{A}$ is characterized as follows: $x_0 = \pi_{\mathcal{A}} x$ iff $x - x_0 \in \mathcal{A}^{\perp}$. Hence each $x \in H$ can be uniquely represented by the formula $x = x_0 + x^{\perp}$, where $x_0 \in \mathcal{A}$ and $x^{\perp} \in \mathcal{A}^{\perp}$, so the entire space H is isometrically isomorphic to the direct sum $\mathcal{A} \oplus \mathcal{A}^{\perp}$ of the closed subspaces $\mathcal{A}$ and $\mathcal{A}^{\perp}$. This fact is known as *decomposition theorem* in Hilbert spaces. In other words, each closed linear subspace $\mathcal{A}$ of H is *topologically complementable*, that is, to each closed subspace $\mathcal{A}$ there is a second closed subspace $\mathcal{B} \subset H$ such that $\mathcal{A} \cap \mathcal{B} = \{0\}$ and $[\mathcal{A} \cup \mathcal{B}] = H$. Of course, a closed subspace $\mathcal{A}$ is topologically complemented by its orthogonal space.

4. *Existence of Orthonormal Bases.* An orthogonal system $\mathfrak{O} \subset H$ is denoted an *orthonormal system*, if $\|x\| = 1$ for all $x \in \mathfrak{O}$. However, if $\mathfrak{O}$ contains only *finitely* many elements, then the span $[\mathfrak{O}]$ forms a *closed* subspace in H, and the projection $\pi_{[\mathfrak{O}]}$ of any element x onto $[\mathfrak{O}]$ can be expressed by the formula

$$\pi_{[\mathfrak{O}]}\, x \;=\; \sum_{i=1}^{n} \langle x, e_i\rangle\, e_i .$$

However, if $\mathfrak{O}$ consists of arbitrarily many elements, the inequality $\langle x, e_\alpha\rangle \neq 0$ can only be valid for mostly countable many elements $e_\alpha \in \mathfrak{O}$, which in conclusion enables to add the coefficients $\langle x, e_\alpha\rangle$ together about all elements of $\mathfrak{O}$.

An orthonormal system $\mathfrak{O}$ is called an *orthonormal base* in H, if its span $[\mathfrak{O}]$ lies dense in H. In this case we may represent each $x \in H$ according to

$$x \;=\; \sum_{e_\alpha \in \mathfrak{O}} \langle x, e_\alpha\rangle\, e_\alpha , \tag{5.3}$$

which is primarily a formal sum, but consists of mostly countable many summands. The inner products $\langle x, e_\alpha\rangle$ are called the *Fourier coefficients* of x with respect to $\mathfrak{O}$. The right hand side of (5.3) consists of *mostly countable many* non-zero summands, and this series is invariant by realignments of the elements $e_\alpha \in \mathfrak{O}$.

There is the need to determine, whether a given orthonormal system $\mathfrak{O}$ forms an orthonormal base. In fact, it can be shown that the following statements are equivalent:

(a) $\mathfrak{O}$ is an orthonormal base in H.

(b) $\mathfrak{O}$ is *complete*, that is, $\langle x, e\rangle = 0$ for all $e \in \mathfrak{O}$ implies $x = 0$.
(c) $\sum_{e_\alpha \in \mathfrak{O}} |\langle x, e_\alpha\rangle|^2 = \|x\|^2$ for all $x \in H$ (*Parseval's equation*).

It is also nearby to ask for the existence of orthonormal bases. First of all, the collection of all orthonormal systems in a Hilbert space H forms a partial ordered set, and under the assumption that ZORN's lemma[4] is valid, it contains a *maximal* element $\mathfrak{O}$ that cannot be increased, hence each Hilbert space $H \neq \{0\}$ owns an orthonormal base.

Furthermore, if $\mathfrak{O}_1$ and $\mathfrak{O}_2$ are orthonormal bases in H, then there is a bijective mapping between them, thus they own identical cardinal numbers. This legitimates us to introduce the notation of *Hilbert space dimension* $|H|$ being the cardinal number of any orthonormal base in H.

The space H is separable if and only if $|H| \leq |\mathbb{N}|$. Indeed, each orthonormal base in H forms a mostly countable set in this case, and H proves to be norm-isomorphic to the Hilbert space ℓ^2 of square summarizable sequences $f : \mathbb{N} \to \mathbb{K}$.

§6. Spaces of Scalar-Valued Functions and Measures

This paragraph forms an enumeration of well-known and important topological linear spaces of functions or measures. It is assumed that the involved functions are either real or complex-valued.

1. *Spaces of continuous functions.*
   For any topological space X the function spaces $C_b(X)$ and $C_0(X)$, both endowed with the *supremum norm*

$$\|f\| := \sup_x |f(x)|,$$

   where x runs through the elements of the underlying space X. Both are complete and hence examples of Banach spaces. If the space X is compact, then $C(X)$, the space of continuous functions $f : X \to \mathbb{C}$, forms a Banach space as well.

2. *The space $\mathscr{M}^-(X)$.*
   Recall from measure theory that a *signed measure*, defined on a measurable space $(X, \mathcal{B})$, is a *σ-additive* set function

[4] The *axion of choice* claims that if $\mathcal{F}$ is a disjoint family F_α $(\alpha \in I)$ of non-empty sets, then there is a set A that shares exactly one element with each other set $F_\alpha \in \mathcal{F}$.

$$\mu : \mathcal{B} \to (-\infty, +\infty] \quad \text{or} \quad \mu : \mathcal{B} \to [-\infty, +\infty).$$

If μ is a *finite signed measure* (the values $+\infty$ und $-\infty$ are excluded in this case), it can be represented as the difference of positive measures μ_+ (*Hahn-Jordan decomposition theorem*). In this case the positive measure $|\mu| := \mu_+ + \mu_-$ is called the *variation* of μ, and the mapping $\|\mu\| := |\mu|(X)$ is said to be the *total variation* of μ. It can be shown that the total variation constitutes a norm on the space $\mathscr{M}^-(X)$ of finite signed measures, which is even complete and hence forms a Banach space.

3. *The space $\mathscr{M}(X)$.*
Similarly to finite signed measures, a *complex measure* on a measurable space $(X, \mathcal{B})$ is a σ-additive set function $\mu : \mathcal{B} \to \mathbb{C}$. The set $\mathscr{M}(X)$ of complex measures on $(X, \mathcal{B})$ actually forms a $\mathbb{C}$-linear space. It is possible to attach to each complex measure μ an arbitrary positive measure $|\mu|$, called the *total variation* of μ. It is the mapping $|\mu| : \mathcal{B} \to [0, \infty]$, which assigns to each measurable set $A \in \mathcal{B}$ the value

$$|\mu|(A) := \sup \sum_{n=1}^{\infty} \| \mu(A_n) \| ,$$

where (A_n) may be any mostly countable and disjoint decomposition of A into measurable subsets of A. Indeed, the mapping $|\mu|$ is a positive measure, and a complex measure μ is called of *bounded variation*, if $|\mu|(X) < \infty$. The mapping

$$\mu \mapsto \|\mu\| := |\mu|(X)$$

satisfies all conditions of a norm, hence the pair $(\mathscr{M}(X), \|\cdot\|)$ forms a normed linear space. Like the space $\mathscr{M}^-(X)$ of finite signed measures, the space $\mathscr{M}(X)$ is complete and thus a Banach space.

4. *The spaces $BV[a,b]$ and $AC[a,b]$.*
Let $BV[a,b]$ be the linear space consisting of all functions of bounded variation. This space can be equipped with the norm

$$\|f\|_{bv} := |f(a)| + \sup_P \sum_{i=0}^{n_P - 1} |f(x_{i+1} - f(x_i)|,$$

where the supremum is taken over all finite partitions of $[a,b]$. The space $BV(a,b)$ forms a Banach space in this norm.

A measurable function $f:[a,b]\to\mathbb{R}$ is called *absolutely continuous*, if for each $\epsilon > 0$ there is $\delta > 0$ such that for a finite decomposition

$$x_0 = a < x_1 < \ldots < x_n = b \text{ mit } \sum_{k=1}^{n}|x_k - x_{k-1}| < \delta$$

of the interval $[a,b]$ the condition $\sum_{k=1}^{n}|f(x_k) - f(x_{k-1})| < \delta$ is valid. Every absolutely continuous function is of bounded variation. The etenge $AC[a,b]$ of absolutely continuous functions on $[a,b]$ endowed with the norm $\|\cdot\|_{BV}$ forms a closed linear subspace of $BV[a,b]$ and hence is a Banach space in its own right.

Let us continue with the consideration of function spaces, where the underling set G is a *domain* in $\mathbb{R}^d$, that is, an open and connected set within an Euclidean space $\mathbb{R}^d$, which may be bounded, unbounded or even the space $\mathbb{R}^d$ itself.

We will also work in the sequel with *multi-indices*, which are nothing but d-tuples $\alpha = (\alpha_1, \ldots, \alpha_d)$ with values $\alpha_i \in \mathbb{N}_0$ and modulus $|\alpha| := \alpha_1 + \ldots + \alpha_d$. They are mostly used to shorten multiple partial derivatives $\partial^\alpha f := \partial_1^{\alpha_1}\partial_2^{\alpha_2}\ldots\partial_d^{\alpha_d} f$ of a sufficiently often differentiable and complex-valued function f, but also to denote the multiple powers $\xi^\alpha := \xi_1^{\alpha_1}\cdot\xi_2^{\alpha_2}\cdot\ldots\cdot\xi_d^{\alpha_d}$ of a vector $\xi = (\xi_1, \xi_2, \ldots, \xi_d) \in \mathbb{C}^d$.

5. *Spaces of continuously differentiable functions.*
The linear subspace $C^k(\overline{G}) \subset C(\overline{G})$ consists of those continuously differentiable functions $f: G\to\mathbb{K}$, whose partial derivatives $\partial^\alpha f$ are bounded on $\overline{G}$ for every multi-index α with $|\alpha| \leq k$. It is complete if endowed with the norm

$$\|f\|_k := \sum_{|\alpha|\leq k} \sup_{x\in\overline{G}} |\partial^\alpha f(x)|.$$

6. *Hölder spaces.*
It has been shown that the spaces $C^k(\overline{G})$ are not very convenient for the investigation of partial differential equations, but the so-called *Hölder spaces* $C^{k,\gamma}(\overline{G})$ are so. Preliminary, a function $f: G\to\mathbb{K}$ is called *Hölder-continuous* with respect to the parameter $\gamma\in(0,1]$, if its *Hölder-norm* is finite, that is,

$$|f|_\gamma := \sup_{x,y\in G,\, x\neq y} \frac{|f(x) - f(y)|}{|x-y|^\gamma} < \infty.$$

Let $C^{k,\gamma}(\overline{G})$ be the linear space of functions $f \in C^k(\overline{G})$, whose partial derivatives $\partial^\alpha f$ are Hölder-continuous for all $|\alpha| = k$. Then the *Hölder norm* $|f|_{k,\gamma}$ of a function $f \in C^{k,\gamma}(\overline{G})$ is defined by the sum of the norm $\|f\|_k$ and the Hölder norms of the partial derivatives $\partial^\alpha f$ for $|\alpha| = k$:

$$|f|_{k,\gamma} := \|f\|_k + \sum_{|\alpha|=k} |\partial^\alpha f|_\gamma .$$

The space $C^{k,\gamma}(\overline{G})$ is complete with respect to this norm. Notice that for $k=0$ and $\gamma=1$ we obtain the space $C^{0,1}(\overline{G})$ of *Lipschitz-continuous* functions.

If G is a bounded domain, we have for $0 < \alpha < \beta \le 1$

$$C^1(\overline{G}) \subsetneqq C^{0,\beta}(\overline{G}) \subsetneqq C^{0,\alpha}(\overline{G}) \subsetneqq C(\overline{G}).$$

7. *Spaces of smooth functions.*
Next we consider three function spaces consisting of indefinitely differentiable functions $f : G \to \mathbb{K}$ that frequently occure in applied analysis and constitute Fréchet spaces respectively complete locally convex linear spaces.
(a) The space $C^\infty(G)$ of functions $f : G \to \mathbb{K}$ being indefinitely differentiable. Unfortunately, there is no norm that makes $C^\infty(G)$ into a complete normed space, but after all it is possible to define a countable family $\mathfrak{F}$ of seminorms to make $C^\infty(G)$ into a Fréchet space. For example, $\mathfrak{F}$ can be given by the mappings

$$p_{N,n} := \max_{x \in K_n} \{ \, |(\partial^\alpha \phi)(x)| : |\alpha| \le N \} \qquad (N, n \in \mathbb{N}),$$

where (K_n) is a sequence of compact subsets of G such that

$$K_1 \subset K_2 \ldots \subset G \text{ for } n \in \mathbb{N}, \quad \bigcup_{n \in \mathbb{N}} K_n = G,$$

and each K_n lies in the interior of K_{n+1} (such a sequence is called a *compact exhaustion* of G). The space $C^\infty(G)$ is often denoted $\mathcal{E}(G)$, when considered in the locally convex topology induced by the totality of seminorms $p_{N,n}$. It is complete with respect to this topology, so it forms a Fréchet space.
(b) The subspace $C_c^\infty(G) \subset C^\infty(G)$ of smooth functions having *compact* support

$$\operatorname{supp}(\phi) := \overline{\{x \in G : \phi(x) \neq 0\}},$$

usually denoted the *space of test functions.* This function space plays an outstanding role in analysis, since it lies dense in a multiplicity of other function spaces. Unfortunately, there is neither a countable family of seminorms making $C_c^\infty(G)$ into a Fréchet space nor is $C_c^\infty(G)$ completely metrizable.

But it is still possible to define a locally convex topology $\mathcal{T}$ on $C_c^\infty(G)$ such that $C_c^\infty(G)$ is complete in the sense of a topological linear space. Instead of saying how the open sets of this topology $\mathcal{T}$ look like, let us define $\mathcal{T}$ by means of a suitable notion of convergence, known as $\mathscr{D}$-convergence in $C_c^\infty(G)$: a sequence (ϕ_n) in $C_c^\infty(G)$ is convergent to $\phi \in C_c^\infty(G)$, if there is a compact set $K \subset G$ such that (i) $\operatorname{supp}(\phi_n) \subset K$ for all $n \in \mathbb{N}$ and (ii) for each multi-index α the sequence $(\partial^\alpha \phi_n)$ converges to $\partial^\alpha \phi$ *uniformly* on K.
The space $C_c^\infty(G)$ equipped with the topology induced by $\mathscr{D}$-convergence is usually denoted $\mathscr{D}(G)$. Notice that we have $\mathscr{D}(G) \hookrightarrow \mathcal{E}(G)$, that is, the space $\mathscr{D}(G)$ is continuously embedded into the space $\mathcal{E}(G)$ with respect to the locally convex topologies of these spaces.

We have $\mathscr{D}(G) \hookrightarrow \mathcal{E}(G)$, that is, the space $\mathscr{D}(G)$ is continuously embedded into the space $\mathcal{E}(G)$ with respect to their locally convex topologies.

(c) The *Schwartz space* $\mathscr{S}(\mathbb{R}^d)$, which is the space of all smooth functions $\phi \in C^\infty(\mathbb{R}^d)$ having the additional property

$$|\partial^\alpha \phi(x) \cdot x^\beta| \to 0 \ \text{ for } \ \|x\|_d \to \infty$$

and arbitrary multi-indices α, β, where $\|x\|_d$ is the Euclidean norm of $x \in \mathbb{R}^d$. The elements of $\mathscr{S}(\mathbb{R}^d)$ are called *rapidly decreasing* or briefly *Schwartz functions.* Hence a Schwartz function and all its derivatives vanish at infinity faster then the reciprocal of any polynomial. The most famous example of a Schwartz function is the Hermite function

$$\mathcal{H}_0(x) \ := \ e^{-\frac{\|x\|^2}{2}} \qquad (x \in \mathbb{R}^d).$$

Rather similar to the space $C^\infty(G)$, it is possible to find for the Schwartz space $\mathscr{S}(\mathbb{R}^d)$ a family of seminorms as follows: define for $\phi \in C^\infty(\mathbb{R}^d)$ and multi-indices α, β

$$p_{\alpha,\beta}(\phi) \ := \ \sup_{x \in \mathbb{R}^d} |\, x^\alpha (D^\beta \phi)(x)\,|. \tag{6.1}$$

Then ϕ is a Schwartz function iff $p_{\alpha,\beta}(\phi) < \infty$ holds for all indices α, β. The collection of mappings $p_{\alpha,\beta} : \mathscr{S}(\mathbb{R}^d) \to \mathbb{R}$ defines a countable family of seminorms, which gives rise to a locally convex topology $\mathcal{T}_{\mathscr{S}}$ on $\mathscr{S}(\mathbb{R}^d)$. Moreover, since the Schwartz space is complete with respect to $\mathcal{T}_{\mathscr{S}}$, it actually forms a *Fréchet space.*

Comparing the Schwartz space $\mathscr{S}(\mathbb{R}^d)$ with the function spaces $\mathscr{D}(\mathbb{R}^d)$ and $\mathcal{E}(\mathbb{R}^d)$, it is easy to see that there is the chain of inclusions

$$\mathscr{D}(\mathbb{R}^d) \hookrightarrow \mathscr{S}(\mathbb{R}^d) \hookrightarrow \mathcal{E}(\mathbb{R}^d).$$

Moreover, by comparing the locally convex topologies of these spaces, it turns out that these inclusions are continuous.

8. *Lebesgue spaces.*

To introduce the spaces $L^p(G)$ for $p \in [1, \infty)$, let $\mathcal{L}^p(G)$ be the $\mathbb{K}$-linear space of measurable functions $f : G \to \mathbb{K}$ such that $|f|^p$ is Lebesgue-integrable.

We define for $f, g \in \mathcal{L}^p(G)$ the equivalence relation $f \sim g$ according to $f = g$ almost everywhere. Then the elements of the space $L^p(G)$ are the equivalence classes in $\mathcal{L}^p(G)$ with respect to $\sim$, that is, $L^p(G) := \mathcal{L}^p(G)/\sim$. The *Lebesgue norm* in $L^p(G)$ is given by

$$\|f\|_p := \Big(\int_G |f|^p \, d\mu \Big)^{\frac{1}{p}}.$$

Finally, the Fischer-Riesz *theorem* claims that the spaces $L^p(G)$ are complete with respect to the Lebesgue norm. It turns out that $L^p(G)$ is separable if and only if $1 \le p < \infty$.

The case $p = 2$ gives rise to a special attention: here it is possible to derive the norm $\|\cdot\|_2$ from the inner product

$$\langle f, g\rangle := \int_G f \, \overline{g} \, d\mu,$$

thus $\|f\|_2 = \langle f, f\rangle^{1/2}$ and $L^2(G)$ becomes into a Hilbert space.

For $p = \infty$ and any measurable function $f : G \to \mathbb{K}$ we define the mapping

$$\|f\|_\infty := \inf\{\, a \ge 0 : |f| \le a \ \mu\text{-almost everywhere}\,\}, \qquad (6.2)$$

taking values in the set $\mathbb{R}^+ \cup \{\infty\}$. Let $\mathcal{L}^\infty(G)$ be the linear space of all measurable functions $f : G \to \mathbb{K}$ with $\|f\|_\infty < \infty$ and

$L^\infty(G)$ be the linear space of all equivalence classes in $\mathcal{L}^\infty(G)$, where $f \sim g$ means $f = g$ μ-almost everywhere. Members of these spaces are called *essentially bounded.* Similar to the case $p < \infty$, the space $L^\infty(G)$ is complete with respect to (6.2), but not separable.

9. *Sobolev spaces.*
A function $f : G \to \mathbb{C}$ is said to belong to the space $L^1_{loc}(G)$, if f is Lebesgue-integrable on each compact subset of G. This space is useful to define the concept of *weak differentiability.* A function $f \in L^1_{loc}(G)$ is said to be *weakly differentiable* with respect to a given multi-index α, if there is $g \in L^1_{loc}(G)$ such that

$$\int_G g(x)\,\phi(x)\,dx = (-1)^{|\alpha|} \int_G f(x)\,\partial^\alpha \phi(x)\,dx \text{ for } \phi \in C_c^\infty(G).$$

We denote $\partial^\alpha f := g$ in this case, and weak differentiability arises as a linear mapping $\partial^\alpha : D_\alpha \subset L^1_{loc}(G) \to L^1_{loc}(G)$, where each D_α is a linear subspace of $L^1_{loc}(G)$. Notice that the concept of weak derivative is compliant with that of the classical derivative: if f has classical derivative $\partial^\alpha f$, then f is also weakly differentiable, and the weak derivative is equal to $\partial^\alpha f$.

(a) The next function spaces under consideration are those, whose elements are p-integrable and weakly differentiable up to order k at the same time. More precisely, for any integer k and real $p \geq 1$, the *Sobolev space* $W^{k,p}(G)$ is the set of functions f in the Lebesgue space $L^p(G)$, whose weak derivatives $\partial^\alpha f$ exist and also belong to $L^p(G)$ for all multi-indices α with $|\alpha| \leq k$, briefly written

$$W^{k,p}(G) := \{ f \in L^p(G) : \partial^\alpha f \in L^p(G),\ |\alpha| \leq k \}.$$

Sobolev spaces are of course linear spaces, and especially for $k = 0$ the spaces $W^{0,p}(G)$ are nothing but the Lebesgue spaces $L^p(G)$ themselves.

For functions $f \in W^{k,p}(G)$ let the mapping $f \mapsto |f|_{k,p}$ be defined according to

$$|f|_{k,p} := \Big(\sum_{|\alpha| \leq k} \int_G |\partial^\alpha f|^p\,dx \Big)^{1/p}. \tag{6.3}$$

It fulfills all the properties of a norm, thus the space $W^{k,p}(G)$ is actually a normed space, and the mapping defined by (6.3) is usually denoted a *Sobolev norm.* With respect to the question of completeness, it is not hard to show that the spaces $W^{k,p}(G)$ are complete and therefore constitute Banach spaces as well.

There is a second way to define Sobolev space. Consider the subspace

$$C^\infty_{k,p}(G) \ := \ \{\, f \in C^\infty(G) : |f|_{k,p} < \infty \,\},$$

and since this space is *not* complete with respect to the norm $|\cdot|_{k,p}$, we set

$$\mathcal{W}^{k,p}(G) \ := \ \overline{C^\infty_{k,p}(G)}.$$

The space $\mathcal{W}^{k,p}(G)$ is defined by a purely topological operation, that is, by taking the closure of a smaller space. Thus for elements in this space there is no instrinsic notion of derivative, and one introduces the notion of *strong derivative* as follows: given $u \in \mathcal{W}^{k,p}(G)$ and a multi-index α ($|\alpha| \leq k$), a further function $\partial^\alpha u$ also belonging to $\mathcal{W}^{k,p}(G)$ is said to be the α-strong derivative of u, if there is a sequence $\{\phi_n\} \in C^\infty_{k,p}(G)$ converging to u such that

$$|\partial^\alpha u - \partial^\alpha \phi_n)|_{k,p} \to 0 \qquad (n \to \infty).$$

Sobolev spaces are reflexive for $1 < p < \infty$, but not for $p = 1$.

The MEYERS-SERRIN *theorem* claims $\mathcal{W}^{k,p}(G) \cong W^{k,p}(G)$ in the sense of Banach space isomorphisms. Moreover, the notions of strong and weak derivative are equivalent when identifying functions in $\mathcal{W}^{k,p}(G)$ and $W^{k,p}(G)$.

One of the most important properties of Sobolev spaces is given by the fact that under certain hypotheses they can continuously be embedded into further spaces and those of type $C^k(\overline{G})$. We define for fixed $1 \leq p < \infty$ the *Sobolev numbers* $\theta_{k,d,p}$ by

$$\theta_{k,d,p} := k - d/p \qquad (k, d \in \mathbb{N}, p \geq 1),$$

which play a crucial role in SOBOLEV*'s embedding theorem*:

THEOREM I.2. Let $G \subset \mathbb{R}^d$ be a domain having Lipschitz-continuous boundary. If k, m are chosen such that $\theta_{k,d,p} > m$, then there are continuous embeddings $W^{k,p}(G) \hookrightarrow C^m(\overline{G})$ and $W^{k,p}(G) \hookrightarrow C^{m,\gamma}(\overline{G})$ for $0 < \gamma < 1$.

Hence each class $f \in W^{k,p}(G)$ has a representative in these spaces, supposed the Sobolev number is sufficiently large.

In case of $\theta_{k,d,p} \le 0$, however, one cannot embed $W^{k,p}(G)$ into $C(\overline{G})$. But it is at least possible to find an embedding of $W^{k,p}(G)$ into some lower-dimensional Lebesgue spaces $L^q(G)$:

THEOREM I.3. For $G \subset \mathbb{R}^d$, $k \in \mathbb{N}$ and $p, q \ge 1$ such that

$$1 < q < \frac{dp}{d - kp},$$

there are continuous embeddings $W^{k,p}(G) \hookrightarrow L^q(G)$.

Finally, there are also embeddings of Sobolev spaces into further Sobolev spaces, where the parameter p is fixed:

THEOREM I.4. If $G \subset \mathbb{R}^d$ has Lipschitz-continuous boundary, $s, s' \in \mathbb{N}$ and $1 < s < s' < \infty$, then $W^{s',p}(G) \hookrightarrow W^{s,p}(G)$.

(b) Let us consider in the family $W^{k,p}(G)$ of Sobolev spaces the case $p = 2$ and denote the space $H^k(G) := W^{k,2}(G)$ for $k \in \mathbb{N}_0$, which is often called a *Hilbert-Sobolev space.*

Each space $H^k(G)$ can be endowed with the inner product

$$\langle f, g \rangle_k := \sum_{|\alpha| \le k} \int_G \partial^\alpha f \, \partial^\alpha \overline{g} \, dx \,, \qquad f, g \in H^k(G)$$

and with norm $|f|_k = \langle f, f \rangle^{1/2}$ for $f \in H^k(G)$, so $H^k(G)$ forms a Hilbert space in its own right.

(c) The space $C_c^\infty(G)$ of test functions can surely be endowed with the norms $|\cdot|_{k,p}$, but this leads to uncomplete normed spaces. Hence it is nearby to introduce the spaces $W_0^{k,p}(G)$ for $k \in \mathbb{N}$ by completion of the space $C_c^\infty(G)$ with respect to the Sobolev norms:

$$W_0^{k,p}(G) := \overline{C_c^\infty(G)}_{|\cdot|_{k,p}}.$$

The elements of these spaces can be identified with functions in $W^{k,p}(G)$. Furthermore, the spaces $W_0^{k,p}(G)$ form closed linear subspaces of the Sobolev spaces $W^{k,p}(G)$. Clearly, the spaces $H_0^k(G) := H_0^{k,2}(G)$ are Hilbert spaces with inner products and norms as defined for $H^k(G)$.

§7. Hierarchy of Topological and Normed Linear Spaces

The following figure shows the inheritance tree of topological and normed linear spaces, where Banach and Hilbert spaces are positioned at the very end of this tree. For the notion of *reflexive space* we refer to section III.

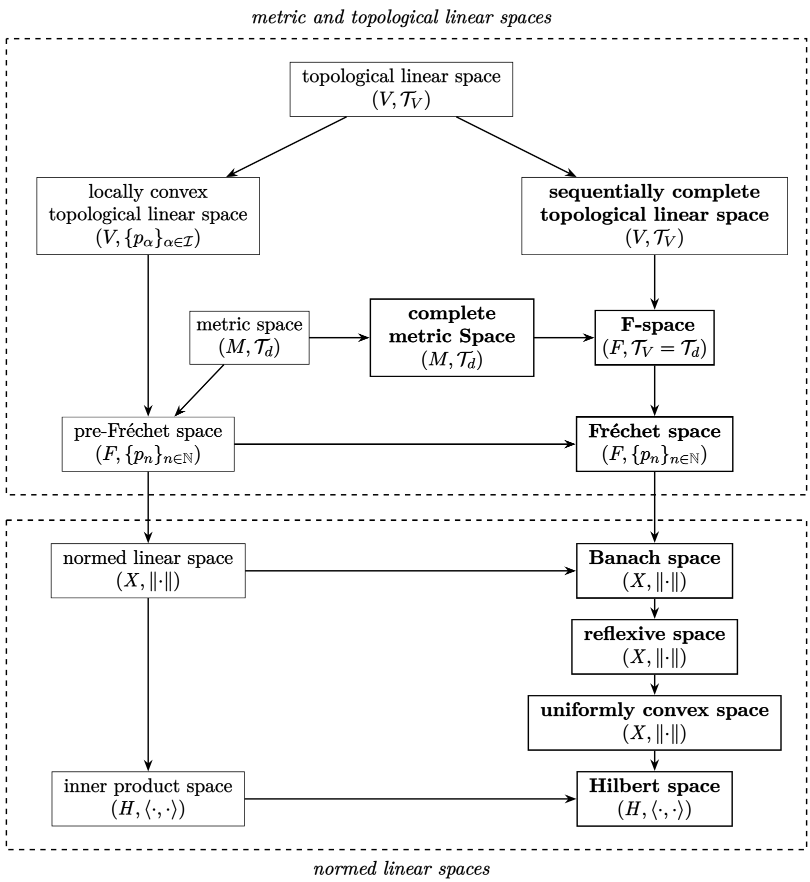

Figure 7.1: The hierarchy of metric and topological linear spaces

References

[1] N. Dunford, J. T. Schwartz: *Linear Operators I.* Wiley & Sons, 1958.

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II.
Linear Operators in Banach Spaces

The aim of this second part is to present some fundamental facts on linear operators acting between Banach spaces. Furthermore, we want to introduce various classes of linear operators by means of the validness of certain topological properties, which to the end lead to the concept of *continuous*, *compact*, *closed* and *closable* operator. It is also our intention to point out certain relationships between these notions and to establish some equivalences.

§1. Fundamentals of Operator Theory

Let X and Y be normed linear spaces (or more general topological linear spaces). By a *linear operator* between X and Y is meant a mapping $A : \mathfrak{D}(A) \subset X \to Y$, where the *domain of definition* $\mathfrak{D}(A)$ forms a linear subspace of X and A is *linear*, that is,

$$A(\lambda x + \mu y) \;=\; \lambda\, Ax + \mu\, Ay$$

holds for all $x, y \in \mathfrak{D}(A)$ and $\lambda, \mu \in \mathbb{K}$. The space X is denoted the *domain space* of A, whereas Y is said to be the *range space* of A.

The domain of definition $\mathfrak{D}(A)$ may of course coincide with the entire space X, but it is often sufficient to require that $\mathfrak{D}(A)$ lies dense in X, so each $x \in X$ can be approximated by a sequence of elements in $\mathfrak{D}(A)$. It is usual to say that A is *densely defined* in X in this case. However, if the set $\mathfrak{D}(A)$ is *not* dense in X, it is nearby to consider the closure $X_0 := \overline{\mathfrak{D}(A)} \subset X$ and to pay attention to the slightly modified but densely defined operator $A_0 : \mathfrak{D}(A) \subset X_0 \to Y$.

Rather the domains of definition for operators A_1 and A_2 with same domain space X and range space Y may be proper linear subspaces of X, the *sum* $A_1 + A_2$ and the *composition* $A_1 \circ A_2$ are well-defined, if their domains of definition are given by

$$\begin{cases} \mathfrak{D}(A_1 + A_2) \;:=\; \mathfrak{D}(A_1) \cap \mathfrak{D}(A_2), \\ \mathfrak{D}(A_1 \circ A_2) \;:=\; \{\, x \in \mathfrak{D}(A_2) : A_2\, x \in \mathfrak{D}(A_1) \,\}. \end{cases}$$

Furthermore, the operator λA is well-defined by $\mathfrak{D}(\lambda A) := \mathfrak{D}(A)$ and $(\lambda A)x := \lambda(Ax)$ for $x \in \mathfrak{D}(\lambda A)$, where $\lambda \in \mathbb{K}$ is arbitrary.

If the domain of definition of a linear operator is a proper subset of the domain space, the concept of *extension* is meaningful. Let A_1

and A_2 be linear operators with domain of definitions $\mathfrak{D}(A_1) \subset X$ and $\mathfrak{D}(A_2) \subset X$, respectively. We introduce the partial ordered relation

$$A_1 \prec A_2 :\Longleftrightarrow \mathfrak{D}(A_1) \subset \mathfrak{D}(A_2) \text{ and } A_1 x = A_2 x \text{ for all } x \in \mathfrak{D}(A_1)$$

and denote A_2 an *extension* of A_1 if $A_1 \prec A_2$. In addition, the operators A_1 and A_2 are said to be *equal* (briefly $A_1 = A_2$) iff $A_1 \prec A_2$ and $A_2 \prec A_1$, or equivalently, if $\mathfrak{D}(A_1) = \mathfrak{D}(A_2)$ and $A_1 x = A_2 x$ for all $x \in \mathfrak{D}(A_1)$.

Moreover, to every linear operator $A : \mathfrak{D}(A) \subset X \to Y$ there are basically assigned three linear subspaces, namely

1. the *kernel* $\mathfrak{N}(A) = \{ x \in \mathfrak{D}(A) : Ax = 0 \} \subseteq X$;
2. the *range* $\mathfrak{R}(A) = \{ Ax : x \in \mathfrak{D}(A) \} \subseteq Y$;
3. the *cokernel* $\mathfrak{C}(A) \subseteq Y$, i. e. the linear-algebraic complement of $\mathfrak{R}(A)$ within the space Y, hence $\mathfrak{R}(A) \oplus \mathfrak{C}(A) = Y$.

A first classification of linear operators between a domain space X and a range space Y can be performed by checking certain properties, which are defined by means of the norm topologies in these spaces. It leads in conclusion to the concepts of *continuous*, *compact*, *closed* and *closable* operator.

§2. Continuous Linear Operators

A linear operator $A : \mathfrak{D}(A) \subset X \to Y$ is called *continuous*, if the preimage $A^{-1}(\mathcal{O})$ of each open set $\mathcal{O} \subset Y$ is open in the domain space X, which of course is the well-known definition of continuity for mappings between arbitrary topological spaces.

For the mappings between normed linear spaces, however, we may replace continuity by *sequential continuity*, which means that a mapping f is continuous in $x \in X$ if for each convergent series (x_n) with limit element $x \in X$ we have $\lim_{n\to\infty} f(x_n) = f(x)$. The equivalence is especially true if f forms a linear operator A and the domain of definition $\mathfrak{D}(A)$ is a proper subset of X.

If X, Y are normed, the domain is dense in X and A is continuous, then A can uniquely be extended to a linear mapping defined on the closure $\overline{\mathfrak{D}(A)} = X$, which proves to be continuous as well. This is the assertion of the *B.L.T. theorem* (written out the *bounded linear transformation theorem*, see [2], page 6). It is given by the fact that any continuous linear operator A is even *uniformly* continuous, that is, for each $\epsilon > 0$ there exists $\delta > 0$, such that $\|x - y\|_X < \delta$ implies $\|Ax - Ay\|_Y < \epsilon$ for *all* $x, y \in \mathfrak{D}(A)$, hence δ can be chosen independently of x and y.

Hence we may basically assume that if A is continuous, then $\mathfrak{D}(A) = X$ by applying this canonical extension, so A appears as a mapping $A : X \to Y$.

Let us turn to the next concept of importance. A linear operator A acting between normed linear spaces X and Y is called *bounded* if and only if it is *norm-bounded*, that is, if there is $c \geq 0$ such that $\|Ax\| \leq c \cdot \|x\|$ for each $x \in X$. In this case we define the *operator norm* of A according to

$$\|A\| := \sup_{\|x\|=1} \|Ax\| . \tag{2.1}$$

The following result on continuous operators is well-known:

THEOREM II.1. For $A : X \to Y$ being a linear operator, the subsequent assertions are equivalent to each other:
(a) A is continuous;
(b) A is uniformly continuous;
(c) A is sequentially continuous in every $x \in X$;
(d) A is sequentially continuous at the origin $0 \in X$;
(e) A is bounded.

Let $\mathscr{L}(X, Y)$ be the linear space of all continuous linear operators $A : X \to Y$, which forms a subspace of $L(X, Y)$, the space of *all* linear but not necessarily continuous mappings. The space $\mathscr{L}(X, Y)$ from a normed linear space X to a Banach space Y and endowed with the norm given by (2.1) is complete, since Y is supposed to be complete, hence $\mathscr{L}(X, Y)$ forms a Banach space in its own right.

In addition, in case of $Y = X$ the space $\mathscr{L}(X) := \mathscr{L}(X, X)$ constitutes a *normed algebra* with composition $(A_1, A_2) \mapsto A_2 \circ A_1$ as its multiplicative operation and owns the property

$$\|A_1 \circ A_2\| \leq \|A_1\| \cdot \|A_2\| .$$

It is called the *operator algebra* associated with X.

§3. Compact Linear Operators

A linear operator $k : X \to Y$ acting between normed linear spaces X, Y is called *compact*, if the image of each bounded set under

k is relative compact with respect to the norm topology in Y, or equivalently, if $\{x_n\}_{n\in\mathbb{N}}$ is a bounded sequence in X, then the sequence $\{k\,x_n\}_{n\in\mathbb{N}}$ in Y owns a convergent subsequence.

For example, a linear operator $k : X \to Y$ is compact, if at least one of the spaces X and Y is finite-dimensional. In case of $Y = X$, the identity operator $1_X : X \to X$ is compact iff X is finite-dimensional.

Further properties of compact linear operators are:

(a) every compact operator is continuous (the opposite assumption leads to a contradiction);
(b) the composition $A_1 \circ A_2$ of continuous operators is compact, if one of them is so;
(c) finite linear combinations of compact linear operators are compact, too;
(d) the limit of a norm convergent sequence of compact operators is compact.
(e) the first and second property together imply that the set $\mathscr{C}(X)$ of compact operators $k : X \to X$ is a *closed* ideal within the operator algebra $\mathscr{L}(X)$, hence it is complete and forms a Banach algebra in its own right;

A linear operator $f : X \to Y$ is said to have *finite rank*, if the range $\mathfrak{R}(f)$ is finite-dimensional. It is obvious that f is compact.

Suppose that (f_n) is a sequence of finite-rank operators converging to the operator A with respect to the norm of $\mathscr{L}(X, Y)$. Then A is compact and the set $\mathscr{C}(X, Y)$ of compact linear operators proves to be a subset of the closure of the set $F(X, Y)$ of finite-rank operators, i. e. $\mathscr{C}(X, Y) \subset \overline{F(X, Y)}$.

Reversely, in the setting of Hilbert spaces each compact linear operator k appears in this way, that is, k is the norm-limit of a sequence of finite-rank operators (but this is not necessarily true in the general setting of Banach spaces).

Let $\mathfrak{B} = \{e_1, e_2, \ldots\}$ be any orthonormal base of a *separable* Hilbert space H and let $A \in \mathscr{L}(H)$. We define the number $S(A)$ according to

$$S(A) := \sum_{i\in\mathfrak{B}} \|A\,e_i\|^2 .$$

Then it turns out that this definition for $S(A)$ does not depend on the choice of the orthonormal base. In case of $S(A) < \infty$ the operator A is called a HILBERT-SCHMIDT *operator*, and the number

$$\|A\|_{HS} := \sqrt{S(A)}$$

is denoted the *Hilbert-Schmidt norm* of A, since the mapping $A \mapsto \|A\|_{HS}$ owns all the properties of a norm.

The set $S(H)$ of Hilbert-Schmidt operators in H is topologically and algebraically closed in $\mathscr{L}(H)$ with respect to the operations $A_1 + A_2$ and λA, hence $S(H)$ endowed with norm $\|\cdot\|_{HS}$ forms a Banach space in its own right. Each Hilbert-Schmidt operator proves to be compact, so $S(H)$ forms a two-sided ideal within the space $\mathscr{C}(H)$ of compact operators acting in H.

§4. Closed and Closable Linear Operators

A linear operator $A:\mathfrak{D}(A) \subset X \to Y$ is called *graph-closed* (or shortly *closed*) if the set

$$\Gamma_A := \{ (x, Ax) : x \in \mathfrak{D}(A) \} \subset X \times Y,$$

denoted the *graph* of A, is closed in the space $X \times Y$ with respect to the product topology $\mathcal{T}_{X \times Y}$. This property especially implies that the kernel $\mathfrak{N}(A)$ of A forms a closed set in X.

It is a matter of fact that an operator A is closed iff it is *sequentially closed*, that is, $x_n \to x$ and $Ax_n \to y$ implies $x \in \mathfrak{D}(A)$ and $Ax = y$, briefly written

$$\left.\begin{array}{r} x_n \to x \\ Ax_n \to y \end{array}\right\} \quad \Longrightarrow \quad x \in \mathfrak{D}(A) \text{ and } Ax = y.$$

This can easily proved by the fact that a subset of a normed linear space is closed if and only if it is sequentially closed. Checking the sequentially closeness is the usual method to verify that a given linear operator is graph-closed. But notice that the equivalence of these both notations is in general not valid, if X and Y are non-metrizable topological linear spaces.

It is often helpful to make the domain of definition $\mathfrak{D}(A)$ of a linear operator A into a normed linear space by its own. This can be done by introducing the so-called *graph norm*

$$\|x\|_A := \|x\|_X + \|Ax\|_Y$$

for $x \in \mathfrak{D}(A)$. In fact, the pair $(\mathfrak{D}(A), \|\cdot\|_A)$ forms a normed linear space, which constitutes a Banach space if and only if A is sequentially closed. This equivalence provides a further method to verify closeness of a given linear operator.

The following theorem uses the possible validness of an abstract SCHAUDER *estimate* to ensure sequentially closenss:

THEOREM II.2. Let $A_0 : X \to Y$ be a continuous linear operator, where the Banach space X is continuously and densely embedded into the Banach space Y by a mapping $\iota : X \to Y$. If the estimate

$$\|x\|_X \le c \cdot \{ \|\iota x\|_Y + \|A_0 x\|_Y \} \tag{4.1}$$

is fulfilled for some $c > 0$ and all $x \in X$, then the linear operator A acting in Y with $\mathfrak{D}(A) := \iota(X)$ and $A\,\iota x := A_0 x$ for $x \in X$ is closed.

In fact, let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $\mathfrak{D}(A)$ with $\|x_n - x\|_X \to 0$ and $\|A\,x_n - y\|_Y \to 0$. It is to verify that $x \in \mathfrak{D}(A)$ and $Ax = y$. Inequality (4.1) implies that $(x_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in X with limit element $x \in X$. Since A_0 is continuous, the element y is also the limit of the sequence $(A\,\iota x_n)_{n\in\mathbb{N}}$, hence A is closed.

Many linear operators acting between Banach or Hilbert spaces are not closed, but can be extended to operators being closed. We shall call such operators *closable* or *pre-closed.* If a linear operator A is closable, then there is a *minimal closed extension* A_c of A, called the *closure* of A. This mapping A_c is given by the fact that each other closed extension of A extends A_c, and A_c is nothing but the intersection of all closed extensions of A. Clearly, each closed operator is closable, since it constitutes a closed extension of itself. Furthermore, given a closed linear operator A with domain of definition $\mathfrak{D}(A)$, any linear subspace $\mathcal{L} \subset \mathfrak{D}(A)$ is called a *core* of A, if the closure of A restricted to $\mathcal{L}$ resembles A.

There is a well-known criterion for closability:

THEOREM II.3. A linear operator A with domain of definition $\mathfrak{D}(A)$ is closable, if each sequence $(x_n, Ax_n)_{n\in\mathbb{N}}$ with $x_n \in \mathfrak{D}(A)$ for $n \in \mathbb{N}$ and converging to $(0, y)$ implies $y = 0$, briefly written

$$\left.\begin{array}{r} x_n \to 0 \\ Ax_n \to y \end{array}\right\} \quad \Longrightarrow \quad y = 0.$$

Let us emphasize that in general the concepts of closeness and continuity of a linear operator A are not comparable. However, if

X and Y are Banach or Hilbert spaces and $\mathfrak{D}(A)$ is a *closed* set in X, then closeness of A is an easy consequence of continuity of A.

This assertion is especially true in the following situation: if A is densely defined and continuous, then it can be extended to a continuous linear operator $A_{ext} : X \to Y$, and since X forms a closed linear space, the extension A_{ext} is closed, too. Hence continuous linear operators of the form $A : X \to Y$ fit in a natural way into the class of densely defined and closed linear operators.

§5. The Closed Graph Theorem

Let us first emphasize that in general the concepts of closeness and continuity of a linear operator A are not comparable. However, if X and Y are Banach or Hilbert spaces and $\mathfrak{D}(A)$ is a *closed* set in X, then closeness of A is an easy consequence of continuity of A. This fact can be applied to the following situation: if A is densely defined and continuous, then it can be extended to a continuous linear operator $A_{ext} : X \to Y$, and since X forms a closed linear space, the extension A_{ext} is closed, too. Hence continuous linear operators of the form $A : X \to Y$ fit in a natural way into the class of densely defined and closed linear operators.

For the rest of this paragraph it is assumed that the spaces X, Y are Banach. The *closed graph theorem* claims that under a certain condition the reverse statement is also true:

THEOREM II.4. If the linear subspace $\mathfrak{D}(A)$ is closed in X, then closeness of A implies continuity, so both concepts are equivalent to each other in this case.

The following application of the closed graph theorem is of outstanding meaning:

THEOREM II.5. If a linear operator A is closed and bijective, then its inverse A^{-1} is also linear and closed, and since the domain of definition $\mathfrak{D}(A^{-1})$ is the entire space Y, the closed graph theorem implies that $A^{-1} : Y \to X$ is continuous[5].

The closed graph theorem is a consequence of BANACH's *open mapping theorem*:

[5] One also says that A is *invertible* if it is bijective and A^{-1} is continuous.

THEOREM II.6. If A is continuous and surjective, then the image $A(\mathcal{O})$ of an open set $\mathcal{O} \subset X$ is open in Y. Moreover, if A is injective, then A^{-1} is continuous and constitutes a linear homeomorphism between the spaces X and Y (this is the *bounded inverse theorem* for continuous operators).

The closed graph theorem, the open mapping theorem and the bounded inverse theorem are equivalent to each other and can be derived from *BAIRE's category theorem*, which is a fundamental result in general topology.

In conclusion, if the domain of definition $\mathfrak{D}(A)$ of a linear operator A is a *closed* subspace of the domain space X, then the following chain of implications is valid:

$$A \text{ compact} \implies A \text{ bounded} \iff A \text{ continuous} \iff A \text{ closed}.$$

Regarding the solvability of $Ax = y$ where A is acting between Banach spaces X and Y, the following concepts are available: the equation $Ax = y$ is called

(a) *densely solvable* ([3]), if $\overline{\mathfrak{R}(A)} = Y$;
(b) *uniquely solvable* ([3]), if A is injective;
(c) *everywhere solvable* ([3]), if A is surjective;
(d) *solvable* ([1]), if it is uniquely and everywhere solvable;
(e) *well-posed*, if it is solvable and A^{-1} is continuous.

Hence well-posedness means that solutions of the linear equations depend continuously on the right hand side. We want to emphasize that if the linear operator A is continuous or closed, then property (d) implies property (e) by the bounded inverse theorem.

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III.
Adjoint Spaces and Operators

It is often useful to combine the study of a topological or normed linear space X with that of its *dual space* X^*, which is nothing but the set of continuous and $\mathbb{K}$-valued linear mappings endowed with a linear-algebraic structure. Dual spaces are sometimes of interest in its own right, since they may constitute extensions of the original space X, which has many helpful applications in higher analysis.

§1. Continuous Linear Functionals

Let X be an arbitrary topological $\mathbb{K}$-linear space. The underlying fields $\mathbb{R}$ or $\mathbb{C}$ surely form topological $\mathbb{K}$-linear spaces again, where the topologies are given by the so-called *standard topologies.*

The main objects in this paragraph are *continuous linear functionals* on X, that is, continuous mappings of the form $\phi : X \to \mathbb{K}$, where ϕ is linear at the same time, so

$$\phi(\alpha x + \beta y) \;=\; \alpha\,\phi(x) + \beta\,\phi(y)$$

holds for all $\alpha, \beta \in \mathbb{K}$ and $x, y \in X$. We define the mappings $\phi_1 + \phi_2$ and $\alpha\phi$ pointwise for $x \in X$, so the set X^* of all continuous linear functionals on X forms a $\mathbb{K}$-linear space, too, called the *adjoint space* or the *dual space* of X. If the topology on X is locally convex, then a specific version of the celebrated *Hahn-Banach theorem* ensures that there are non-trivial continuous linear mappings on X, thus we may assume $X^* \neq \{0\}$.

In case of $\mathbb{K} = \mathbb{C}$ there are often reasons to consider continuous mappings $\phi : X \to \mathbb{K}$, which are *antilinear* in the sense that

$$\phi(\alpha x + \beta y) \;=\; \overline{\alpha}\,\phi(x) + \overline{\beta}\,\phi(y)$$

for all $\alpha, \beta \in \mathbb{C}$ and $x, y \in X$. Let us denote X_a^* the linear space of continuous antilinear functionals on X, usually called the *conjugate linear space* or the *antidual space* of X.

§2. Weak Topologies, Bidual Spaces and Reflexivity

Let us first remember that if a linear space X is normed, then the adjoint space X^* is also normed, where the norm of an element $\phi \in X^*$ is defined by

$$\|\phi\| := \sup_{x\in X,\,\|x\|\leq 1} |\phi(x)|.$$

Since the fields $\mathbb{R}$ and $\mathbb{C}$ are complete, the adjoint space X^* is complete, too, which can easily be proved.

Recall also from section I that a normed linear space X owns a specific topology, called the *norm topology*, which is the coarsest one to make the norm into a continuous mapping. But there is a further and useful topology on X, denoted the *weak topology* $\mathcal{T}_w$. It is the coarsest topology on X such that all continuous linear functionals $\phi\in X^*$ remain continuous. The notion of convergence with respect to this topology is known as *weak convergence*. More precisely, a sequence $\{x_n\}_{n\in\mathbb{N}}$ of elements in X is called *weakly convergent* to $x\in X$ (notation: $x_n \rightharpoonup x$), if $\lim_{n\to\infty}\phi(x_n)=\phi(x)$ for each $\phi\in X^*$.
The weak topology is also Hausdorff. But it does not contain more open sets than the norm topology, so $\mathcal{T}_w$ is weaker than $\mathcal{T}_{\|\cdot\|}$. In other words, norm convergence of any sequence in X implies weak convergence. If X is finite-dimensional, however, both topologies coincide, but this is in general not true in arbitrary normed linear spaces.

Let for a given normed linear space X the *bidual space* defined by $X^{**} := (X^*)^*$, which forms a complete normed linear space again. Then there is a natural embedding

$$\begin{cases} \iota : X \hookrightarrow X^{**}, \\ \iota(x) := \phi_x \text{ for } \phi_x \in X^*, \end{cases}$$

where $\phi_x(f) := f(x)$ for $f\in X^*$. The mapping ι is linear, continuous and injective. In other words, each element $x\in X$ gives rise to a mapping $\phi_x\in X^{**}$, where $\|x\| = \|\iota_x\|$ holds, so X can be seen as a linear subspace of X^{**}.
It can in general nothing be said whether the mapping ι is onto. But if this is the case, then ι constitutes a norm-isomorphism between X and X^*. Many normed linear spaces considered in functional analysis own this remarkable property, which is indicated by the concept of *reflexivity*: the space X is called *reflexive*, if ι is onto and consequently $X \cong X^{**}$ holds.

We collect some propositions concerning reflexivity:

1. Since adjoint spaces of normed spaces are throughout Banach spaces, every reflexive space must be complete.

2. If a Banach space X is norm-isomorphic to a reflexive space, then X must be reflexive, too.
3. Dual spaces of reflexive spaces are reflexive.
4. Reflexivity of a Banach space X is strongly related to reflexivity of *all* its adjoint spaces, that is, X is reflexive iff one of the spaces $X^*, X^{**}, X^{***}, \ldots$ is reflexive. On the other hand, if X is not reflexive, then all these spaces prove to be pairwise unisomorphic to each other.

Important examples of reflexive Banach spaces are given by the *Milman-Pettis theorem*, which claims that *each uniformly convex Banach space is reflexive.*

We finally introduce the *weak**-topology on the adjoint X^* of a locally convex topological linear space X: it is the smallest topology in X^* such that for each $x \in X$ the linear functional ϕ_x remains continuous.

The weak*-topology on the adjoint space X^* may possibly contain fewer open sets than the weak topology and is therefore coarser. But X^* is at least a Hausdorff space with respect to the weak*-topology. The main result concerning weak*-topologies in adjoint spaces in the setting of normed linear spaces is given by the *Alaoglu theorem*:

THEOREM III.1. The closed unit ball $B_1(0) := \{f \in X^* : \|f\| \leq 1\}$ is compact with respect to the weak*-topology on X^*.

§3. Representation Theorems for Adjoint Spaces

In this paragraph we describe the adjoint spaces for a couple of normed linear spaces X arising in applied analysis by quoting a linear homeomorphism $X^* \cong Y$, where the representation space Y has to be chosen in a suitable way. Indeed, it is one of the basic tasks in functional analysis to identify the adjoint space X^* of a given normed linear space X with another known Banach space Y by finding a suitable norm-isomorphism $\Phi : X^* \to Y$ in order to obtain $X^* \cong Y$.

1. *The adjoint space of $C_c(G)$.*

   It is known from measure theory that a continuous function $f : G \to \mathbb{R}$ is measurable, but in general not Lebesgue-integrable. If $f \in C_c(G)$, however, then f is bounded and clearly Lebesgue-integrable.

Every Borel measure μ gives rise to a positive and continuous linear functional ϕ_μ on the space $C_c(G)$ endowed with the supremum norm. Conversely, the question arises whether each continuous linear functional $\phi \in C_c(G)^*$ appears in this way. The answer is given by the RIESZ-MARKOV *theorem*, quoting that the representation of positive linear functionals on G can just be done by the *Radon measures* on G:

THEOREM III.2. Let ϕ be a positive and continuous linear functional on the normed linear space $C_c(G)$. Then there is exactly one Radon measure $\mu : \mathcal{B}_G \to [0, \infty]$ such that

$$\phi(f) := \int_G f \, d\mu, \qquad f \in C_c(G).$$

2. *The adjoint space of* $C_0(G)$.

A *complex* measure $\mu : \mathcal{B}_G \to \mathbb{C}$ is said to be *Radon*, if the so-called *variation* $|\mu|$ is Radon in the sense of a positive measure on G.

Notice that $|\mu|$ is always finite. The space of complex Radon measures $\mathscr{M}_r(G)$ proves to be complete with respect to the norm $\|\mu\| := |\mu|$ for $\mu \in \mathscr{M}_r(G)$.

The following theorem of RIESZ-MARKOV type describes the adjoint space of $C_0(G)$:

THEOREM III.3. Let ϕ be a complex-valued and continuous linear functional defined on the space $C_0(G)$. Then there is exactly one complex Radon measure μ such that

$$\phi(f) := \int_G f \, d\mu, \qquad f \in C_0(G).$$

In conclusion, $C_0(G)^* \cong \mathscr{M}_r(G)$, so $C_0(G)^*$ is norm-isomorphic to the Banach space $\mathscr{M}_r(G)$ of complex Radon measures.

3. *The adjoint space of* $C[a, b]$.

Especially in the case $G = [a, b]$ we may quote that *every* continuous linear functional G on $C[a, b]$ arises as Riemann-Stieltjes integral with respect to a certain function g of bounded variation, that is, we have $C[a, b]^* \cong BV(a, b)$. In fact, functions of bounded variation correspond to Radon measures on the interval $[a, b]$, hence the norm-isomorphism is an immediate consequence of theorem III.2.

4. *The adjoint space of $L^p(X)$.*
   If $1 < p < \infty$ and $(X, \mathcal{B}, \mu)$ is an ordinary measure space, we have $L^p(X)^* \cong L^q(X)$, where q is conjugated to p: $p^{-1} + q^{-1} = 1$. For $p = 1$ and $(X, \mathcal{B}, \mu)$ being a σ-finite measure space, however, $L^1(X)^* \cong L^\infty(X)$ holds.

5. *The adjoint space of $L^p(a, b; X)$.*
   If $1 < p < \infty$ and the Banach space X is reflexive and separable, we have $L^p(a, b; X)^* \cong L^q(a, b, X^*)$, where q is conjugated to p.

6. *The conjugate space of a complex Hilbert space.*
   Due to the properties of the inner product, each element $x \in H$ gives rise to a continuous and conjugate linear functional $\phi_x \in H_a^*$ according to

   $$\phi_x(y) \ := \ \langle x, y \rangle \qquad \text{for all } y \in H.$$

   The mapping $\Phi : H \to H_a^*$ defined by $x \mapsto \phi_x$ is linear, called the RIESZ *map* of H. Since the inner product is positive definite, Φ turns out to be injective.

   Riesz's representation theorem claims that the mapping Φ is actually surjective, so there is a canonical correspondence between the elements of H and those of H_a^*:

   THEOREM III.4. If H is a complex Hilbert space, then $\Phi : H \to H_a^*$ is bijective and norm-preserving, i. e. $\|\Phi(x)\| = \|x\|$ for $x \in H$.

   This theorem implies that to each $\phi \in H_a^*$ there is a unique element $x_\phi \in H$, which fulfills the equation $\langle x_\phi, y\rangle = \phi(y)$ for every $y \in H$. Moreover, the RIESZ map gives rise to introduce an inner product on the conjugate space H_a^* by

   $$\langle \phi, \psi \rangle \ := \ \langle \Phi^{-1}(\phi), \Phi^{-1}(\psi) \rangle \qquad (\phi, \psi \in H_a^*),$$

   hence Φ is not only norm-preserving, but also *isometric*, that is, it preserves the inner products $\langle x, y\rangle$ for $x, y \in H$.

7. *The adjoint space of a real Hilbert space.*
   On the other hand, if H is a *real* Hilbert space, we may replace H_a^* by H^* and verify that the RIESZ map Φ is a norm-isomorphism between H and H^*, since in this case the inner product in H is also linear in the second argument.

The adjoint spaces of smooth function spaces will be treated in the next paragraph and lead us to so-called *distributional spaces*.

§4. Spaces of Distributions

The notion of *distribution* forms a generalization of an ordinary function $f : G \to \mathbb{K}$. The corresponding subject in functional analysis, called the *theory of distributions*, was developed by L. SCHWARTZ in the midth of the 20th century to describe ideas and concepts of theoretical physics by an exact mathematical formalism.

Distribution theory can be seen as a direct application of the abstract theory of locally convex linear spaces and their dual spaces. In the following let us enumerate the most important spaces of distributions:

1. *The adjoint space of* $\mathscr{D}(G)$. Remember the space $\mathscr{D}(G)$, consisting of test functions and furnished with a locally convex topology, in which this space is complete but not metrizable. Then every continuous linear functional $F : \mathscr{D}(G) \to \mathbb{C}$ is called a *Schwartz distribution*. We denote $\mathscr{D}'(G) := \mathscr{D}(G)^*$ the adjoint space of $\mathscr{D}(G)$ respectively the space of Schwartz distributions on G. It is also possible to characterize a Schwartz distribution with respect to $\mathscr{D}$-convergence, when applied to test functions: a linear functional $F : C_c^\infty(G) \to \mathbb{C}$ is a Schwartz distribution iff $F(\phi_n) \to 0$ for any $\mathscr{D}$-convergent sequence (ϕ_n) converges to the zero function.

   Notice that the linear functional F_f derived from a locally integrable function f by

   $$F_f(\phi) \ := \ \int_G f\,\phi\,dx, \qquad \phi \in C_c^\infty(G)$$

   is continuous, and this is also true for the linear functional induced by the Radon measure μ according to

   $$F_\mu(\phi) \ := \ \int_G \phi\,d\mu, \qquad \phi \in C_c^\infty(G).$$

   Thus locally integrable functions as well as Radon measures may be regarded as Schwartz distributions. Notice that the space $\mathscr{D}'(G)$ is indeed extremely extensive and contains nearly all functions being important in applications, amongst others the functions belonging to the well-known spaces $L^p(G)$, $L^\infty(G)$, $C(G)$, $C(\overline{G})$ and $C_0(G)$, but also measures on G, which are regular enough in some sense.

   A Schwartz distribution is said to be a *regular distribution*, if it is the continuous linear functional F_f induced by a locally integrable function $f \in L^1_{loc}(G)$, otherwise it is called *singular*.

Well-known examples of singular distributions are the *Dirac distributions*

$$\delta_a(\phi) \;:=\; \phi(a) \quad \text{for all } \phi \in C_c^\infty(G) \qquad (a \in G).$$

2. *The adjoint space of* $\mathcal{E}(G)$. Regarding the Fréchet space $\mathcal{E}(G)$, each element $F \in \mathcal{E}'(G) \;:=\; \mathcal{E}(G)^*$ is called a *distribution with compact support.* This notion is founded by the fact that for such F there is a compact set $K_F \subset G$, called the *support* of F such that $F\phi = 0$ for all $\phi \in \mathcal{E}(G)$ having support in $G \setminus K_F$. On the other hand, each measurable function f with compact support gives rise to a compactly supported distribution $F_f \in \mathcal{E}'(G)$.

   Since it is always possible to increase an adjoint space by decreasing the underlying base space, there is a continuous inclusion $\mathcal{E}'(G) \hookrightarrow \mathscr{D}'(G)$, which is dual to embedding $\mathscr{D}(G) \hookrightarrow \mathcal{E}(G)$. So we basically may regard each compactly supported distribution as a Schwartz distribution.

3. *The adjoint space of* $\mathscr{S}(\mathbb{R}^d)$. Similar to the spaces $\mathscr{D}(G)$, we introduce the notion of *tempered distribution* as a continuous linear functional $T : \mathscr{S}(\mathbb{R}^d) \to \mathbb{C}$, which generalizes that of bounded and slow-growing locally integrable functions. The space of tempered distributions is usually denoted $\mathscr{S}'(\mathbb{R}^d)$.

   Comparing the known kinds of distributional spaces, we obtain a chain of inclusions, which is dual to that of the base spaces:

$$\mathcal{E}'(\mathbb{R}^d) \subset \mathscr{S}'(\mathbb{R}^d) \subset \mathscr{D}'(\mathbb{R}^d).$$

   IF we endow each of these spaces with the *weak*-topology*, then these embeddings prove to be continuous. Hence a distribution having compact support can also be regarded as a tempered distribution, which in turn gives rise to a Schwartz distribution.

The usage of distributions for the solution theory of linear partial differential equations can only be exhausted in full strength if the notions of *convolution* and *tensor product* between elements of the spaces $\mathscr{D}'(\mathbb{R}^d)$ and $\mathcal{E}'(\mathbb{R}^d)$ has been introduced as far as possible.

1. *Convolutions.* The convolution $\phi * \psi$ of functions $\phi, \psi \in \mathscr{S}(\mathbb{R}^d)$ is defined according to

$$(\phi * \psi)(x) \;:=\; \int_{\mathbb{R}^d} \phi(y)\, \psi(x-y)\, dy$$

   and always exists, hence the space of Schwartz functions forms an algebra with respect to convolution as multiplicative operation. Furthermore, convolution between a regular distribution $F_f \in \mathscr{D}'(\mathbb{R}^d)$ and a test function $\phi \in \mathscr{D}(\mathbb{R}^d)$ is similarly defined:

$$(F_f * \phi)(x) := \int_{\mathbb{R}^d} f(y)\, \phi(x-y)\, dy.$$

More general, the convolution $F * G$ $F, G \in \mathscr{D}'(\mathbb{R}^d)$ exists, if the *support condition* is fulfilled, meaning that for each compact subset $K \subset \mathbb{R}^d$ the set

$$K_{F,G} := \{(x,y) : x \in \mathrm{supp}(F), y \in \mathrm{supp}(G), x+y \in K\} \subset \mathbb{R}^d \times \mathbb{R}^d$$

is compact, too. In this case we may define the convolution $F * G$ according to

$$(F * G)\, \phi := \iint_{K_{F,G}} F(x)\, G(y)\, \phi(x+y)\, dx\, dy, \quad \phi \in C_c^\infty(\mathbb{R}^d).$$

The support condition is sufficient but not necessary for the existence of the convolutional distribution $F * G$. Nevertheless, the condition is always fulfilled if at least one of the distributions F, G is compactly supported. Notice also that convolution on the space $\mathscr{D}'(\mathbb{R}^d)$ is commutative, if exists.

Especially the convolution between $F \in \mathscr{D}'(G)$ and the Dirac distribution δ_0 always exists and satisfies the equation $F * \delta_0 = F$, thus δ_0 serves as identity element of this operation. But convolution is also associative, hence for distributions $F, G, H \in \mathscr{D}'(\mathbb{R}^d)$ the equality

$$(F * G) * H = F * (G * H)$$

holds, provided that the involved convolutions exist.

2. *Tensor products.* Besides the notation of convolution one also requires that of *tensor product* between distributions. Tensor products are useful to transform an inital value of a distributional differential equation into an inhomogeneous part of this equation. Based on the fact that for $\phi \in \mathscr{D}(\mathbb{R}^{d_1})$ and $\psi \in \mathscr{D}(\mathbb{R}^{d_2})$ the tensor product $\phi \otimes \psi$ defined by

$$(\phi \otimes \psi)(x,y) = \phi(x) \cdot \psi(y), \quad (x,y) \in \mathbb{R}^{d_1+d_2}$$

belongs to the space $\mathscr{D}(\mathbb{R}^{d_1+d_2})$, there is a unique distribution $F \otimes G \in \mathscr{D}'(\mathbb{R}^{d_1+d_2})$ for $F \in \mathscr{D}'(\mathbb{R}^{d_1})$ and $G \in \mathscr{D}'(\mathbb{R}^{d_2})$, which for all test functions $\phi \in \mathscr{D}(\mathbb{R}^{d_1})$ and $\psi \in \mathscr{D}(\mathbb{R}^{d_2})$ fulfills

$$F \otimes G\,(\phi \otimes \psi) \;=\; F(\phi) \cdot G(\psi).$$

It is called the *tensor product* between F and G. Performing the tensor product is a commutative and associative operation, i. e.

$$F \otimes G = G \otimes F \quad \text{and} \quad (F \otimes G) \otimes H \;=\; F \otimes (G \otimes H).$$

§5. Adjoint Operators in Locally Convex Spaces

The concept of adjoint operator is helpful to generalize the domains of linear partial differential operators and Fourier transformation to the very extensive spaces of distributions, but one first has to introduce this notion in the setting of locally convex spaces.

Let X, Y be locally convex topological linear spaces and X^*, Y^* their adjoint spaces. Given a continuous linear operator $A : X \to Y$, each linear functional $f \in X^*$ gives rise to a linear functional $g \in Y^*$ by $g := f \circ A$. The assignment $f \mapsto g$ induces a continuous mapping $A^* : Y^* \to X^*$, called the *adjoint operator* of A. One often says that the functional g is the *pullback* of f along the operator A. The adjoint operator A^* is also linear.

The operator A^* is also linear. If $A : V \to W$ is continuous, then A^* is injective iff if $\mathfrak{R}(A)$ is dense in W. Moreover, if A is bijective, then A^* is bijective, too, and its inverse $(A^*)^{-1}$ is the adjoint operator of A^{-1}.

Finally, if X and Y are Fréchet spaces, then the solvability of the linear equation $Ax = y$ is related to certain properties of the adjoint operator: the operator $A : X \to Y$ is surjective iff A^* is injective and $\mathfrak{R}(A^*)$ is closed in Y^*.

We use the notion of adjoint operator to define derivatives of Schwartz distributions in the space $\mathscr{D}'(G)$. Let ∂^α be the elementary differential expression given by the multi-index α. Then ∂^α is a continuous linear operator $\partial^\alpha : \mathscr{D}(G) \to \mathscr{D}(G)$, due to the fact that the space $\mathscr{D}(G)$ is endowed with a very fine topology.

The action $\partial^\alpha_{\mathscr{D}}$ of the expression ∂^α on the distributional space $\mathscr{D}'(G)$ is defined by the *formally adjoint* expression $(\partial^\alpha)^* := (-1)^\alpha \partial^\alpha$, or equivalently

$$(\partial^\alpha_{\mathscr{D}} F)\,\phi \;:=\; F((\partial^\alpha)^* \phi) \;=\; (-1)^{|\alpha|}\, F(\partial^\alpha \phi)$$

for all $F \in \mathscr{D}'(G)$ and $\phi \in \mathscr{D}(G)$.

This definition is chosen in order to obtain compatibility with possible classical derivatives of *regular* distributions. In fact, if

$F_f \in \mathscr{D}'(G)$ is the distribution generated by a differentiable function $f : G \to \mathbb{C}$, then $\partial^\alpha_{\mathscr{D}} F_f = F_{\partial^\alpha f}$ holds.

The mapping $F \mapsto \partial^\alpha_{\mathscr{D}} F$ proves to be continuous on the Schwartz space $\mathscr{D}'(G)$. We also have the following compatibility rules between convolutions and tensor products:

$$\partial^\alpha_{\mathscr{D}}(F * G) \;=\; (\partial^\alpha_{\mathscr{D}} F) * G \;=\; F * (\partial^\alpha_{\mathscr{D}} G)\,,$$
$$\partial^\alpha_{\mathscr{D}}(F \otimes G) = (\partial^\alpha_{\mathscr{D}} F) \otimes G = F \otimes (\partial^\alpha_{\mathscr{D}} G).$$

§6. Adjointness in Banach Spaces

Adjoint operators are also useful to study properties of a densely defined linear operator acting between Banach spaces. Indeed, it is profitable to combine the study of a linear operator with that of its *adjoint operator.*

To introduce this concept, let X, Y be Banach spaces, X^* Y^* be the adjoint spaces and $A : \mathfrak{D}(A) \subset X \to Y$ be a linear operator with domain of definition $\mathfrak{D}(A)$ lying dense in X. Then the domain of definition $\mathfrak{D}(A^*)$ of the *adjoint operator* A^* is defined by

$$\mathfrak{D}(A^*) \;:=\; \{\, g \in Y^* : g \circ A : \mathfrak{D}(A) \to \mathbb{K} \text{ is continuous} \,\}.$$

For $g \in \mathfrak{D}(A^*)$ let $A^* g \;:=\; \overline{g \circ A}$, the unique extension of the continuous linear functional $g \circ A$ to the whole of X, which implies $A^* g \in X^*$. Thus the linear operator A^* has domain space Y^* and range space X^*, and the domain of definition $\mathfrak{D}(A^*)$ of the adjoint operator A^* forms a linear subspace of Y^*, on which A^* is linear.

In the following we denote $\langle \cdot, \cdot \rangle$ the natural pairing between the linear spaces X and X^*, i. e. $\langle x, x^* \rangle \;:=\; x^*(x)$ for $x \in X$ and $x^* \in X^*$. Then it is possible to define the adjoint operator A^* by the validness of the equations

$$\langle A\,x, y^* \rangle \;=\; \langle x, A^* y^* \rangle \qquad (x \in X,\; y^* \in Y^*).$$

Hence $y^* \in \mathfrak{D}(A^*)$, if there is $x^* \in X^*$ such that $\langle A\,x, y^* \rangle = \langle x, x^* \rangle$ for all $x \in X$, which gives rise to define $A^* y^* \;:=\; x^*$.

Adjoint operators enjoy a couple of nice properties:

Theorem III.5. Let $A : \mathfrak{D}(A) \subset X \to Y$ be densely defined.

(a) The adjoint operator A^* is closed.

(b) If $Y = X$ and A is closable, then A^* is densely defined and the bi-adjoint $A^{**} := (A^*)^*$ is well-defined.

(c) If $Y = X$ is reflexive and A is closable, then $A^{**} = \overline{A}$, which implies that if A is closed, then $A^{**} = A$.

Let the *annihilator* $M^\perp$ and the *pre-annihilator* ${}^\perp N$ of $M \subset X$ and $N \subset X^*$ be defined by

$$\begin{cases} M^\perp := \{\, x^* \in X^* : \langle x, x^* \rangle = 0 \text{ for all } x \in M \,\}, \\ {}^\perp N := \{\, x \in X : \langle x, x^* \rangle = 0 \text{ for all } x^* \in N \,\}. \end{cases}$$

We then obtain the both fundamental relations

$$\overline{\mathfrak{R}(A^*)} = \mathfrak{N}(A)^\perp \quad \text{and} \quad \overline{\mathfrak{R}(A)} = {}^\perp\mathfrak{N}(A^*) \tag{6.1}$$

between the kernels and ranges of the linear operators A and A^*.

With respect to solvability of the equation $A\,x = y$, the relationships (6.1) imply that A has dense range in Y iff A^* is injective.

In the case, where A is *continuous* and $\mathfrak{D}(A) = X$, the adjoint of A is a continuous linear operator $A^* : Y^* \to X^*$ and the following assertions can be done:

1. If A is bijective, then A^* proves to be continuous and bijective, too, and $(A^*)^{-1} = (A^{-1})^*$ holds.

2. Taking the adjoint is a linear and injective operation in the normed linear space $\mathscr{L}(X, Y)$, that is,
$$(\lambda A_1 + \mu A_2)^* = \lambda\, {A_1}^* + \mu {A_2}^*$$
for $\lambda, \mu \in \mathbb{K}$, where the norm is preserved, i. e. $\|A^*\| = \|A\|$ for $A \in \mathscr{L}(X, Y)$.

3. If Z is a further Banach space, then for $A_1 \in \mathscr{L}(Y, Z)$ and $A_2 \in \mathscr{L}(X, Y)$ the equality $(A_1 \circ A_2)^* = {A_2}^* \circ {A_1}^*$ holds.

4. With respect to compactness of A, SCHAUDER's theorem is valid: the operator A is compact if and only if A^* is compact.

To the end we introduce the concept of *adjoint pair* of linear operators. The densely defined linear operators $A : \mathfrak{D}(A) \subset X \to Y$ and $B : \mathfrak{D}(B) \subset Y^* \to X^*$ are said to be *formally adjoint* to each other or to be a *formally adjoint pair* (A, B) if

$$\langle A\,x, f \rangle_{Y \times Y^*} = \langle x, Bf \rangle_{X \times X^*}$$

holds for all $x \in \mathfrak{D}(A)$ and $f \in \mathfrak{D}(B)$. In addition, we define the *restricted adjoint* B_r of the linear operator B according to $\mathfrak{D}(B_r) :=$

$\{x \in X : \text{there is } y \in Y \text{ such that } (Bf)\,x = f(y) \text{ for all } f \in \mathfrak{D}(B)\}$ and $B_r\,x := y$. The mapping B_r acts between the Banach spaces X and Y and proves to be linear. We may reformulate the notation of formally adjoint pair by saying that A and B are formally adjoint to one another if B_r is an extension of A.

Finally, by a *realization* of a formally adjoint pair (A, B) we mean any densely defined and closed linear operator R lying between A and B_r, so $A \prec R \prec B_r$ holds.

§7. Hilbert Adjoints and Normal Operators

We finally give a more direct definition of the adjoint operator in the Hilbert space setting, that in conclusion leads to the concept of *normal operator.*

Remember that the RIESZ $\tilde{\Phi}$ map is a norm-isomorphism between a complex Hilbert space H and its conjugated space $H^\times$ consisting of continuous and antilinear functionals $\phi : H \to \mathbb{C}$. Then, given a linear operator A with dense domain $\mathfrak{D}(A)$ in H, the operator A^*_H, often called the *Hilbert adjoint* of A, is nothing but the adjoint A^* conjugated by the Riesz map, that is,

$$A^*_H \;:=\; \tilde{\Phi}^{-1} \circ A^* \circ \tilde{\Phi}. \tag{7.1}$$

Hence A^*_H constitutes a linear mapping in H.

In case of a real Hilbert space, however, we may replace $\tilde{\Phi}$ in (7.1) by the *linear* RIESZ map $\Phi : H \to H^*$, so $A^*_H := \Phi{-}1 \circ A^* \circ \Phi$. As a rule for the sequel, whenever we consider an adjoint operator in a Hilbert space, we will write A^* instead of A^*_H, so A^* has to be understood as an operator acting in H.

In the following we use a direct approach to define the operator A^* that does not rely on the concept of adjoint operator in the Banach space setting.

If A is linear and densely defined in H, the domain of definition $\mathfrak{D}(A^*)$ of the adjoint operator A^* consists of those elements $y \in H$, for which there is $y^* \in H$ fulfilling $\langle A\,x, y\rangle = \langle x, y^*\rangle$ for all $x \in \mathfrak{D}(A)$. We then define $A^*y := y^*$ in this case, and A^* proves to be linear.

The set $\mathfrak{D}(A^*)$ coincides with the set of elements $y \in H$, for which the linear functional $f_y : \mathfrak{D}(A) \to \mathbb{K}$ given by $f_y(x) := \langle A\,x, y\rangle$ is continuous. By the B. L. T. theorem, we may extend f_y to a linear mapping $\tilde{f}$ defined on the whole of H, and due to RIESZ's representation theorem there is a unique element $y^* \in H$ such that $\tilde{f}_y(x) = \langle x, y^* \rangle$ for all $x \in H$. Thus we may define $A^*y := y^*$ for $y \in \mathfrak{D}(A^*)$.

Notice that the operators A and A^* are coupled by the formula

$$\langle A\,x, y \rangle \;=\; \langle x, A^*y \rangle, \tag{7.2}$$

where $x \in \mathfrak{D}(A)$ and $y \in \mathfrak{D}(A^*)$. Moreover, equation (7.2) gives rise to the following relationships between the kernels and ranges of A and A^*, which are variants of (6.1) for operators in Hilbert space:

$$\overline{\mathfrak{R}(A^*)} \;=\; \mathfrak{N}(A)^{\perp} \quad \text{and} \quad \overline{\mathfrak{R}(A)} \;=\; \mathfrak{N}(A^*)^{\perp}.$$

They are useful to investigate the possible property of surjectivity of the operator A. More precisely, if one can prove that A^* is injective, then the range of A is dense in H, and if in addition A has closed range (for example if A is bounded below), then we can conclude that A is even surjective.

We collect the main properties of adjoint operators to the following

THEOREM III.6. Let A be a linear and densely defined operator in Hilbert space H.

(a) A^* is closed. But A^* is not necessarily densely defined. In fact, it can happen that $\mathfrak{D}(A^*) = \{0\}$.

(b) If B is an extension of A, then A^* is an extension of B^*, that is, $A \prec B$ implies $B^* \prec A^*$. Indeed, let $y \in \mathfrak{D}(B^*)$, then there is $y^* \in H$ such that

$$\langle Bx, y \rangle \;=\; \langle A\,x, y \rangle \;=\; \langle x, y^* \rangle \quad \text{for all } x \in \mathfrak{D}(B)\,,$$

and it is easy to see that $y \in \mathfrak{D}(A^*)$ with $A^*y = B^*y$.

(c) We especially have $(A - \lambda 1_H)^* = A^* - \overline{\lambda} 1_H$ for $\lambda \in \mathbb{C}$, where 1_H is the identity operator in H. The domain of definition of the operator $(A - \lambda 1_H)^*$ is given by

$$\mathfrak{D}((A - \lambda 1_H)^*) \;=\; \mathfrak{D}(A^* - \overline{\lambda} 1_H) \;=\; \mathfrak{D}(A^*).$$

(d) $(\lambda A)^* = \overline{\lambda} A^*$ for $\lambda \in \mathbb{C} \setminus \{0\}$ (but this is not true in case of $\lambda = 0$).

(e) If A is closable with closure $\overline{A}$, then taking the adjoint of both operators leads to the same result, that is, $A^* = \overline{A}^{\,*}$.

(f) A is closable if and only if $\overline{\mathfrak{D}(A^*)} = H$. In this case the operator $A^{**} := (A^*)^*$ is well-defined and forms the closure of A, that is, $A^{**} = \overline{A}$. Thus the closure of a closable linear operator can be obtained by taking the adjoint twice.

Statement (f) is also true in the Banach space setting under the condition that the involved Banach spaces are reflexive.

Furthermore, in the Hilbert space setting the densely defined and closed linear operators A, B constitute a formally adjoint pair, if $\langle Ax, y\rangle = \langle x, By\rangle$ for all $x \in \mathfrak{D}(A)$ and $y \in \mathfrak{D}(B)$. Equivalently, the pair of operators (A, B) is formally adjoint, if $A \prec B^*$ and $B \prec A^*$.
For example, if A is densely defined and closed, the pairs (A, A^*) and (A^*, A) are formally adoint to each another, but also the operators $A + \lambda$ and $A^* + \overline{\lambda}$ are so.

Let us consider the Hilbert adjoints of *continuous* and everywhere defined operators more in detail. If A is continuous and densely defined, then the adjoint A^* is continuous and everywhere defined, too. Hence both operators belong to the Banach algebra $\mathscr{L}(H)$, which proves to be closed with respect to taking the adjoint. But the mapping $* : \mathscr{L}(H) \to \mathscr{L}(H)$ has some more salient properties:
(a) $A \mapsto A^*$ is antilinear, that is, $(\alpha A_1 + \beta A_2)^* = \overline{\alpha}\, A_1^* + \overline{\beta}\, A_2^*$.
(b) $\|A\| = \|A^*\|$, so $A \mapsto A^*$ is norm-preserving in $\mathscr{L}(H)$.
(c) $(A_1 \circ A_2)^* = A_2^* \circ A_1^*$.
(d) $A^{**} = A$, i. e. taking the adjoint is an involutional operation.
(e) $\|A \circ A^*\| = \|A\|\,\|A^*\|$, called the *$C^*$-property* of $\mathscr{L}(H)$, which makes $\mathscr{L}(H)$ into a unital *C^*- algebra* (see section VII).

The concept of Hilbert adjoint gives rise to introduce the class of *normal* linear operators: a densely defined linear operator A is denoted *normal*, if

$$A \circ A^* = A^* \circ A.$$

This definition of course implies the identity of the sets $\mathfrak{D}(A)$ and $\mathfrak{D}(A^*)$. Notice that a normal operator is closed (since A^* is so), the normed space $(\mathfrak{D}(A^*), \|\cdot\|_{A^*})$ is complete and the norm $\|\cdot\|_A$ is equivalent to the norm $\|\cdot\|_{A^*}$.

The next theorem characterizes normality of a linear operator A by other conditions:

THEOREM III.7. The subsequent statements are equivalent:
(a) A is normal;
(b) A^* is normal;
(c) $\mathfrak{D}(A) = \mathfrak{D}(A^*)$ and $\|A\,x\| = \|A^* x\|$ for each $x \in \mathfrak{D}(A)$.

Let us also mention that the class of *normal* and *bounded* linear operators in Hilbert space includes a couple of important subclasses:

1. A *unitary* operator U distinguishes by the property $U \circ U^* = 1$ respectively $U^{-1} = U^*$;
2. a *symmetric operator* A is defined by the property $A \prec A^*$, and
3. by a *projection* P is meant a linear operator, which is symmetric and fulfills $P \circ P = P$.

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IV.
Normally Solvable Operators

Given a densely defined linear operator A acting between Banach spaces X and Y, the orthogonality relations in section IV can be improved if the ranges of A and A^* are *closed* subspaces of Y and X^*, so in these cases we actually would have

$$\mathfrak{R}(A) = {}^{\perp}\mathfrak{N}(A^*) \quad \text{and} \quad \mathfrak{R}(A^*) = \mathfrak{N}(A)^{\perp}. \tag{0.1}$$

One also says that a linear operator A is *normally solvable* if it is densely defined and $\mathfrak{R}(A) = {}^{\perp}\mathfrak{N}(A^*)$ holds, which means that A has closed range.

§1. Linear Operators with Closed Range

In this paragraph we shall mention two important theorems, which give us sufficient conditions for a linear operator to own closed range.

The *closed range theorem* is one of the classical results in functional analysis and provides equivalent conditions for a densely defined and closed linear operator to have closed range:

Theorem IV.1. Let X, Y be Banach spaces and $A : \mathfrak{D}(A) \subset X \to Y$ be densely defined and closed. Then it is equivalent to quote:

(a) $\mathfrak{R}(A)$ is closed.
(b) $\mathfrak{R}(A^*)$ is closed.
(c) $\mathfrak{R}(A)$ is the pre-annihilator of $\mathfrak{N}(A^*)$, i. e. $\mathfrak{R}(A) = {}^{\perp}\mathfrak{N}(A^*)$.
(d) $\mathfrak{R}(A^*)$ is the annihilator of $\mathfrak{N}(A)$, i. e. $\mathfrak{R}(A^*) = \mathfrak{N}(A)^{\perp}$.

For a proof of this theorem see for example [3], pp. 205-207. The statement was originally proved by S. Banach for continuous linear operators and have later been extended to closed linear operators, see [1], theorem 1.1 and theorem 1.2.

Besides theorem IV.1 it is natural to search for further conditions that imply closeness of the range of an operator. One of them is given as follows: a linear operator A is called *bounded below* if there is $\theta > 0$ such that

$$\|Ax\|_Y \geq \theta \, \|x\|_X \text{ for all } x \in \mathfrak{D}(A). \tag{1.1}$$

The *bounded-below theorem* is a useful tool to verify the closeness of $\mathfrak{R}(A)$:

THEOREM IV.2. If A fulfills inequality (1.1), then A is injective and the operator $A^{-1}:\mathfrak{R}(A)\subset Y\to X$ is bounded with operator norm $\|A^{-1}\|\leq\theta^{-1}$. Furthermore, the range $\mathfrak{R}(A)$ is closed if and only if A is closed.

Indeed, injectivity of the operator A can easily proved, supposed (1.1) holds, and closeness of $\mathfrak{R}(A)$ is a consequence of sequentially closeness of A.

§2. The Fredholm Index

Given the normed linear spaces X, Y and a linear operator

$$A:\mathfrak{D}(A)\subset X\to Y,$$

let $\alpha(A)$ (the *nullity index* of A) and $\beta(A)$ (the *deficiency index* of A) be the Hamel dimensions of the linear subspaces $\mathfrak{N}(A)\subset X$ and $\mathfrak{C}(A)\subset Y$.

Both integers together can be used to measure the deviation of A from bijectivity. Moreover, if at least one of them is finite, then the so-called *Fredholm index*

$$\chi(A) := \begin{cases} +\infty, & \alpha(A)=\infty \\ \alpha(A)-\beta(A), & \alpha(A)<\infty,\ \beta(A)<\infty \\ -\infty, & \beta(A)=\infty \end{cases}$$

is well-defined and takes values in the set $\{-\infty\}\cup\mathbb{N}\cup\{+\infty\}$. In the special case, where $\alpha(A)<\infty$, $\beta(A)<\infty$ and consequently $\chi(A)\in\mathbb{Z}$, the operator A is said to have *finite index.*

The linear operator $A:\mathfrak{D}(A)\subset X\to Y$ is called *semi-Fredholm*, if

(F-1) the range $\mathfrak{R}(A)$ is closed,

(F-2) the kernel $\mathfrak{N}(A)$ or the cokernel $\mathfrak{C}(A)$ is finite-dimensional (rsp. $\alpha(A)<\infty$ or $\beta(A)<\infty$).

Notice the remarkable fact that if A is continuous with $\mathfrak{D}(A)=X$ and $\beta(A)<\infty$, then the range $\mathfrak{R}(A)$ is necessarily closed, since it constitutes the topological complement of the finite-dimensional linear subspace $\mathfrak{C}(A)\subset Y$, that is, $X=\mathfrak{R}(A)\oplus\mathfrak{C}(A)$.

We refine the concept of semi-Fredholm operator as follows: a linear operator $A:\mathfrak{D}(A)\subset X\to Y$ with closed range $\mathfrak{R}(A)$ is called

(F-3) *upper semi-Fredholm*, if $\alpha(A) < \infty$ rsp. $\chi(A) < \infty$,
(F-4) *lower semi-Fredholm*, if $\beta(A) < \infty$ rsp. $\chi(A) > -\infty$.

Examples of upper semi-Fredholm operators are the bounded-below ones. In fact, theorem IV.2 implies that these operators are upper semi-Fredholm with vanishing nullity index.

If a continuous linear operator A satisfies an abstract *Schauder estimate*, one may quote a sufficient condition for A to be upper semi-Fredholm, which is known as PEETRE*'s lemma*:

THEOREM IV.3. Let X, Y, Z be three Banach spaces, where X is compactly embedded into Z, that is, there is an injective and compact linear mapping $\iota : X \to Z$. Then a continuous operator $A : X \to Y$ is upper semi-Fredholm iff there is $c > 0$ such that

$$\|x\|_X \leq c\,(\|\iota x\|_Z + \|A\,x\|_Y) \qquad (x \in X).$$

For a proof of this theorem, see [3], theorem 12.12, page 180.

§3. Perturbations of Semi-Fredholm Operators

Nowadays there is a well-elaborated *perturbation theory* for semi-Fredholm operators available, where conditions are studied, under which a linear operator of the form $A+B$ is semi-Fredholm, assumed A is semi-Fredholm and B fulfills certain properties. In this case the operator B is regarded as a perturbation of A.

We next quote two theorems with respect to conditions on B, which ensure the semi-Fredholm property of the operator $A+B$. We basically assume that X and Y are Banach spaces.

THEOREM IV.4. Let $A : X \to Y$ be semi-Fredholm and $B : X \to Y$ be linear and bounded with sufficiently small norm $\|B\|$. Then the operator $A+B$ remains semi-Fredholm with $\chi(A+B) = \chi(A)$.

In other words, this theorem asserts that the set of semi-Fredholm operators is open in the set $\mathscr{L}(X, Y)$. Especially linear operators of the form $A+\lambda$ for $|\lambda|$ small enough are semi-Fredholm again. But more can be said on the values $\alpha(A+\lambda)$, $\beta(A+\lambda)$ and $\chi(A+\lambda)$ by GOHBERG*'s punctured neighbourhood theorem*:

THEOREM IV.5. Let the continuous linear operator $A : X \to X$ be semi-Fredholm. Then there is $\lambda_0 > 0$ such that for all $\lambda \in (0, \lambda_0)$ we have $\alpha(A+\lambda) \leq \alpha(A)$, $\beta(A+\lambda) \leq \beta(A)$ and $\chi(A+\lambda) = \chi(A)$.

§4. Basic Theory on Fredholm Operators

Let us refine the definition of semi-Fredholm operator to a stronger concept: in addition to the properties (F-1) to (F-4), a continuous linear operator $A : X \to Y$ acting between normed linear spaces X and Y is called

(F-5) *Fredholm*, if $\chi(A)$ is finite, i. e. $\alpha(A) < \infty$ and $\beta(A) < \infty$.

The relationships (0.1) imply that iff A is densely defined and Fredholm, then the adjoint operator A^* is also Fredholm and $\chi(A^*) = -\chi(A)$ holds. Surely, if A is self-adjoint and Fredholm at the same time, then $\chi(A) = 0$ must hold.

Next we collect a couple of properties of Fredholm operators:

1. A^* is Fredholm if and only if A is so.
2. If one of A or A^* is Fredholm, then the dimensions of the kernels and ranges of A and A^* are related to each other by $\alpha(A^*) = \beta(A)$ and $\beta(A^*) = \alpha(A)$, which implies $\chi(A^*) = -\chi(A)$.
3. Fredholm operators acting in X and belonging to one and the same connected component within the algebra $\mathscr{L}(X)$ have identical index. Thus, if A is Fredholm and the linear operator B can be reached by a continuous path in $\mathscr{L}(X)$ with starting point A, then B is Fredholm, too, and $\chi(B) = \chi(A)$ holds.

Our next subject is the representation of Fredholm operators using compact linear operators. More precisely, a Fredholm operator can be represented by means of a *left* and *right parametrix*, respectively.

Let $A : X \to Y$ be a continuous linear operator. Then a linear mapping $P_l : Y \to X$ is called a *left parametrix* of A if $A \circ P_l = 1_X + k_l$ for some compact operator $k_l \in \mathscr{L}(X)$. In a similar way, a linear mapping $P_r : Y \to Y$ is said to be a *right parametrix* of A, if $P_r \circ A = 1_Y + k_r$ for some compact operator $k_r \in \mathscr{L}(Y)$. Every Fredholm operator owns a left and right parametrix, where the involved compact operators k_l, k_r even are of *finite-rank*, that is, they have finite-dimensional ranges.

Atkinson's theorem quotes that a Fredholm operator does not lose the Fredholm property if perturbed by a compact linear operator:

THEOREM IV.6. Let A be Fredholm and k be compact. Then $A + k$ is Fredholm again, and $\chi A + k = \chi A$ holds.

We have introduced the Fredholm property of linear operators under the assumption that these are continuous and everywhere defined in their domain spaces. If a linear operator A is closed instead of continuous, however, we define the possible Fredholm property by regarding A as a continuous linear operator $A : \mathfrak{D}(A) \to Y$, where the domain of definition $\mathfrak{D}(A)$ is endowed with the *graph norm*

$$\|x\|_A := \|x\|_X + \|Ax\|_Y$$

for $x \in \mathfrak{D}(A)$. The pair $(\mathfrak{D}(A), \|\cdot\|_A)$ forms a Banach space due to the fact that the operator A is assumed to be closed.

§5. Fredholm's Alternative Theorem

In the following we cite the FREDHOLM *alternative theorem* as the outstanding result in Fredholm theory with respect to the solvability of the linear equation $Ax = y$. Remember the *rank-nullity theorem* known from linear algebra, claiming that if a linear space V is finite-dimensional, then a linear mapping $A : V \to V$ is surjective if and only if it is injective.

The alternative theorem is a considerable generalization of the rank-nullity theorem to arbitrary Banach spaces and Fredholm operators with vanishing index:

THEOREM IV.7. Let X, Y be Banach spaces and the continuous linear operator $A : X \to Y$ be Fredholm with index $\chi(A) = 0$.

According to the solvability of the linear equation $A\,x = y$, exactly one of the following cases occurs:

1. $A\,x = y$ is uniquely solvable for all $y \in X$, so A is bijective,
2. $A\,x = y$ is not for all $y \in X$ solvable, and A is neither injective nor surjective in this case. However, if the equation is solvable for an element $y_0 \in X$, then the set of solutions forms a finite-dimensional subspace of X. Moreover, the equation $A\,x = y$ owns a solution $x \in X$ if and only if $y \in {}^{\perp}\mathfrak{N}(A^*)$.

§6. Concepts of Solvability for Linear Equations

Let us finally illustrate the interplay between Fredholm theory and the theory of bijective linear operators having compact inverse.

Remember that if X is a Banach space and $k : X \to X$ is compact, then the operator $1_X + k$ has finite nullity index and closed range, hence it is upper semi-Fredholm. This is also true for the operator

$1_{X^*}+k^*$, since k^* is compact, too, and both properties together imply that 1_X+k is Fredholm with index $\chi(1_X+k)=0$.

The following computation is of central meaning and for example useful in solving the *Dirichlet problem* for linear-elliptic differential operators. Consider the linear equation $Ax=y$, where X is a Banach space and $A:\mathfrak{D}(A)\subset X\to X$ is bijective linear operator with compact inverse for some $\zeta\in\mathbb{C}$. Then for $\zeta\in\rho(A)$ we surely have

$$A\,x \;=\; (A-\zeta)\,x+\zeta\,x \;=\; y,$$

and applying the mapping $(A-\zeta)^{-1}$ from the left yields

$$x\;+\;\zeta(A-\zeta)^{-1}\,x \;=\; (A-\zeta)^{-1}\,y \;=:\; \tilde{y}$$

or equivalently $x+\tilde{k}\,x=\tilde{y}$, where $\tilde{k}:=\zeta(A-\zeta)^{-1}$ is compact. The equations $A\,x=y$ and $(1_X+\tilde{k})\,x=\tilde{y}$ are called *equivalent to each other* in the sense that the element x is a solution of the first one if and only if it is a solution of the second one.

We summarize the result of the previous computation to the following

THEOREM IV.8. Let $A:\mathfrak{D}(A)\subset X\to X$ be an operator such that $(A-\zeta)^{-1}$ exists for some $\zeta\in\mathbb{C}$ and is compact. Then the equation $Ax=y$ is equivalent to an equation of the form $Fx=(1_X+\tilde{k})x=\tilde{y}$, where the operator F is Fredholm rsp. the operator $\tilde{k}$ is compact.

Clearly, FREDHOLM's alternative theorem IV.7 holds for the operator F and therefore provides assertions also on the solutions of the original equation $Ax=y$.

We want to finalize our elaborations on linear equations with a complete collection of concepts regarding the solvability of $A\,x=y$ where again A is assumed to be closed and acting between Banach spaces X and Y. The equation $A\,x=y$ is called

(a) *densely solvable* ([3]), if $\overline{\mathfrak{R}(A)}=Y$;
(b) *normally solvable* ([3]), if $\overline{\mathfrak{R}(A)}=\mathfrak{R}(A)$;
(c) *correctly solvable* ([3]), if A is bounded below;
(d) *semi-solvable* ([1]), if A is Fredholm;
(e) *almost solvable* or *regularly solvable* ([1]), if A is semi-solvable and $\chi(A)=0$;
(f) *uniquely solvable* ([3]), if A is injective;
(g) *everywhere solvable* ([3]), if A is surjective;

(h) *solvable* ([1]), if it is uniquely and everywhere solvable;
(i) *well-posed*, if it is solvable and A^{-1} is continuous, or equivalently, if $0 \in \rho(A)$;
(j) *completely solvable* ([1]) or *compactly solvable* if it is solvable and A^{-1} is compact.

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V.
Calculus of Vector-Valued Functions

In section I we have considered topological and normed linear spaces, whose elements are real or complex-valued functions. There are good reasons to extend the definitions of some of these spaces to contain so-called *vector-valued* functions, which are nothing but mappings $f:(a,b)\to X$, where $(a,b)\subset\mathbb{R}$ is a finite interval and X is a real or complex Banach space.

The most important spaces containing such functions are the vector-valued Hölder spaces, the Bochner spaces and the Bochner-Sobolev spaces. For the latter one the notion of weak differentiability of a vector-valued function is required.

§1. Spaces of Classical Differentiable Functions

If X is an arbitrary Banach space and $(a,b)\subset\mathbb{R}$, then a vector-valued function $x:[a,b]\to X$ is called *classical differentiable* in $t_0\in(a,b)$, if there is an element $x'(t_0)\in X$ such that

$$\lim_{t\to t_0}\left\|\frac{x(t)-x(t_0)}{t-t_0}-x'(t_0)\right\|=0.$$

Moreover, the function x is called classical differentiable in (a,b), if x is so in each $t\in(a,b)$. The linear space consisting of functions having *continuous* classical derivatives in (a,b) is denoted $C^1(a,b;X)$.

The spaces $C^k(a,b;X)$ $(k\in\mathbb{N})$ are defined recursively as usual, and the space $C^\infty(a,b;X)$ is defined by taking the intersection of all spaces $C^k(a,b;X)$, $k\in\mathbb{N}_0$.

We continue with the introduction of vector-valued Hölder spaces. First of all, let $0<\alpha\le 1$ and $C^\alpha(a,b;X)$ be the linear space of *Hölder-continuous* functions $f:[a,b]\to X$, defined by the property

$$\|f\|_\alpha := \sup_{x,y\in[a,b],\,x\neq y}\frac{\|f(x)-f(y)\|}{\|x-y\|^\alpha}<\infty. \tag{1.1}$$

The mapping $f\in C^\alpha(a,b;X)\mapsto\|f\|_\alpha$ is a norm, in which the space $C^\alpha(a,b;X)$ is complete. Then for $k\in\mathbb{N}$ and $0<\alpha\le 1$ we define $C^{k,\alpha}(a,b;X)$ as the space of functions $f:[a,b]\to X$ such that

(a) $f\in C^k(a,b;X)$;

(b) the derivatives $f^{(1)},f^{(2)},\dots,f^{(k)}$ can continuously be extended from (a,b) to the closure $[a,b]$;

(c) The highest derivative $f^{(k)}$ belongs to $C^\alpha(a,b;X)$.

In fact, the mapping $\|\cdot\|_{k,\alpha}$ defined by

$$\|f\|_{k,\alpha} := \sum_{i=0}^{k} \sup_{x\in[a,b]} \|f^{(i)}(x)\| + \|f^{(k)}\|_\alpha$$

constitutes a norm on $C^{k,\alpha}(a,b;X)$, in which this space is complete.

§2. Riemann Integration

It is the purpose in the current paragraph to sketch the setup of the *Riemann integral* for vector-valued functions $f:[a,b]\to X$. Like in the case $X=\mathbb{R}$, one usually starts with a partition

$$\mathcal{Z} = \{x_0 := a, x_1, \ldots, x_{n-1}, x_n := b\}, \quad x_{i-1} < x_i \quad (i = 1,\ldots,n),$$

where the length $\|\mathcal{Z}\|$ of the largest subinterval is called the *grid size* of $\mathcal{Z}$:

$$\|\mathcal{Z}\| := \max_{1\le i\le n} (x_i - x_{i-1}).$$

Furthermore, a *tagged partition* $\mathcal{Z}(t_1,\ldots,t_n)$ of $[a,b]$ is a partition $\mathcal{Z}$ together with a sequence of intermediate points $t_1,\ldots,t_n$, where each t_i belongs to the subinterval $[x_{i-1},x_i]$. Then the *Riemannian sum* of $f:[a,b]\to X$ with respect to the tagged partition $\mathcal{Z}(t_1,\ldots,t_n)$ is given by

$$R(f,\mathcal{Z}) := \sum_{i=1}^{n} f(t_i)\cdot(x_i - x_{i-1}).$$

The *Riemann integral* of f on $[a,b]$ is defined as the possible limit of Riemannian sums $R(f,\mathcal{Z})$ for $\|\mathcal{Z}\|\to 0$. Alternatively, a function f is called *Riemann-integrable* on $[a,b]$ to the value $R_f\in X$, if to each $\epsilon>0$ there is a number $\delta>0$ such that

$$\|R_f - R(f,\mathcal{Z})\| < \epsilon$$

for each tagged partition $\mathcal{Z}(t_1,\ldots,t_n)$ with $|\mathcal{Z}|<\delta$, and in this case the vector R_f is said to be the *Riemann integral* of f, denoted

$$\int_a^b f\,dx := R_f.$$

Continuity of a vector-valued function f is a sufficient condition for the existence of the Riemann integral for f. Indeed, every continuous function $f:[a,b]\to X$ is Riemann-integrable. The Riemann integral constitues a linear map on $C([a,b],X)$, satisfying the estimate

$$\left\| \int_a^b f(t)\,dt \right\| \le \int_a^b \|f(t)\|\,dt.$$

Moreover, any Riemann-integrable function fulfills the *mean value theorem*

$$\int_a^b f(t)\,dt = (b-a)\,\overline{f}\,,$$

where the function $\overline{f}$ belongs to the closed convex hull of the set $\{\, f(t) : t\in[a,b]\,\}$.

As in the scalar-valued case, in order to commute integration and taking the limit of a sequence $\{f_n\}_{n\in\mathbb{N}}$ of continuous vector-valued functions $f_n:[a,b]\to X$, it suffices that $\{f_n\}$ is *uniformly convergent* to a continuous function $f:[a,b]\to X$, and under this condition

$$\lim_{n\to\infty}\int_a^b f_n(t)\,dt = \int_a^b f(t)\,dt.$$

Like in classical calculus, we also introduce the notion of the *improper* Riemann integral. If f is defined on the real line and is R-integrable on each compact interval, then by the unproper Riemann integral of f on $\mathbb{R}$ is meant the possible limit

$$\int_{-\infty}^{+\infty} f(t)\,dt := \lim_{a\to\infty}\int_{-a}^{+a} f(t)\,dt.$$

If $f:\mathcal{U}\to X$ is defined on a domain $\mathcal{U}$ within the complex plane and $\gamma:[a,b]\to\mathcal{O}$ is a continuous *path*, then we define

$$\int_\gamma f(z)\,dz := \int_a^b (f\circ\gamma)dt,$$

and if γ is continuously differentiable, the path integral may be computed by

$$\int_a^b f(\gamma(t))\,\gamma(t)\,dt.$$

The following statement claims that Riemann integration of vector-valued functions is commutable with the action of closed linear operators:

THEOREM V.1. Let $A:\mathfrak{D}(A)\to X$ be a closed linear operator acting in a Banach space X and $f:[a,b)\to X$ $(a<b\le\infty)$ continuous.

Suppose that

1. $f(t) \in \mathfrak{D}(A)$ for all $t \in [a, b]$,
2. the mapping $t \mapsto Af(t)$ is continuous in $[a, b)$,
3. the possibly improper integral values x and y exist, where

$$x := \int_a^b f(t)\,dt \quad \text{and} \quad y := \int_a^b Af(t)\,dt.$$

Then $x \in \mathfrak{D}(A)$ and $Ax = y$ holds.

We finally want to mention the concept of *Riemann-Stieltjes integral* for a vector-valued function $f : [a, b] \to X$. If $g : [a, b] \to \mathbb{R}$ is a further function and

$$\mathcal{Z} := \{t_0 = a, t_1, \ldots, t_n = b; \xi_0, \ldots, \xi_{n-1}\}$$

a tagged partition of $[a, b]$, then f is called *Riemann-Stieltjes-integrable*, if there is an element $R_f \in X$ such that

$$\sum_{i=0}^{n-1} f(\xi_i)\,[g(t_{i+1}) - g(t_i)], \quad \xi_i \in [t_{i+1}, t_i]$$

converges to R_f for $|\mathcal{Z}| \to 0$, and in this case we define

$$\int_a^b f\,dg := R_f.$$

The mapping g plays the role of a weight function. A sufficient criterion for Riemann-Stieltjes integrability of a function $f : [a, b] \to X$ is for example, that f is piecewise continuous and g is monotonly growing in $[a, b]$.

Improper Riemann-Stieltjes integrals may be defined in a similar manner. In fact, such integrals exist if and only if the following limits exist:

$$\lim_{\substack{a \to -\infty \\ b \to +\infty}} \int_a^b f\,dg$$

§3. Linear Evolution Equations

By an *evolution equation* is meant an ordinary or partial differential equation that describes the behaviour of a dynamical system. More precisely, it describes the propagation of one or more physical magnitudes along the time, where their initial states are given in a starting point $t_0 \in \mathbb{R}$.

An elementary example of an evolution equation is given by the differential equation $x' + \lambda x = 0$ for $x(0) = x_0 \in \mathbb{R}$ and $\lambda > 0$. It is well known that this problem is solved by the exponential function $x(t, x_0) = x_0\, e^{-\lambda t}$ for $t > 0$.

Functional analysis provides a rich theory on *homogeneous* and *linear* evolution equations of the form

$$\begin{cases} x'(t) + A\,x(t) = 0, \\ x(0) = x_0 \in X, \end{cases} \tag{3.1}$$

where A is a linear operator acting in a Banach space X and with domain of definition $\mathfrak{D}(A)$. The task is to find *vector-valued* solutions $x : [0, \tau) \to X$ of the equation $x' + Ax = 0$ satisfying the initial value condition $x(0) = x_0$. The evolution equation (3.1) with prescribed initial value is denoted the *Cauchy problem of first order.*

The homogeneous Cauchy problem of *second order* consists of solving the initial value problem

$$\begin{cases} x''(t) + A\,x(t) = 0, \\ x(0) = x_0, \\ x'(0) = x_1, \end{cases} \tag{3.2}$$

where A is densely defined and closed. We also suppose $x_0 \in \mathfrak{D}(A)$, but the element x_1 may belong to any other linear subspace of X.

In the following we want to introduce some concepts of solution for the first order Cauchy problem:

1. A vector-valued function $x \in C([0, \tau), \mathfrak{D}(A)) \cap C^1((0, \tau), X)$ is called a *classical solution* of (3.1), if it satisfies the initial condition $x(0) = x_0$ and the differential equation in each point $t \in (0, \tau)$.
2. a continuous function $x : [0, \tau) \to X$ is said to be a *mild solution* of (3.1) if

$$(1)\ \ \tilde{x}(t) := \int_0^t x(s)\, ds \in \mathfrak{D}(A) \qquad \text{and} \qquad (2)\ \ A\,\tilde{x}(t) = x(t) - x_0$$

holds for all $0 < t < \tau$, where the integral on the right-hand side of (1) is meant in the Riemannian sense. One is led to the notion of mild solution in a natural way by integrating both sides of the equation $x' + A\,x = 0$, which then becomes into

$$\int_0^t x' ds \; = \; x(t) - x(0) \; = \; - \int_0^t A\,x(s)\, ds.$$

If A is closed, then the action of A and performing the integration commutes for continuous functions $x:[0,t]\to\mathfrak{D}(A)$, that is,

$$\int_0^t A\,x(s)\,ds \;=\; A\int_0^t x(s)\,ds,$$

and to the end we obtain the equation

$$x(t) \;=\; x(0) - A\int_0^t x(s)\,ds \qquad (t\in(0,\tau)).$$

Notice that each classical solution is a mild solution, and reversely a mild solution is classical iff it is continuously differentiable in $(0,\tau)$. Thus classical and mild solutions of homogeneous Cauchy problems only differ with respect to this regularity condition.

After having the notions of classical and mild solution for the homogeneous Cauchy problem available, we want to introduce the concepts of *correct posedness* and *well-posedness*.

First, the homogeneous Cauchy problem (3.1) is called *correctly posed*, if there is a unique classical solution x for any initial value $x_0\in\mathfrak{D}(A)$. Second, it is denoted *well-posed*, if it is correctly posed and the solution x depends continuously on the initial value x_0, or in other words, if for each $\tau>0$ there is $c_\tau>0$ such that

$$\|x(t)\| \le c_\tau\,\|x_0\|$$

for all $t\in(0,\tau)$ and all $x_0\in X$.

In [2] the continuous dependency of the solutions from the initial values is expressed as follows: if $\{x_n\}_{n\in\mathbb{N}}$ is a converging sequence in $\mathfrak{D}(A)$ with limit element $0\in X$, then the related solutions $t\mapsto x_n(t)$ are uniformly convergent to the zero function in all compact intervals $[0,t_0]$ for $0<t_0<\tau$ (similar definitions for correct posedness and well-posedness can also be quoted for the second-order homogeneous Cauchy problem (3.2)).

Let us finally mention that problem (3.1) governed by a densely defined and closed linear operator A is well-posed iff it is correctly posed and in addition A is regular, that is, $\rho(A)\neq\emptyset$ holds.

§4. Bochner Integration

To introduce a concept of integration for vector-valued functions, which generalizes the Lebesgue integral for scalar-valued functions, we first have to introduce a suitable notation of measurability for vector-valued functions.

A function $f:\Omega\to X$ is called *weakly measurable*, if $\phi\circ f:\Omega\to\mathbb{K}$ is measurable for every $\phi\in X^*$. Then a weakly measurable function $f:\Omega\to X$ is called *Bochner-integrable*, if

$$\int_\Omega \|f\|\ d\mu\ <\ \infty.$$

For any $p\geq 1$ we denote $L^p(\Omega,X)$ the space of functions $f:\Omega\to X$ having Bochner-integrable powers $|f|^p$. The elements of this space have also to be understood as equivalence classes of functions being identical almost everywhere in Ω. Endowed with the norm

$$f\mapsto \|f\|_p\ :=\ (\int_\Omega \|f\|^p\ d\mu)^{1/p},$$

the Bochner spaces $L^p(\Omega,X)$ are complete and thus constitute Banach spaces.

It should be mentioned that if we suppose the Banach space X to be separable, then a function $f:\Omega\to X$ is weakly measurable if and only if f is strongly (or Pettis-) measurable. The latter notion of measurability uses approximations of f by sequences $\{e_n\}_{n\in\mathbb{N}}$ of elementary functions $e_n:\Omega\to X$, and the Bochner integral of f is obtained by taking the limit of the Bochner integrals of the functions e_n, that is,

$$\int_\Omega f\ d\mu\ :=\ \lim_{n\to\infty}\int_\Omega e_n\ d\mu.$$

This definition proves to be independent of the specific choices of the approximating sequences for f.

With respect to applications in partial differential equations, Bochner spaces are of interest on the one hand for the choices $\Omega=G$ and $X=\mathbb{K}\in\{\mathbb{R},\mathbb{C}\}$, where G is a domain in one of the Euclidean spaces $\mathbb{R}^d$ and $L^p(G,\mathbb{K})$ is nothing but the Lebesgue space $L^p(G)$. On the other hand, in the weak solution theory of evolution equations we chose Ω to be a finite or infinite interval $I\subset\mathbb{R}$, and the Banach space X may be any complete function space $\mathbb{F}$ consisting of differentiable mappings $f:G\to\mathbb{K}$, so we are dealing with the Bochner spaces $L^p(I,\mathbb{F})$ in this case.

§5. Inhomogeneous Cauchy Problems

In this paragraph we deal with the solution theory of the *inhomogeneous* Cauchy problem

$$\begin{cases} x'(t) + A\,x(t) \;=\; y(t) \\ x(0) \;=\; x_0 \in X, \end{cases} \tag{5.1}$$

where y is a function $y:[0,\tau) \to X$ and the solution $x:(0,\tau) \to X$ of the problem (5.1) has to be searched.

Let us first remember DUHAMEL *'s principle*, also called the *method of variation of constants*: if x is a classical solution of (5.1), the right hand side $y:[0,\tau) \to X$ is continuous and the linear operator A is bounded, then

$$x(t) \;=\; S(t)\,x_0 + \int_0^t A\,y(s)\,ds, \quad (0 < t \leq \tau), \tag{5.2}$$

where $S(t)$ is the solution operator of the homogeneous problem (3.1) at time $t > 0$.

The basic problem in solving the inhomogeneous Cauchy problem is to find conditions on the function y, which ensure existence and uniqueness of a solution. We first review the both notions of classical and mild solution:

1. A function $x:[0,\tau] \to X$ is called a *classical solution* for problem (5.1), if $u \in C([0,\tau)) \cap C^1((0,\tau))$, fulfills the differential equation $x' + Ax = y$ in every $t \in (0,\tau)$ and the initial condition $x(0) = x_0$.
2. A function $x:[0,\tau] \to X$ is called a *mild solution*, if x fulfills equation (5.2).

While in the homogeneous case a mild solution is classical if and only if it is continuously differentiable, the validness of this assertion remains true for the inhomogeneous problem only under some additional regularity properties of the right hand side y.

To be precise, we have to check that y belongs to one of the following function spaces:

1. $y \in L^1([0,\tau), X) \cap C((0,\tau), X)$,
2. $y(t) \in \mathfrak{D}(A)$ for each $t \in (0,\tau)$,
3. $Ay \in L^1((0,\tau), X)$.

§6. Weak Derivatives and Bochner-Sobolev Spaces

We next introduce so-called *Bochner-Sobolev spaces*, which differ by the property that their members are at the same time Bochner-integrable and *weakly differentiable*.

Let H be a separable Hilbert space and $L^2(a,b;H)$ be the Bochner space consisting of functions $f:(a,b)\to H$, that is, we have

$$\int_a^b \|f(t)\|^2\, dt < \infty$$

for $f\in L^2(a,b,;H)$. We endow this space with the inner product

$$\langle f,g\rangle := \int_a^b \langle f(t),g(t)\rangle\, dt, \qquad f,g\in L^2(a,b;H)$$

such that the norm is given as usual by $\|f\|_2 = \langle f,f\rangle^{1/2}$. Thus $L^2(a,b;H)$ is Hilbertian, and if (a,b) is a *finite* interval, the inclusion $L^2(a,b;H)\subset L^1(a,b;H)$ holds. We shall denote $L^2(a,b;H)$ a *Hilbert-Bochner space.*

For $f\in L^2(a,b;H)$, a further function $g:(a,b)\to H$ is said to be the *weak derivative* of f, if

$$\int_a^b g\phi\, dx = -\int_a^b f\phi'\, dx$$

holds for each $\phi\in C_c^\infty(a,b,X)$, that is, ϕ is an indefinitely differentiable function with compact support in (a,b). If such a function g exists, we denote $f' := g$ in this case, which should not be confused with the classical derivative of f. But notice that if the weak derivative of f is present, then f is also classical differentiable.

§7. Hilbert-Bochner-Sobolev Spaces

Having the notion of weak derivative available, we finally introduce the *Hilbert-Bochner-Sobolev space*

$$H^1(a,b;H) := W^{1,2}(a,b,H) := \{f\in L^2(a,b;H) : f'\in L^2(a,b;H)\},$$

which is a Hilbert space as well with inner product given by

$$\langle f,g\rangle_1 := \int_a^b \langle f,g\rangle\, dt + \int_a^b \langle f',g'\rangle\, dt\,, \quad f,g\in H^1(a,b;H).$$

Finally, let H_1, H_2 be *real* Hilbert spaces, where H_1 is continuously embedded into H_2. Then there is also a continuous embedding

$$H^1(a,b;H_1) \overset{\kappa}{\hookrightarrow} H^1(a,b;H_2).$$

Consequently, given a Hilbert space triple $V\overset{\iota_1}{\hookrightarrow} H \overset{\iota_2}{\hookrightarrow} V^*$, we may regard the space $H^1(a,b;V)$ as a linear subspace of $H^1(a,b;V^*)$, which enables us to define the *mixed Hilbert-Bochner-Sobolev space*

$$H^1(a,b;\, V,\, V^*) := \{f \in L^2(a,b;\, V) : f' \in L^2(a,b;\, V^*)\}$$

or equivalently

$$H^1(a,b;\, V,\, V^*) := L^2(a,b;\, V) \cap H^1(a,b;\, V^*).$$

Then the mapping

$$(f,g) \mapsto \langle f,g\rangle_1 := \int_a^b \langle f(t), g(t)\rangle_V \, dt + \int_a^b \langle f'(t), g'(t)\rangle_{V^*} \, dt$$

fulfills the properties on an inner product and makes $H^1(a,b;\, V,\, V^*)$ into a Hilbert space having the following properties:

(a) Each $f \in H^1(a,b;\, V,\, V^*)$ is λ-nearly everywhere identical to a continuous function $f_c : [a,b] \to H$, and by identifying the functions f and f_c the space $H^1(a,b;\, V,\, V^*)$ can continuously be embedded into the space $C([a,b], H)$.

(b) The space $C^\infty([a,b],\, V)$ is dense in $H^1(a,b;\, V,\, V^*)$.

(c) The following product rule is valid for each $t \in (a,b)$:

$$\frac{d}{dt}\, \|u(t)\|^2 = 2\,\mathrm{Re}\,\langle u'(t), u(t)\rangle_1 \text{ for all } u \in H^1(a,b;\, V,\, V^*).$$

§8. Holomorphic Vector-Valued Functions

To the end we introduce the concept of differentiability for functions $x : \mathcal{O} \to X$, where $\mathcal{O}$ is an open and connected subset of the complex plane and X is a complex Banach space. The function x is called *complex differentiable* in a point $z_0 \in \mathcal{O}$, if

$$\lim_{z \to z_0} \frac{x(z) - x(z_0)}{z - z_0}$$

exists, and this limit is usually denoted $x'(z_0)$, called the *complex derivative* of x in z_0. Furthermore, x is said to be *holomorphic* or *analytic* in $z_0 \in \mathcal{O}$, if there is an open neighbourhood U of z_0 such that x has complex derivative in each $z \in U$. Finally, x is said to be holomorphic or analytic in $\mathcal{O}$, if x is so in each $z_0 \in \mathcal{O}$.

The property of x being holomorphic in $\mathcal{O}$ can be characterized as follows: x is holomorphic if and only if the functions $f \circ x : \mathcal{O} \to \mathbb{C}$ are holomorphic considered as complex-valued functions for each $f \in X^*$. As in complex analysis, a holomorphic function $x : \mathcal{O} \to X$ can be locally represented by a power series with coefficients in X. It should also be mentioned that the concepts just introduced also apply to *operator-valued* functions, where the Banach space X is nothing but the space $\mathscr{L}(Y)$ for some further Banach space Y.

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VI.
General Spectral Theory

The aim in this part is to give an outline on the spectral theory of a closed or continuous linear operator A acting in a complex Banach space X.

Let us start with the remark that each real Banach space X can be made into a complex Banach space $X_{\mathbb{C}}$, and the related procedure is denoted the *complexification* of X. It can be performed as follows: let $X_{\mathbb{C}} := X + iX = \{x + iy : x, y \in X\}$ and furnished with the addition and multiplication derived from those in X. The norm in $X_{\mathbb{C}}$ is defined according to

$$\|x + iy\| := \sup\{\|x\| \cos\theta + \|y\| \sin\theta : \theta \in [0, 2\pi)\},$$

in which this space is complete.

Let $\mathbf{1}_X : X \to X$ be the identity operator in X. In the sequel we shall consistently use the abbreviation $A - \zeta$ instead of $A - \zeta\mathbf{1}_X$. Given a linear operator $A : \mathfrak{D}(A) \subset X \to X$, the complex number $\zeta \in \mathbb{C}$ is called a

1. *regular value* of A, if $A - \zeta : \mathfrak{D}(A) \to X$ is bijective and the inverse operator $(A - \zeta)^{-1} : X \to \mathfrak{D}(A)$ is continuous. The set $\rho(A)$ of regular values is denoted the *resolvent set* of A;
2. *spectral value* of A, if $\zeta \notin \rho(A)$. The set $\sigma(A)$ of spectral values of A is said to be the *spectrum* of A, thus $\sigma(A) = \mathbb{C} \setminus \rho(A)$.

Having these concepts available, some further questions arise with respect to the location and shape of the resolvent set and the spectrum within the complex plane. Moreover, various classes of linear operators acting in X can be defined by requiring specific properties of these both fundamental sets.

§1. Resolvent Sets and Resolvent Functions

Let us continue with a discussion of the resolvent set and the resolvent functions. If A is a linear operator acting in a complex Banach space X and $\zeta \in \rho(A)$, then the inverse operator

$$R(A, \zeta) := (A - \zeta)^{-1} : X \to \mathfrak{D}(A)$$

is everywhere defined in X. Since we want to assume A to be closed, the operators $A - \zeta$ and $R(A, \zeta)$ are closed, too, and the *closed graph* *theorem* implies that $R(A,\zeta)$ is continuous, that is, $R(A,\zeta)\in\mathscr{L}(X)$[6]. The operator $R(A,\zeta)$ is called the *resolvent* of A with respect to $\zeta\in\rho(A)$, and the operator-valued mapping

$$\begin{cases} R(A,\cdot):\rho(A)\to\mathscr{L}(X)\,, \\ \zeta\mapsto(A-\zeta)^{-1} \end{cases} \tag{1.1}$$

is denoted the *resolvent function* (or shortly the *resolvent*) of A.

But the resolvents of a linear operator A are not independent from each other. In particular, for all $\zeta_1,\zeta_2\in\rho(A)$ the *first resolvent equation*

$$R(A,\zeta_1)-R(A,\zeta_2) \;=\; (\zeta_2-\zeta_1)\,R(A,\zeta_1)\circ R(A,\zeta_2) \tag{1.2}$$

holds, which implies that the resolvents $R(A,\zeta_1)$ and $R(A,\zeta_2)$ commute with each other.

Let us turn to the resolvent function of a linear operator A with non-empty resolvent set $\rho(A)$. In fact, if $\zeta_0\in\rho(A)$, we have $\zeta\in\rho(A)$ for all $\zeta\in\mathbb{C}$ satisfying

$$|\zeta-\zeta_0| \;<\; r \;:=\; \|R(A,\zeta_0)\|^{-1}\,,$$

thus $\rho(A)$ is an open set in $\mathbb{C}$. For values ζ in the disk with center ζ_0 and radius r one may represent the resolvent function $R(A,\cdot)$ as a power series with coefficients in the operator algebra $\mathscr{L}(X)$:

$$R(A,\zeta) \;=\; \sum_{n=0}^{\infty} R(A,\zeta_0)^{n+1}\,(\zeta_0-\zeta)^n.$$

This implies that $R(A,\cdot)$ is locally holomorphic in $\rho(A)$.

The question, whether a given complex number ζ belongs to the resolvent set or the spectrum of A, may be answered by the following criterion: let $\{\zeta_n\}_{n=1}^{\infty}$ be a sequence in $\rho(A)$ with $\lim_{n\to\infty}\xi_n=\zeta$. Then $\zeta\in\rho(A)$ if and only if

$$\sup_{n\in\mathbb{N}}\|R(A,\zeta_n)\|<\infty.$$

On the other hand, if ζ is an element of the boundary of $\rho(A)$, then $\|R(A,\zeta_n)\|\to\infty$ holds for each sequence $\{\zeta_n\}_{n=1}^{\infty}$ with $\zeta_n\in\rho(A)$ and $\zeta_n\to\zeta$.

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[6] If A is *not* closed, one has in addition to require that for $\zeta\in\rho(A)$ the inverse operator $R(A,\zeta)$ is bounded.

We finally emphasize that if $\rho(A) \neq \emptyset$, then A is closed. In fact, let $\zeta \in \rho(A)$ such that $A - \zeta$ is bijective with domain of definition $\mathfrak{D}((A-\zeta)^{-1}) = Y$ and range $\mathfrak{R}((A-\zeta)^{-1}) = \mathfrak{D}(A)$. Let $x_n \to x$ with $x_n \in \mathfrak{D}(A)$ for $n \in \mathbb{N}$ and $Ax_n \to y$. Then

$$x = \lim_{n\to\infty} x_n = \lim_{n\to\infty} (A-\zeta)^{-1}(A-\zeta)x_n = (A-\zeta)^{-1}(y-\zeta x). \quad (1.3)$$

Thus $x \in \mathfrak{R}((A-\zeta)^{-1}) = \mathfrak{D}(A)$. Applying $(A-\zeta)^{-1}$ from left to (1.3), we obtain $Ax - \zeta x = y - \zeta x$, hence $Ax = y$, which is nothing but closeness of A.

Thus a reasonable spectral theory makes only sense for linear operators being at least closed. Operators, which are closed and in addition own non-empty resolvent sets, are often called *regular.*

§2. Different Kinds of Spectra

We next turn to the spectrum of A. First of all, the set $\sigma(A)$ is closed, since $\rho(A)$ is open. It may happen that $\sigma(A)$ degenerates to the empty set. In the special case, where A is everywhere defined and *bounded*, the spectrum of A proves to be non-empty and forms a bounded subset of the complex plane.

We next will have a look at the spectral values of a closed linear operator A and study possible reasons, which for $\lambda \in \sigma(A)$ prevent the operator $A - \lambda$ to be bijective.

1. Recall from linear algebra that a complex number λ is called an *eigenvalue* of a linear operator A, if there is an element $x \in X$ such that $x \neq 0$ and $Ax = \lambda x$. In this case the element x is called an *eigenvector* of A with respect to λ, and all these eigenvectors form a linear subspace of X, called the *eigenspace* of A for λ. The *point spectrum* of A is defined to be the set $\sigma_p(A)$ of all eigenvalues of A. Clearly, for $\lambda \in \sigma_p(A)$ the operator $A - \lambda$ is not injective, since its kernel is non-trivial.

   If $A - \lambda$ is injective, but not surjective, then $(A-\lambda)^{-1}$ is defined on the set $\mathfrak{R}(A - \lambda)$. We distinguish between two cases:
2. if $\mathfrak{R}(A - \lambda)$ is dense in X, then λ is said to be an element of the set $\sigma_c(A)$, called the *continuous spectrum* of A, otherwise
3. the number λ belongs to the *residual spectrum* $\sigma_r(A)$. There is an interesting identity between the spectrum of a densely defined and closed linear operator A and that of its adjoint A^*, namely $\sigma_r(A) = \sigma_p(A^*)$ (see [1], p. 161).

In summary, for each closed linear operator $A:\mathfrak{D}(A)\subset X\to X$ the set of complex numbers forms a disjoint decomposition of the resolvent set and the different kinds of spectra, namely

$$\mathbb{C} = \rho(A) \sqcup \sigma_p(A) \sqcup \sigma_c(A) \sqcup \sigma_r(A).$$

The *approximate point spectrum* $\sigma_{ap}(A)$ of a closed linear operator A is the set of $\lambda\in\mathbb{C}$, for which $A-\lambda$ is not injective or $\mathfrak{R}(A-\lambda)$ is not closed in X. The elements of $\sigma_{ap}(A)$ are called the *approximate eigenvalues* of A. Any number $\lambda\in\mathbb{C}$ belongs to $\sigma_{ap}(A)$ if and only if there is a sequence $\{x_n\}_{n\in\mathbb{N}}\subset\mathfrak{D}(A)$ such that $\|x_n\|=1$ for all $n\in\mathbb{N}$ and $\lim_{n\to\infty}\|Ax_n-\lambda x_n\|=0$. Since A is assumed to be closed, the topological boundary $\partial\sigma(A)$ of the spectrum forms a subset of $\sigma_{ap}(A)$.

It is easy to see that each eigenvalue of A belongs to $\sigma_{ap}(A)$, so this set generalizes the point spectrum $\sigma_p(A)$. But $\sigma_{ap}(A)$ can be empty only in the case, where $\sigma(A)=\emptyset$ or $\sigma(A)=\mathbb{C}$.

Recall that the adjoint of a closable linear operator in Hilbert space is closed, so the basic concepts of spectral theory are applicable as well. Let $A:\mathfrak{D}(A)\subset H\to H$ be densely defined and closed. Then the following identities hold:

1. $\rho(A^*) = \{\,\overline{\zeta}\in\mathbb{C} : \zeta\in\rho(A)\,\}$.
2. $\sigma(A^*) = \{\,\overline{\lambda}\in\mathbb{C} : \lambda\in\sigma(A)\,\}$.
3. $\sigma_c(A^*) = \{\,\overline{\lambda}\in\mathbb{C} : \lambda\in\sigma_c(A)\,\}$.
4. If $\lambda\in\sigma_p(A)\setminus\sigma_c(A)$, then $\overline{\lambda}\in(\sigma_p(A^*)\setminus\sigma_c(A^*))\cup\sigma_r(A^*)$.
5. If $\lambda\in\sigma_r(A)$, then $\overline{\lambda}\in\sigma_p(A^*)\setminus\sigma_c(A^*)$.

There are a couple of relationships between the spectrum of a linear operator A and those of its resolvents, that are known as and is known as *spectral mapping theorems* for resolvents, see for example [1], pp. 161-162:

THEOREM VI.1. For fixed $\zeta\in\mathbb{C}$, various kinds of spectra of the operator $R(A,\zeta)$ are related to the spectra of A by the identities

1. $\sigma(R(A,\zeta))\setminus\{0\} = \{(\lambda-\zeta)^{-1} : \lambda\in\sigma(A)\}$,
2. $\sigma_p(R(A,\zeta))\setminus\{0\} = \{(\lambda-\zeta)^{-1} : \lambda\in\sigma_p(A)\}$,
3. $\sigma_{ap}(R(A,\zeta))\setminus\{0\} = \{(\lambda-\zeta)^{-1} : \lambda\in\sigma_{ap}(A)\}$,
4. $\sigma_r(R(A,\zeta))\setminus\{0\} = \{(\lambda-\zeta)^{-1} : \lambda\in\sigma_r(A)\}$.

If A is bounded, the *spectral radius* of A is defined according to

$$r(A) := \sup\{|\lambda| : \lambda \in \sigma(A)\}.$$

It can be shown that $r(A) \leq \|A\|$, and since each resolvent $R(A,\zeta)$ is bounded, we have

$$\|R(A,\zeta)\| \geq r(R(A,\zeta)) \geq \delta(\zeta,\sigma(A))^{-1} \tag{2.1}$$

for any $\zeta \in \rho(A)$, where $\delta(\zeta,\sigma(A))$ is the minimal distance between the singleton $\{\zeta\}$ and the spectrum of A. Notice that inequality (2.1) is a consequence of the spectral mapping theorem VI.1.

§3. Riesz-Schauder Spectral Theory

We next turn to the spectral theory of compact linear operators developed by F. RIESZ and J. SCHAUDER in the early decades of the 20$^{\text{th}}$ century. In advance, their results are known as the RIESZ-SCHAUDER *spectral theorem*, which resemble those of (d,d)-matrices considered as linear mappings in the finite-dimensional spaces $\mathbb{C}^d$.

Before quoting the main theorem of this paragraph, let us have a short look to this case, which can completely be covered by the theory of square matrices. In finite-dimensional linear algebra, the spectral theory begins with the study of eigenvalues and eigenvectors of a square matrix $\mathcal{A} \in \mathbb{C}^{d\times d}$. Remember that an eigenvalue λ of $\mathcal{A}$ is a complex number satisfying $\mathcal{A}x = \lambda x$ for some nonzero vector $x \in \mathbb{C}^d$, called an eigenvector corresponding to λ. The set of all eigenvalues of $\mathcal{A}$, counting multiplicities, forms the spectrum $\sigma(\mathcal{A})$. Moreover, the eigenvalues of $\mathcal{A}$ are precisely the roots of its characteristic polynomial $p_{\mathcal{A}}(\lambda) = det(\lambda - \mathcal{A})$, a monic polynomial of degree d. The algebraic multiplicity of an eigenvalue λ is the multiplicity of λ as a root of $p_{\mathcal{A}}(\lambda)$. The geometric multiplicity of λ, however, is the dimension of the eigenspace $\mathfrak{N}(\mathcal{A}-\lambda)$, which is at most the algebraic multiplicity.

A matrix $\mathcal{A}$ is diagonalizable if and only if for every eigenvalue geometric multiplicity equals the algebraic multiplicity. When $\mathcal{A}$ is not diagonalizable, its structure is captured by the *Jordan canonical form*, which decomposes $\mathbb{C}^d$ into generalized eigenspaces. For an eigenvalue λ, the generalized eigenspace is nothing but $\mathfrak{N}(\mathcal{A}-\lambda)^m$, where m is the *index* of λ, defined as the smallest positive integer such that $\mathfrak{N}(\mathcal{A}-\lambda)^m = \mathfrak{N}(\mathcal{A}-\lambda)^{m+1}$. This index equals the size of the largest Jordan block associated with λ. The space $\mathbb{C}^n$ then decomposes as a direct sum of these generalized eigenspaces over all distinct eigenvalues.

The *Jordan canonical form theorem* states that every square matrix $\mathcal{A}$ over an algebraically closed field such as $\mathbb{C}$ is similar to a unique (up to permutation of blocks) Jordan matrix J, meaning there exists an invertible matrix P such that $\mathcal{A}=PJP^{-1}$. The matrix J is block-diagonal, consisting of Jordan blocks $J_k(\lambda)$ for each eigenvalue λ, where a $k\times k$ Jordan block has the value λ on the main diagonal and the value 1 on the superdiagonal. This form was established by C. JORDAN in his 1870 treatise. A simple example of a non-diagonalizable matrix is

$$\mathcal{A}=\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix},$$

which has the eigenvalue 0 with algebraic multiplicity 2 but geometric multiplicity 1.

In summary, the JORDAN decomposition of a (d,d)-matrix propagates the existence of k pairwise different complex eigenvalues $\lambda_1,\dots,\lambda_k$ and associated eigenspaces $E_1,\dots,E_k$ such that $\mathbb{C}^d=E_1\oplus E_2\oplus\dots\oplus E_k$. Moreover, the poles of the resolvent function $R(A,\cdot)$ coincides with the eigenvalues of the operator A.

The *spectral theorem* for compact linear operators shows that the spectral theory of such operators in principle does not differ from that of linear transformations acting in the spaces $\mathbb{C}^d$:

THEOREM VI.2. Let X be a complex Banach space and $k:X\to X$ be compact. Then the following assertions hold:
(a) The spectrum $\sigma(k)\subset\mathbb{C}$ is mostly countable.
(b) $\lambda=0$ is the only possible limit point of $\sigma(k)$.
(c) If X is infinite-dimensional, then $0\in\sigma(k)$.
(d) Each $0\neq\lambda\in\sigma(k)$ is an eigenvalue of k with finite-dimensional eigenspace.
(e) Every $0\neq\lambda\in\sigma(k)$ is a pole of the resolvent function $R(k,\cdot)$.

Notice that the forth assertion in theorem VI.2 is nothing but the second part of FREDHOLM's alternative theorem (see paragraph IV.5). Indeed, this is true since for each compact operator k the operators $k-\lambda$ $(\lambda\in\mathbb{C})$ are semi-solvable, that is, they are Fredholm operators with vanishing index, see theorem IV.6.

§4. Linear Operators with Pure Point Spectrum

In this paragraph we consider a closed linear operator acting in a Banach space, whose spectrum consists of eigenvalues only and does not have accumulation points in the complex plane. By this reason such an operator is called to *have pure point spectrum.*

In other words, we study linear operators A having the property $\sigma(A) = \sigma_p(A)$, that is, the continuous and residual spectrum of A are both empty and $\sigma(A)$ consists of eigenvalues only. Such operators are of outstanding meaning in applied analysis, since they appear as realizations of linear-ellipic partial differential operators in Sobolev spaces for *bounded* domains $G \subset \mathbb{R}^d$. Recall also from paragraph IV.6 that in this case the operator equation $Ax = y$ is equivalent to an operator equation of the form $(1_X + k)x = y$ with k being compact, hence FREDHOLM's alternative theorem may be applied.

Furthermore, a closed linear operator A with domain of definition $\mathfrak{D}(A) \subset X$ is said to *own compact resolvents*, if at least one resolvent $(A - \zeta)^{-1}$ with $\zeta \in \rho(A)$ is compact. The first resolvent equation (1.2) implies that in this case all other resolvents of A are compact. It is easy to verify that A has pure point spectrum if and only if it has compact resolvents.

The eigenvalues of a linear operator A with pure point spectrum are mostly countable and can be ordered by their moduli, that is, there is a sequence $\{\lambda_n\}_{n \in \mathbb{N}}$ consisting of eigenvalues of A such that

$$|\lambda_1| \leq |\lambda_2| \leq |\lambda_3| \leq \dots .$$

More can be said in the "symmetric" case, see the *discrete spectral theorem* (paragraph IX.3).

There is a useful criterion to check whether a linear operator owns the property of having pure point spectrum:

THEOREM VI.3. Let the domain of definition $\mathfrak{D}(A) \subset X$ of a densely defined operator A in X be equipped with the graph norm induced by A. Then A has compact resolvents if and only if the embedding $\iota : (\mathfrak{D}(A), \|\cdot\|_A) \hookrightarrow (X, \|\cdot\|)$ is compact.

§5. The Riesz-Dunford Functional Calculus

Our subject in this paragraph is the exposition of the *holomorphic functional calculus*, sometimes also called the RIESZ-DUNFORD *calculus* for bounded linear operators acting in a Banach space X. It

allows to define functions $f(A)$ for a given linear operator A.
To introduce the Riesz-Dunford functional calculus, first recall that a function $f: G \subset \mathbb{C} \to \mathbb{C}$ is called *analytic* or *holomorphic*, if $f(\zeta)$ for ζ close to ζ_0 can locally be represented by a power series

$$f(\zeta) = \sum_{k=0}^{\infty} a_k(\zeta_0)\,(\zeta - \zeta_0)^k \quad \text{for } |\zeta - \zeta_0| < R(\zeta_0).$$

Let $H(G)$ be the space of analytic functions in G. We denote $H^\infty(G)$ the subspace of all *bounded* functions in $H(G)$, which is a Banach space with respect to the norm

$$\|f\|_\infty := \sup_{\zeta \in G} |f(\zeta)|.$$

Let $\mathcal{O} \subset \mathbb{C}$ be an open and connected subset of the complex plane, $g: [a,b] \to \mathcal{O}$ be a rectifiable path and $f: \mathcal{O} \to \mathbb{C}$ be a complex-valued function. Then

$$\int_\gamma f(\zeta)\,d\zeta := \int_a^b f\,dg$$

is called the *contour integral* of f with respect to the function g, where γ is the image of $[a,b]$ under g.

Cauchy*'s integral theorem* is a corner stone in complex analysis and quotes that if f is *analytic* in $\mathcal{O}$, then for *every* closed curve γ we have

$$\int_\gamma f(\zeta)\,d\xi = 0.$$

Furthermore, Cauchy*'s integral formula* says that for each $z_0 \in \mathcal{O}$ belonging to the interior of a closed curve γ the value $f(z_0)$ is nothing but

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - z_0}\,dz\,, \tag{5.1}$$

where the *winding number* of γ has value 1. This representation of $f(z_0)$ does not depend on the shape of γ.

If we consider $z_0 \in \mathbb{C}$ as a linear operator $Z_0: \mathbb{C} \to \mathbb{C} \in \mathscr{L}(\mathbb{C})$ defined by $Z_0\,\zeta := z_0 \cdot \zeta$ for $\zeta \in \mathbb{C}$, then the spectrum of $\mathbb{Z}_0$ consists of the single point z_0 and the resolvent adopts the form $R(Z_0, \zeta) = (\zeta - z_0)^{-1}$. Using (5.1), we obtain

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma f(\zeta)\,R(Z_0, \zeta)\,d\zeta. \tag{5.2}$$

The formula (5.2) serves as a pattern for the *holomorphic* or *analytic functional calculus* in unital and complex Banach algebras like the operator algebra $\mathscr{L}(X)$. Let A be a bounded linear operator acting in X (so the spectrum $\sigma(A)$ is non-emtpy and bounded), γ be a closed and rectificable path with winding number 1 containing $\sigma(A)$ and $f: G \to \mathbb{C}$ be a bounded and analytic function defined on an open set $G \supset \sigma(A)$. Then the linear operator $f(A)$ is defined by the *Riesz-Dunford integral*

$$f(A) := \frac{1}{2\pi i} \int_{\gamma} f(\zeta) R(A, \zeta)\, d\zeta. \tag{5.3}$$

The right hand side is a vector-valued Riemann-Stieltjes integral, and convergence of the intermediate sums to the element $f(A)$ is understood in the sense of the norm topology in $\mathscr{L}(X)$. The value $f(A)$ is well-defined, since f is bounded and analytic.

Recall that the set $H^\infty(\mathcal{O})$ of bounded and analytic functions $f: \mathcal{O} \to \mathbb{C}$ defined in an open neighbourhood $\mathcal{O}$ of $\sigma(A)$ forms a Banach algebra. Thus the mapping

$$\begin{cases} \Phi_A : H^\infty(\mathcal{O}) \to \mathscr{L}(B), \\ f \mapsto f(A) \text{ as defined in (5.3)} \end{cases}$$

forms a *Banach algebra homomorphism* from $H^\infty(\mathcal{O})$ to $\mathscr{L}(B)$, that is, the mapping Φ_A is linear, norm-preserving and respects the multiplicative operations in both algebras. But Φ_A also maps polyomials $p \in H^\infty(\mathcal{O})$ to polynomials $p(A) \in \mathscr{L}(B)$. Due to these properties, Φ_A is called a *functional calculus*, which in addition deserves the appendix "analytic", since the space $H^\infty(\mathcal{O})$ consists of analytic functions.

It should be mentioned that for this kind of functional calculus the *spectral mapping theorem* is valid, that is, $\sigma(f(A)) = f(\sigma(A))$ for $A \in \mathscr{L}(B)$ and $f \in H^\infty(\mathcal{O})$. The resulting operator $f(A) \in \mathscr{L}(X)$ defined by (5.3) is called the RIESZ-DUNFORD *operator* associated with A and $f \in H^\infty(G)$.

In the following we suppose that λ is an *isolated* point of the spectrum $\sigma(A)$, so $\sigma(A) \cap \mathcal{U}_\lambda = \{\lambda\}$ holds for every sufficiently small neighbourhood $\mathcal{U}_\lambda$ of λ. Let γ_λ be a closed and rectificable path, which surrounds only λ, but not any other element of $\sigma(A)$. Then for $x \in X$ the contour integral

$$\pi_\lambda(x) := \frac{1}{2\pi i}\int_{\gamma_\lambda} R(A,\zeta)x\,d\zeta$$

gives rise to a further linear mapping π_λ, called the RIESZ *projection* of A with respect to the spectral value λ.

To the end, we collect the properties of the mapping π_λ to the following list:

1. π_λ is linear;
2. π_λ owns the *projection property* $\pi_\lambda \circ \pi_\lambda = \pi_\lambda$;
3. The spectrum of this mapping only consists of the value λ_0 itself;
4. π_λ commutes with A, so $A \circ \pi_\lambda = \pi_\lambda \circ A$ holds;
5. The range Π_λ of π_λ is a closed subspace in X;
6. There is a closed subspace $\Pi_\lambda^\perp \subset X$ such that $X = \Pi_\lambda \oplus \Pi_\lambda^\perp$, where $\Pi_\lambda^\perp$ is defined by $\Pi_\lambda^\perp := 1_H - \Pi_\lambda$;
7. Both subspaces Π_λ and $\Pi_\lambda^\perp$ are A-invariant, so $A\,\Pi_\lambda \subset \Pi_\lambda$ and $A\,\Pi_\lambda^\perp \subset \Pi_\lambda^\perp$. One also says that the subspaces Π_λ and $\Pi_\lambda^\perp$ *reduces* the entire space X;
8. The mapping $\pi_\lambda^\perp := 1_H \pi_\lambda$ is a projection again, and both projections π_λ and $\pi_\lambda^\perp := 1_H - \pi_\lambda$ are *orthogonal* to each other, that is, $\pi_\lambda \circ \pi_\lambda^\perp = \pi_\lambda^\perp \circ \pi_\lambda = 0$.

The concept of Riesz projection may be extended into two directions. First, the linear operator A may be assumed to be closed instead of being continuous, and second the subset $\{\lambda\} \subset \sigma(A)$ may be replaced by any connected component $\sigma_0(A)$ of the set $\sigma(A)$.

§6. Essential and Discrete Spectrum

The concept of semi-Fredholm operator (see section IV) can be used to characterize the *essential spectrum* $\sigma_{ess,1}(A)$ of a closed operator A acting in a complex Banach space X: a number $\lambda \in \mathbb{C}$ belongs to $\sigma_{ess,0}(A)$ if and only if $A - \lambda$ is *not* semi-Fredholm. But there are further approaches to define the notion of essential spectrum:

$$\begin{aligned}
\sigma_{ess,1}(A) &:= \mathbb{C} \setminus \{\lambda \in \mathbb{C} : A - \lambda \text{ is upper semi-Fredholm}\},\\
\sigma_{ess,2}(A) &:= \mathbb{C} \setminus \{\lambda \in \mathbb{C} : A - \lambda \text{ is lower semi-Fredholm}\},\\
\sigma_{ess,3}(A) &:= \mathbb{C} \setminus \{\lambda \in \mathbb{C} : A - \lambda \text{ is semi-Fredholm}\},\\
\sigma_{ess,4}(A) &:= \mathbb{C} \setminus \{\lambda \in \mathbb{C} : A - \lambda \text{ is Fredholm}\},\\
\sigma_{ess,5}(A) &:= \mathbb{C} \setminus \{\lambda \in \mathbb{C} : A - \lambda \text{ is Fredholm with } \chi(A-\lambda) = 0\},\\
\sigma_{ess,6}(A) &:= \{\lambda \in \sigma_{ess,5}(A) : \text{ the scalars near } \lambda \text{ are in } \rho(A)\}.
\end{aligned}$$

The sets $\sigma_{ess,4}(A)$ and $\sigma_{ess,5}(A)$ are called the *Wolf spectrum* and the *Schechter spectrum*, respectively, whereas the set $\sigma_{ess,6}(A)$ is denoted the *Browder spectrum.*

All sets $\sigma_{ess,k}(A)$ are closed subsets of $\sigma(A)$, and by definition we have the chain of inclusions

$$\sigma_{ess,3}(A) \subset \sigma_{ess,1}(A) \cup \sigma_{ess,2}(A) \subset \sigma_{ess,4}(A) \subset \sigma_{ess,5}(A) \subset \sigma_{ess,6}(A).$$

In the Hilbert space setting, however all sets $\sigma_{ess,k}(A)$ coincide and form the largest subset $\sigma_{ess}(A) \subset \sigma(A)$ such that for every compact linear operator $k \in \mathscr{L}(X)$

$$\sigma_{ess}(A+k) = \sigma_{ess}(A).$$

On the other hand, the *discrete spectrum* $\sigma_d(A)$ is defined to be the relative complement of $\sigma_{ess,5}(A)$ with respect to $\sigma(A)$, hence we have a disjoint decomposition of the spectrum of A according to

$$\sigma(A) = \sigma_d(A) \sqcup \sigma_{ess,6}(A).$$

The set $\sigma_d(A)$ contains those points $\lambda \in \sigma(A)$, which are isolated in $\sigma(A)$ and for which the rank of the *Riesz projection* P_λ is finite, where P_λ is defined by the *Dunford integral*

$$P_\lambda(x) := \frac{1}{2\pi i} \int_{\gamma_\lambda} R(A,\zeta)x \, d\zeta \quad (x \in X) \tag{6.1}$$

and γ_λ is a closed continuous path enclosing λ but no other values in $\sigma(A)$. Elements of the set $\sigma_d(A)$ are called *normal eigenvalues.* One may also introduce the set $\sigma_d(A)$ as the collection of isolated points $\lambda \in \sigma(A)$ having finite algebraical multiplicity (this magnitude is the Hamel dimension of the linear space $E_{gen}(A,\lambda) := \cup_{n\in\mathbb{N}} E(A^n,\lambda)$ of generalized eigenvalues of A).

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VII.
Commutative Banach Algebras

The aim in this part is to review commutative Banach algebras and their applications to other topics in functional analysis and operator theory. They are of own interest due to the celebrated *Gelfand transformation.* This presence of this fundamental mapping lead us on the one hand to the *Fourier transformation* in certain Lebesgue spaces and on the other hand to the continuous functional calculus for *normal* linear operators acting in Hilbert space.

§1. Normed Algebras

By a *normed algebra* (rsp. *Banach algebra*) is meant a normed linear space (rsp. Banach space) together with a multiplicative operation, which is compliant with the norm in some sense.

More specifically, let $(B,+,\circ)$ be a $\mathbb{K}$-algebra, that simultaneously constitutes a normed space $(B,\|\cdot\|)$. Then B is called a *normed algebra*, if the following conditions are fulfilled:

(A-1) $\|x\circ y\| \le \|x\|\cdot\|y\|$ for each pair of elements $x,y\in B$.

(A-2) If B has an identity $1\in B$, then $\|1\|=1$.

Condition (A-1) implies that the map $(x,y)\mapsto x\circ y$ is continuous. If (A-2) is valid, then B is called a *unital* algebra. Finally, B is said to be *commutative*, if

(A-3) $x\circ y=y\circ x$ for all $x,y\in B$.

As for normed linear spaces, completeness of normed algebras is of interest, thus a normed algebra is denoted a *Banach algebra*, if it is complete as a normed linear space.

The most prominent example of a non-commutative Banach algebra is given by the $\mathbb{C}$-algebra $\mathscr{L}(X)$ of bounded linear operators $A:X\to X$, where X is a real or complex Banach space. It forms a unital Banach algebra with respect to composition of mappings and the operator norm $\|A\|$ of $A\in\mathscr{L}(X)$.

One of the key results in the theory of complex Banach algebras is the *Gelfand-Mazur theorem* (see also [1], p. 14):

Theorem VII.1. If each element $x\neq 0$ in a complex Banach algebra B is invertible, then $B\cong\mathbb{C}$.

In other words, theorem VII.1 quotes that the field of complex numbers is essentially the only complex Banach algebra, in which each element $x \neq 0$ is invertible.

Let B be a complex and unital Banach algebra. By the *Neumann series* of $x \in B$ is meant the infinite sum $s_x := \sum_{n=0}^{\infty} x^n$. If s_x is convergent in B, then the element $\mathbf{1} - x$ is invertible, and a short computation proves $(\mathbf{1} - x)^{-1} = s_x$. Like in the case $B = \mathbb{C}$, the Neumann series is convergent for all $x \in B$ fulfilling $\|x\| < 1$.

If $x \in B$ is invertible, then any other element $x' \in B$ having the property

$$\|x - x'\| < \frac{1}{\|x^{-1}\|}$$

proves to be invertible, too. Thus all elements in a sufficiently small neighbourhood of an invertible element have convergent Neumann series and are invertible, which implies that the set $\mathbf{GL}(B)$ of invertible elements in B is open. It is called the *group of units* in B, since $x \circ y$ and x^{-1} are invertible for $x, y \in \mathbf{GL}(B)$. Both operations $(x, y) \mapsto x \circ y$ and $x \mapsto x^{-1}$ are continuous, so $\mathbf{GL}(B)$ forms a topological group, when considered in the subset topology of B.

In this section we are especially interested in *commutative* Banach algebras. Here are some important examples with respect to applications:

1. The field $\mathbb{C}$ of complex numbers may be regarded as a complex unital Banach algebra in its own right.
2. For any topological space X the function spaces $C_b(X)$ and $C_0(X)$, both endowed with the supremum norm $\|f\| := \sup_x |f(x)|$.

   If X is *compact*, then the function space $C(X)$ consisting of continuous functions $f : X \to \mathbb{C}$ forms a Banach algebra, which in addition is unital with identity $\mathbf{1}(x) := 1$ for $x \in X$.
3. The algebra $L^1(G, \mathcal{B}_G, \mu)$ of Lebesgue-integrable functions with respect to the Haar measure μ on a locally compact topological group G, where multiplication is defined by *convolution*

   $$(f * g)(x) := \int_{\mathbb{R}} f(y)\, g(x - y)\, d\mu$$

   for $f, g \in L^1(G)$. Important examples for G are the classical groups $\mathbb{R}^d$, $\mathbb{Z}$ and $\mathbb{T} = \{z \in \mathbb{C} : \|z\| = 1\}$.
4. The Banach algebra $L^\infty(X)$ of essentially bounded functions

$f: X \to \mathbb{R}$, where $(X, \mathcal{B}_X, \mu)$ may be any measure space. Addition and multiplication in this algebra are defined pointwise.

5. The algebra $\mathcal{M}^r(G)$ of regular complex Borel measures defined on a locally compact topological group G. The multiplication is defined by convolution of measures:

$$(\mu * \nu)(B) := (\mu \times \nu)(\{(x, y) \in G \times G : x + y \in B\})$$

for any Borel set $B \subset G$. This Banach algebra is even unital, since the so-called *Dirac measure* μ_0 owns the property $\mu_0 * \nu = \nu * \mu_0 = \nu$ for all $\nu \in \mathcal{M}^r(G)$.

§2. Resolvent and Spectrum of Banach Algebra Elements

The basic concepts of spectral theory, as introduced for bounded linear operators acting in a Banach space, are also well-defined for elements in a unital complex Banach Algebra B. Let $\mathbf{1} \in B$ be the identity and $x \in B$ be arbitrary, then

1. any number $\zeta \in \mathbb{C}$ is called *regular* with respect to x, if $x - \zeta\mathbf{1}$ is invertible;
2. the set of regular numbers $\rho(x) := \{\zeta \in \mathbb{C} : (x - \zeta\mathbf{1})^{-1} \text{ exists }\}$ is said to be the *resolvent set* of x;
3. the mapping $R_x : \zeta \in \rho(x) \mapsto (x - \zeta\mathbf{1})^{-1}$ is denoted the *resolvent function* and $R_x(\zeta)$ is called the *resolvent* of x with respect to $\zeta \in \rho(x)$;
4. the complementary set $\sigma(x) := \mathbb{C} \setminus \rho(x)$ is called the *spectrum* of x, and any element $\lambda \in \sigma(x)$ is called a *spectral value* of x.

We just reformulate some known results for resolvent sets and spectra of bounded linear operators, which are also true for elements in a unital complex Banach algebra B. Because the set $\mathbf{GL}(B)$ of invertible elements is open, the resolvent set $\rho(x)$ of $x \in B$ forms an open subset in $\mathbb{C}$. Moreover, the resolvent function $R(x, \cdot) : \rho(x) \to B$ is continuous and even analytic, that is, $R(x, \cdot)$ can locally be represented by a convergent power series with coefficients in B.

On the other hand, the spectrum $\sigma(x)$ is closed and *non-empty*, which is a generalization of the fact that $\sigma(A) \neq \emptyset$ for any bounded linear operator $A : X \to X$.

§3. Gelfand Representation

Given the Banach algebras B_1, B_2, a linear mapping $\Phi : B_1 \to B_2$ is denoted a *Banach algebra homomorphism*, if it preserves multiplication according to

$$\Phi(x \circ y) = \Phi(x) \circ \Phi(y) \text{ for all } x, y \in B_1.$$

Moreover, B_1 and B_2 are said to be *isomorphic* to each other, if there is a bijective homomorphism $\Phi : B_1 \to B_2$ fulfilling $\|\Phi(x)\|_{B_2} = \|x\|_{B_1}$ for all $x \in B_1$, which in this case is called a *Banach algebra isomorphism* (notation: $B_1 \cong B_2$).

The *Gelfand representation* is a powerful mapping in the setting of commutative Banach algebras and provides a Banach algebra homomorphism between an arbitrary commutative Banach algebra B and the algebra $C_0(X)$ of continuous functions vanishing at infinity, where X is assumed to be a locally compact Hausdorff space.

A *character* on a complex Banach algebra B is a *non-zero* Banach algebra homomorphism $\chi : B \to \mathbb{C}$. Characters (often also called *multiplicative functionals*) are of great importance in the theory of unital commutative Banach algebras, as we will see in the following.

By the multiplication property of a Banach algebra homomorphism, each character χ on B satisfyies the inequality $\|\chi\| \leq 1$, so χ is a continuous mapping and thus an element of the dual space B'. The set Γ of all characters is called the *structure space* or the *spectrum* of the Banach algebra B.

Due to the inequality $\|\chi\| \leq 1$ for all $\chi \in \Gamma$, the set Γ is a subset of the closed unit ball $B_1(0) \subset B'$. Thus we may endow Γ with the subset topology on $B_1(0)$ inherited from the weak-*-topology on B', which is called the *Gelfand topology* on the structure space. Now, by the *Alaoglu-Bourbaki theorem* the weak-*-topology on B' is weak*-compact, implying that the Gelfand topology must be locally compact and Hausdorff. Hence the structure space Γ becomes into a locally compact Hausdorff space. Moreover, if B is unital, it can be shown that the set Γ is even compact.

From now on let B be a *commutative* Banach algebra. We consider the natural mapping $\gamma : B \to C(\Gamma)$, where γ assigns to each $x \in B$ the function $\hat{x} \in C(\Gamma)$ by way of $\hat{x}(\chi) := \chi(x)$, that is, the function values of γ are given by evaluating the characters χ at the element x. The mapping γ is called the *Gelfand representation* of B.

We collect the main properties of the Gelfand representation for a commutative unital Banach algebra B to the following list:

1. γ is a Banach algebra homomorphism from B to $C(\Gamma)$ or $C_0(\Gamma)$, depending whether B is unital or not.
2. If B is unital, then $\hat{1}$ is the constant function 1 on Γ.
3. If B is unital, then $x \in B$ is invertible iff $\hat{x}$ nowhere vanishes.
4. $\|\hat{x}\|_\infty \leq \|x\|$ for each $x \in B$.

The Gelfand representation is injective iff B is a *semisimple* Banach algebra, that is, if the intersection of all maximal ideals in B, called the *Jacobsen radical* of B, is equal to the set $\{0\}$.

§4. Involutional Algebras and C*-Algebras

We now will have a look at the theory of C*-algebras and especially at the Banach algebra $\mathscr{L}(H)$ of bounded linear operators in a complex Hilbert space H. This algebra contains an important subclass of linear operators, which are called *normal* and are defined by satisfying a certain commutativity relation.

We first define the concept of an *involution* for an arbitrary complex Banach algebra $\mathcal{A}$ as follows: a bijective algebra-homomorphism $*: \mathcal{A} \to \mathcal{A},\ x \mapsto x^*$ is called an *involution*, if for all $\lambda, \mu \in \mathbb{C}$ and $x, y \in \mathcal{A}$

$$\begin{cases} (\lambda x + \mu y)^* = \overline{\lambda} \cdot x^* + \overline{\mu} \cdot y^* \,, \\ (x^*)^* = x \,, \\ (xy)^* = y^* x^* . \end{cases}$$

An involution acting on a normed algebra $\mathcal{A}$ let the norms of the algebra elements unchanged, that is, $\|x\| = \|x^*\|$ holds for all $x \in \mathcal{A}$, which implies that involutions are always isometric. Furthermore, an algebra homomorphism $\Phi: \mathcal{A}_1 \to \mathcal{A}_2$ between involutional algebras $\mathcal{A}_1, \mathcal{A}_2$ is called a **-homomorphism*, if it is compliant with the involution operations in both algebras, that is, $\Phi(x^*) = \Phi(x)^*$ holds for all $x \in \mathcal{A}_1$.

If a normed algebra $\mathcal{A}$ is furnished with an involution, one may define some additional properties for the elements of $\mathcal{A}$. In particular, an element $x \in \mathcal{A}$ is called

1. *normal*, if $xx^* = x^*x$.
2. *self-adjoint*, if $x^* = x$.

3. *skew-adjoint*, if $x^* = -x$.
4. *projection*, if $xx = x = x^*$.
5. *unitary*, if $\mathcal{A}$ is unital, x is invertible and $x^* = x^{-1}$.

It can easily be verified that all unitary and self-adjoint elements in $\mathcal{A}$ are normal. Furthermore, each element $x \in \mathcal{A}$ may be written as $x = x_1 + ix_2$, where x_1, x_2 are self-adjoint elements in $\mathcal{A}$ defined by the formulas

$$x_1 := \frac{1}{2}(x + x^*), \qquad x_2 := \frac{1}{2i}(x - x^*).$$

The following concept is crucial in the theory of involutional Banach algebras: an involution $* : \mathcal{A} \to \mathcal{A}$ of a normed algebra $\mathcal{A}$ is said to have the *C*-property*, if

$$\|x^* x\| = \|x\|^2 \quad \text{for all } x \in \mathcal{A}.$$

This property leads to the notion of *C*-algebra*, which is defined to be an involutional Banach algebra satisfying the C*-property.

Examples of C*-algebras are the function algebras $C_b(X)$ and $C_0(X)$, where X is an ordinary topological space, and the algebra $L^\infty(X)$ for a measure space (X, Σ, μ). In all three cases the involution is given by conjugation $f \mapsto \overline{f}$ of functions. But for the convolution algebra $L^1(\mathbb{R}, +, *)$ the involution does not satisfy the C*-property, hence this algebra cannot be a C*-algebra.

There are a couple of important properties, which differ a C*-algebra from an arbitrary involutional algebra. The first one is given by the fact that the spectrum of any element $x \in C$ does not increase if it is considered as an element of a proper C*-subalgebra $\mathcal{U}$ of C. In other words, we have

$$\sigma_{\mathcal{U}}(x) = \sigma_C(x) \text{ for all } x \in C.$$

So the spectrum $\sigma(x)$ of $x \in C$ is unique, anyway to which specific subalgebra of C it is computed. Second, for a normal element x in a C*-algebra the spectral radius can easily be computed by the formula $r(x) = \|x\|$. Third, self-adjoint and unitary elements of a C*-algebra can be characterized by properties of their spectra:

1. x is self-adjoint iff $\sigma(x) \subseteq \mathbb{R}$.
2. x is unitary iff $\sigma(x) \subseteq \tau = \{z \in \mathbb{C} : |z| = 1\}$.

§5. The Gelfand-Naimark Theorem

In this paragraph we want to elaborate the Gelfand-Naimark theorem and the continuous functional calculus based on this theorem.

Preliminary, two further concepts are meaningful when dealing with C*-algebras:

1. A subset $\mathcal{U}$ of a C*-algebra is called a C*-subalgebra, if $\mathcal{U}$ is a *closed* subalgebra, which is also closed with respect to the involution, that is, $x^* \in \mathcal{U}$ for all $x \in \mathcal{U}$.
2. Let $\mathcal{S}$ be a non-empty subset of the C*-algebra C. Then the intersection of all C*-subalgebras containing $\mathcal{S}$ is called the C*-algebra *generated* by $\mathcal{S}$ (notation: $[\mathcal{S}]$), and S itself is said to be a *generator* of $[\mathcal{S}]$.

Recall the Gelfand transformation γ for a commutative Banach algebra, which is a Banach algebra homomorphism from an arbitrary commutative Banach algebra B to one of the Banach algebras $C(\Gamma)$ and $C_0(\Gamma)$. In this context, the topological space Γ is the structure space of B, which turned out to be a locally compact Hausdorff space. It was also quoted that the Gelfand transformation is injective for each semisimple Banach algebra, and this statement is especially true in case of a C*-algebra. Note that in general the Gelfand representation of a semisimple commutative Banach algebra is not surjective, so the range of the structure space γ may be only a subalgebra of $C(\Gamma)$.

Now, the question arises under which conditions the Gelfand representation is bijective and therefore is a Banach algebra isomorphism between the Banach algebra B and the function algebras $C_0(\Gamma)$ and $C(\Gamma)$, respectively. As already mentioned, a necessary condition is that of B being semisimple, and a sufficient condition is given by the fact that B is a C*-algebra, which legitimates the meaning of this class of algebras.

We collect these results to the *Gelfand-Naimark theorem*:

THEOREM VII.2. Let $(C, +, \circ)$ be a commutative C*-algebra.

(a) The Gelfand representation γ is surjective.
(b) If C is unital, then C is isometrically isomorphic to the function algebra $C(\Gamma)$.
(c) If C is *not* unital, then C is isometrically isomorphic to the function algebra $C_0(\Gamma)$.

Thus, by this theorem, each element of a commutative C*-algebra can be considered as a continuous function defined on a locally compact Hausdorff space. The theorem emphasizes the meaning of the Banach algebras $C_0(X)$ and $C(X)$ in case of X being a locally compact Hausdorff space.

Conversely, for a fixed normal element $x \in C$, one may send each function $f \in C(\sigma(x))$ to a C*-algebra element $f(x)$ by the inverse mapping of the Gelfand representation. Given a normal element $x \in C$, the family of mappings $f \mapsto f(x)$ is called a *continuous functional calculus* for x.

The following *spectral mapping theorem* establishes a link between the spectrum $\sigma(x)$ of a normal element x of a C*-algebra and the range of the continuous function f associated by the Gelfand representation:

THEOREM VII.3. Let f be a continuous complex-valued function, which is defined on the spectrum Γ of the C*-algebra C. Then for any normal element $x \in C$ the equation $\sigma(f(x)) = f(\sigma(x))$ holds.

Thus, if C' is the C*-algebra generated by a normal element x and the identity, that is, $C' := [\{x, 1\}]$, which is of course commutative, then C' is *-isomorphic to the commutative C*-algebra $C(\sigma(x))$. In this case any isomorphism maps 1 to $\chi_{\sigma(x)}$ and x to the identity function $id : z \mapsto z$.

We shall apply the Gelfand-Neumark theorem to the commutative C*-algebra $x]$ generated by a normal element $x \in C$, its adjoint element x^* and the identity element 1.

Recall that if Γ_x is the structure space of $[x]$, then the Gelfand representation γ_x is a C*-isomorphism between the C*-algebras $[x]$ and $C(\Gamma_x)$, that is, one can assign to each function $f \in C(\Gamma_x)$ a unique element $f(x) \in [x]$ by applying the inverse Gelfand representation $\Phi_x := \gamma_x^{-1}$.

Hence it is possible to define a bijective and continuous mapping $h_x : \Gamma_x \to \sigma(x)$, where $\sigma(x)$ is the spectrum of x. Since Γ_x is compact and the set $\sigma(x)$ is Hausdorff with respect to its metric topology induced by that in $\mathbb{C}$, a well-known theorem from point-set topology claims that Γ_x and $\sigma(x)$ are homeomorphic to each other. As a consequence, there is also a homeomorphism

$$g_x : C(\Gamma_x) \to C(\sigma(x))$$

between the associated function spaces $C(\Gamma_x)$ and $C(\sigma(x))$, and finally there is a C*-isomorphism

$$\begin{cases} \Phi_x : C(\sigma(x)) \to [x] \,, \\ f \mapsto (\gamma_x^{-1} \circ g_x^{-1})(f) \,, \end{cases}$$

called the *continuous functional calculus* of the normal element x.

The following diagram shows the isomorphism in detail:

$$\begin{tikzcd} {[x] \subset C} \arrow[r, "\gamma_x"] & C(\Gamma_{[x]}) \arrow[d, "g_x"] \\ & C(\sigma(x)) \arrow[ul, "\Phi_x"] \end{tikzcd}$$

Figure 5.1: continuous functional calculus Φ_x for a normal element $x \in C$.

The importance of C*-algebras is justified by the fact that for each Hilbert space H the associated Banach algebra $\mathscr{L}(H)$ of bounded linear operators forms a C*-algebra, where the involution mapping is given by

$$* : \mathscr{L}(H) \to \mathscr{L}(H), \quad A \mapsto A^*.$$

If H is a Hilbert space and $A \in \mathscr{L}(H)$ normal, then for each continuous function $f : \sigma(A) \to \mathbb{C}$ the operator $f(A) \in \mathscr{L}(H)$ is well defined. This is especially true for the operators $|S|$ and $\sqrt{|S|}$.

§6. Locally Compact Abelian Groups

Before discussing abstract Fourier transformations, it is useful to review some concepts coming from general topology, especially that of a topological group, which plays a fundamental role in the theory of Fourier transforms.

There is a subclass within that of general topological spaces, which is of great importance in higher analysis and whose members are called *topological groups.* They are mathematical objects owning the algebraic structure of a group and a topological structure together. Both have to be compatible to each other in the following sense: let $(G, \circ)$ be an ordinary group and a topological space $(G, \mathcal{T})$ at once. Then G is called a *topological group*, if the group operation $(x, y) \mapsto x \circ y^{-1}$ is continuous.

Instead to require continuity of the mapping $(x, y) \mapsto x \circ y^{-1}$, one can equivalently require that the mappings $(x, y) \mapsto x \circ y$ and $x \mapsto x^{-1}$ have to be continuous.

Well-known examples of topological groups are:
1. each group G with its discrete topology 2^G,
2. the group $\mathbb{R}^\times := \mathbb{R} - \{0\}$,
3. the group $\mathbb{C}^\times := \mathbb{C} - \{(0,0)\}$,
4. the group $\mathbb{R}^+ := \{x \in \mathbb{R} : x > 0\}$,
5. the additive groups $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$,
6. the *circle group* $\mathbb{T} := \{z \in \mathbb{C} : |z| = 1\}$,
7. each subgroup U of a topological group G endowed with the subspace topology and its closure $\overline{U}$ are topological groups.

Like for purely algebraic groups, a *homomorphism* between topological groups is defined to be a *continuous* group homomorphism, and such a homomorphism ι is called an *isomorphism*, if it is a group isomorphism and ι^{-1} is continuous, too (we use the notation $G \cong H$ in this case).

A well-known example of a group isomorphism is the *exponential function* $t \mapsto e^t$, which maps the topological group $(\mathbb{R}, +)$ to the group $(\mathbb{R}^+, \cdot)$. Rather similar, the function $t \in [0, 1) \mapsto e^{2\pi it}$ forms a group isomorphism between the topological groups $\mathbb{R}/\mathbb{Z}$ and $\mathbb{T}$.

Because in a topological group G the left translation $x \mapsto g \circ x$ and the right translation $x \mapsto x \circ g$ are continuous mappings, they are both examples for *automorphisms*, that is, isomorphisms from the group G to itself. Hence the topological properties of a topological group are completely determined by the topology in a neighbourhood of the identity element $1 \in G$, which gives rise to call a topological group to be *homogeneous* in this case.

Moreover, the kind, how the identity 1 of a topological group G lies in a given subgroup U, determines the topology of that subgroup. In particular,
1. $U \subset G$ is open iff 1 is an inner point of U.
2. U is discrete iff 1 is an isolated point of U.

Remember that the circle group $\mathbb{T}$ is the group consisting of all complex numbers of modulus 1 with group operation

$$e^{i\alpha} \circ e^{i\beta} := e^{i\alpha} \cdot e^{i\beta} := e^{i(\alpha+\beta)}, \quad \alpha, \beta \in \mathbb{R}.$$

Endowed with the subset topology inherited from the standard topology on the field $\mathbb{C}$, the unit circle $\mathbb{T}$ forms a topological group. Consider the set $C(G, \mathbb{T})$ consisting of all continuous functions $f : G \to \mathbb{T}$. By furnishing this set with the compact-open topology, it becomes into a Hausdorff space.

A *character* χ is a (topological) group homomorphism from a topological group G to the circle group $\mathbb{T}$ and hence a continuous mapping. We denote G^* the set of characters $\chi : G \to \mathbb{T}$, which forms a subset of the Hausdorff space $C(G, \mathbb{T})$. We can make G^* into a topological space by taking the subset topology with respect to the compact-open topology on $C(G, \mathbb{T})$.

Performing multiplication of characters $\chi_1, \chi_2 \in G^*$ by

$$\chi_1 \circ \chi_2 : g \mapsto \chi_1(g) \cdot \chi_2(g),$$

we obtain a further character $\chi_1 \circ \chi_2 \in G^*$. It can be shown that this operation has all the properties of a group, and since it is commutative, G^* has the structure of an Abelian group, called the *character group* or the *dual group* of G. Moreover, the group operation in G^* is continuous, hence the dual group of any topological group forms a topological group as well.

In the following table we list three classical topological Abelian groups together with their dual groups and characters:

G	G^*	characters
$\mathbb{R}$	$\mathbb{R}^* \cong \mathbb{R}$	$\chi_x(t) := e^{itx}$, $x \in \mathbb{R}$
$\mathbb{Z}$	$\mathbb{Z}^* \cong \mathbb{T}$	$\chi_z(n) := e^{inz}$, $z \in \mathbb{T}$
$\mathbb{T}$	$\mathbb{T}^* \cong \mathbb{Z}$	$\chi_n(z) := z^n$, $n \in \mathbb{Z}$

Table 1: classical topological groups and dual groups.

Concerning cartesian products of topological groups, for each topological group G and H we have

$$(G \times H)^* \cong G^* \times H^*.$$

As a consequence, the group isomorphisms $\mathbb{R}^{d^*} \cong \mathbb{R}^d$ hold for any integer d.

A topological space is called *locally compact*, if every subset owns a compact neighbourhood. The abbreviation *LCA group* usually stands for a *locally compact Abelian group*. The meaning of locally compact groups is based on the fact that these groups have non-trivial characters, that is, we have $G^* \neq \{\chi_1\}$ for such a group G, where χ_1 is the character with $\chi_1(x) = 1$ for all $x \in G$.

Suppose that the Abelian group G is *locally compact*. Then it can be shown (for example by the *Arzelá-Ascoli theorem*) that the dual group G^* is also locally compact with respect to its compact-open topology. Let G^{**} denote the *bidual group* of a topological group G, which is defined to be the dual group of its character group.

One of the main results in the theory of LCA groups is the famous *Pontryagin duality theorem*:

THEOREM VII.4. If G is a locally compact Abelian group, then G and G^{**} are isomorphic to each other.

Notice that each group element $g \in G$ gives rise to a character χ_x on the dual group G^* by $\chi_x(\xi) := \xi(x)$ for all $\xi \in G^*$. Thus the mapping $x \mapsto \chi_x \in G^{**}$ is an injective group homomorphism, and Pontryagin duality means that the canonical embedding $\iota : G \to G^{**}$, $x \mapsto \chi_x$ is also surjective.

It is easy to see that the groups listed in table 1 are examples of Pontryagin duality. This kind of duality also implies remarkable relationships between the topological properties of a LCA group G and its dual group G^*:

1. G is compact if and only if G^* is discrete.
2. G is discrete if and only if G^* is compact.

Finally, let us mention that it is possible to endow the convolution algebra $L^1(G, +, *)$ of any LCA group G with an involutional mapping $* : L^1(G) \to L^1(G)$, defined for $f \in L^1(G)$ by

$$f^*(x) := \overline{f(-x)}, \qquad (x \in G).$$

§7. Abstract Fourier Transformation

Since $L^1(G, +, *)$ forms a commutative Banach algebra for every LCA group G, we may study its Gelfand transformation in detail, which in this case is denoted *Fourier transformation*. If Γ_G is the structure space of the algebra $L^1(G, +, *)$, then Fourier transformation maps each function $f \in L^1(G)$ to a function $\hat{f} \in C_0(\Gamma_G)$. It constitutes a Banach algebra homomorphism between $L^1(G)$ and $C_0(\Gamma_G)$, where convolution in $L^1(G)$ corresponds to multiplication in $C_0(\Gamma_G)$.

Moreover, each character $\chi \in G^*$ gives rise to a continuous multiplicative functional $\gamma \in \Gamma_G$, that is defined according to

$$\gamma(\chi) := \int_G f \, \overline{\chi} \, d\mu,$$

and reversely each continuous multiplicative functional $\gamma \in \Gamma_G$ can be represented in this way, Thus, we identify the structure space of the convolution algebra $L^1(G, +, *)$ with the dual group G^*, and Fourier transformation appears as the Gelfand transformation up to this identification:

1. For $f \in L^1(G)$ the mapping $\hat{f} : \widehat{G} \mapsto \mathbb{C}$ according to

$$\hat{f}(\chi) := \int_G f \, \overline{\chi} \, d\mu, \quad \chi \in \widehat{G}$$

   is called the *Fourier transform* of f.
2. The mapping $\mathscr{F} : f \in L^1(G) \to \hat{f} \in C_0(G^*)$ is called *Fourier transformation.*

Let us repeat some properties of Fourier transformation that are already known from the general Gelfand transformation:

1. For every $f \in L^1(G)$ the mapping $\hat{f}$ is continuous on G^*.
2. the function $\hat{f}$ is vanishing at infinity (*Riemann-Lebesgue theorem*), that is, it maps absolutely integrable functions to elements of the function space $C_0(G^*)$.
3. $\mathscr{F}$ is injective, that is, $\mathscr{F}(f) \neq \mathscr{F}(g)$ for $f \neq g$.
4. $\mathscr{F}$ is norm-preserving, that is, $\|\mathscr{F}(f)\|_\infty \leq \|f\|_1$.
5. $\mathscr{F}(\alpha f + \beta g) = \alpha \mathscr{F}(f) + \beta \mathscr{F}(g)$, $\alpha, \beta \in \mathbb{R}$.
6. $\mathscr{F}(f * g) = \sqrt{2\pi} \, \mathscr{F}(f) \cdot \mathscr{F}(g)$.

In addition, the image of $L^1(G)$ under $\mathscr{F}$ forms a separating subalgebra of $C_(\Gamma_G)$, which lies even dense in $C_0(\Gamma_G)$.

By *inverse Fourier transformation* is meant the procedure, how certain functions in the algebra $L^1(G)$ can be retrieved from the function values of the corresponding Fourier transforms.

Since the adjoint group G^* of a LCA group G with Haar measure μ is LCA again, it owns a Haar measure as well, hence the convolution algebra $L^1(G^*)$ is well-defined. Let f be an element of $L^1(G)$ fulfilling the additional condition $\hat{f} \in L^1(G^*)$. Then there exists the Fourier transform of $\hat{f}$,

$$\hat{\hat{f}}(x) = \int_{G^*} \hat{f}(\gamma)\overline{x(\gamma)}\, d\nu,$$

which forms a complex-valued mapping defined on the biadjoint group G^{**}. Due to Pontryagin duality, G^{**} is isomorphic to G, and we may define the *inverse Fourier transform* F according to

$$F(x) = \int_{G^*} \hat{f}(\gamma)\,\overline{\gamma(x)}\, d\mu, \quad (x \in G).$$

It can be shown that the functions f and F are μ-nearly everywhere identical on G.

Let X be any topological space and μ be a measure define on the Borel σ-algebra $\mathcal{B}_X$. A Borel set $B \subseteq X$ is called *inner regular* if

$$\mu(B) = \sup\{\mu(K) : K \subset B,\, K \text{ kompact}\}.$$

Then a measure μ is denoted as *inner regular*, if all Borel sets are so. Hence inner regular measures defined on X are already uniquely defined by their values for compact subsets of X. Each inner regular measure on a topological space is said to be *Radon* if it adopts finite values for compact sets.

Rather similar, a measure μ is called *outer regular*, if for each Borel set $B \subseteq X$ the equality

$$\mu(B) = \inf\{\mu(\mathcal{O}) : B \subset \mathcal{O},\, \mathcal{O} \text{ open}\}.$$

holds. Finally, a measure is called *regular*, if it is inner and outer regular at the same time.

Let $\mathscr{M}_r(G)$ be the Banach space of finite regular Borel measures, where G is a LCA group. Let μ_1, μ_2 be measures in $\mathscr{M}_r(G)$ and let $\mu_1 \times \mu_2$ be the product measure defined on the group $G \times G$. If $B \in \mathcal{B}_G$ is measurable, then the set B' is defined by

$$B' := \{(x,y) \in G \times G : x + y \in B\}.$$

This set is measurable with respect to the σ-Algebra in $G \times G$. Moreover, the *convolution* operation $\mu_1 * \mu_2$ of two measures μ_1 und μ_2 is defined to be the measure

$$(\mu_1 * \mu_2)(B) := (\mu_1 \times \mu_2)(B') \quad (B \in \mathcal{B}_G),$$

which owns the following properties:
1. $\mu_1 * \mu_2 \in \mathscr{M}_r(G)$,
2. $(\mu_1, \mu_2) \mapsto \mu_1 * \mu_2$ is commutative and associative,
3. the inequality $\|\mu_1 * \mu_2\| \leq \|\mu_1\| \cdot \|\mu_2\|$ holds.

Hence the Banach algebra $\mathscr{M}_r(G, +, *)$ is unital and commutative. Its unit element is the *Dirac measure* δ_0, which distinguishes from other measures in this algebra by the property that it is concentrated to the element $0 \in G$.

In case of $G = (\mathbb{R}^d, +)$, we may regard the algebra $\mathscr{M}_r(\mathbb{R}^d, +, *)$ as the Banach algebra, that is norm-isomorphic to the completion of the non-unital convolution algebra $L^1(\mathbb{R}^d, +, *)$.

Fourier-Stieltjes transformation is another phrase for Gelfand transformation on the unital and commutative Banach algebra $\mathscr{M}_r(G, +, *)$ of finite and regular complex Borel measures on a LCA group G. In fact, if $\mu \in \mathscr{M}_r(G, +, *)$, then the mapping $\hat{\mu}$ defined on the dual group G^* by

$$\hat{\mu}(\gamma) := \int_G -\gamma(x)\, d\,\mu(x), \qquad \gamma \in G^*$$

is called *Fourier-Stieltjes transform* of μ.

Let $B(G^*)$ be the set of functions $\hat{\mu}$ with $\mu \in \mathscr{M}_r(G, +, *)$. Then the following properties are valid:
1. Each function $\hat{\mu} \in B(G^*)$ is bounded and uniformly continuous.
2. If $\sigma = \mu * \nu$, then $\hat{\sigma} = \hat{\mu} \cdot \hat{\nu}$ holds, that is, the mapping $\mu \mapsto \hat{\mu}(\gamma)$ is a Banach algebra endomorphism on $\mathscr{M}_r(G, +, *)$ for every $\gamma \in X^*$.
3. The set $B(G^*)$ is invariant under translation, multiplication and complex conjugation.

§8. Examples of Fourier Transformation

It is nearby to consider Fourier transformation on $L^1(G, +, *)$ for the classical groups $\mathbb{R}^d$, $\mathbb{Z}$ and $\mathbb{T}$:

1. If $G = \mathbb{R}^d$, then the character group of $(\mathbb{R}^d, +)$ is the group $(\mathbb{R}^d, +)$ itself. Hence a function $f \in L^1(\mathbb{R}^d)$ owns the Fourier transform $\hat{f} : \mathbb{R}^d \to \mathbb{C}$, whose function values are defined as follows:

$$\hat{f}(x) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t)\, e^{-itx}\, dt, \quad (x \in \mathbb{R}^d).$$

Recall that for elements $f, g \in L^1(\mathbb{R}^d)$ a multiplication $f * g$ is defined by

$$(f * g)(x) := \int_{\mathbb{R}^d} f(t) \cdot g(x - y)\, dy$$

and denoted the *convolution* in $L^1(\mathbb{R}^d)$. This mapping is commutative, associative and fulfills for all $f, g \in L^1(\mathbb{R}^d)$ the estimate

$$\|f * g\|_1 \leq \|f\| \cdot \|g\| .$$

Since $L^1(\mathbb{R}^d)$ is complete, the algebra $(L^1(\mathbb{R}^d), +, *)$ constitutes a commutative but non-unital Banach algebra.

L^1-Fourier transformation in the Euclidean space $\mathbb{R}^d$ is nothing but the Gelfand mapping

$$\mathscr{F} : f \in L^1(\mathbb{R}^d) \to \hat{f} \in C_0(\mathbb{R}^d),$$

which sends each function $f \in L^1(\mathbb{R}^d)$ to the function $\hat{f} \in C_0(\mathbb{R}^d)$, called the *Fourier transform* of f and defined by

$$\hat{f}(\xi) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} f(x)\, e^{-2\pi i \langle \xi, x \rangle}\, dx \qquad (\xi \in \mathbb{R}^d), \tag{8.1}$$

where $\langle \xi, x \rangle$ is the inner product of $\xi, x \in \mathbb{R}^d$.

Let us again collect the remarkable properties of the mapping $\mathscr{F}$ in the special case $G = G^* = \mathbb{R}$:

1. For each $f \in L^1(\mathbb{R}^d)$ the function $\hat{f}$ is continuous on $\mathbb{R}^d$, so $\hat{f} \in C(\mathbb{R}^d)$.
2. $\hat{f}$ is vanishing at infinity, so $\hat{f} \in C_0(\mathbb{R}^d)$.
3. $\mathscr{F}$ is injective, that is, $\mathscr{F}(f) \neq \mathscr{F}(g)$ for $f \neq g$.
4. $\|\mathscr{F}(f)\|_\infty \leq \|f\|_1$ for $f \in L^1(\mathbb{R}^d)$.

5. $\mathscr{F}(\alpha f + \beta g) = \alpha\, \mathscr{F}(f) + \beta\, \mathscr{F}(g)$ for $\alpha, \beta \in \mathbb{C}$, thus $\mathscr{F}$ is linear.
6. $\mathscr{F}(f * g) = \mathscr{F}(f) \cdot \mathscr{F}(g)$, so $\mathscr{F}$ is an algebra homomorphism between the Banach algebras $L^1(\mathbb{R}^d, +, *)$ and $C_0(\mathbb{R}^d, +, \cdot)$.
7. If $f \in L^1(\mathbb{R}^d)$ and $g : x \in \mathbb{R}^d \mapsto |x| f(x) \in L^1(\mathbb{R}^d)$, then $\hat{f} \in C^1(\mathbb{R}^d)$ and
$$\partial_j f(x) \;=\; -\widehat{i\, \xi_j f(\xi)} \quad (j = 1, \ldots, d).$$

Especially for the LCA group $(\mathbb{R}^d, +)$, where the dual group of $\mathbb{R}^d$ may be identified with $\mathbb{R}^d$ itself, inverse Fourier transform of $\hat{f} \in L^1(\mathbb{R}^d)$ is obtained according to

$$F(x) \;=\; \int_R d\hat{f}(t)\, e^{itx}\, dt, \quad (x \in \mathbb{R}^d).$$

2. Let $G = \mathbb{Z}$. The normed Haar measure on this group is nothing but the counting measure, and the character group $(\mathbb{Z}, +)$ is isomorphic to the multiplicative group $(\mathbb{T}, \cdot)$. Hence the Fourier transform of a sequence $\{f_n\}_{n \in \mathbb{N}}$ is the mapping $\hat{f} : \mathbb{T} \to \mathbb{C}$, which is defined pointwise by the *Fourier series*

$$\hat{f}(\omega) \;:=\; \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} f_n\, e^{-in\omega}, \quad (\omega \in \mathbb{T}).$$

3. The character group of the group $(\mathbb{T}, \cdot)$ is isomorphic to the group $(\mathbb{Z}, +)$, hence the Fourier transform of a function $f : \mathbb{T} \to \mathbb{R}$ is the sequence $\hat{f} : \mathbb{Z} \to \mathbb{C}$ with members

$$\hat{f}_n \;:=\; \frac{1}{\sqrt{2\pi}} \int_{\mathbb{T}} f(z)\, e^{-inz}\, dz, \quad (n \in \mathbb{Z}).$$

§9. Fourier-Plancherel Transformation

We want to extend the Fourier transformation to the Hilbert space $L^2(\mathbb{R}^d)$. Recall that this space consists of the measurable functions $u : \mathbb{R}^d \to \mathbb{C}$ fulfilling

$$\int_{\mathbb{R}^d} |u|^2\, dx \;<\; \infty,$$

and the inner product in $L^2(\mathbb{R}^d)$ is given by

$$\langle u, v \rangle \;=\; \int_{\mathbb{R}^d} u\, \overline{v}\, dx.$$

If the functions $u, v : \mathbb{R}^d \to \mathbb{C}$ are Lebesgue-integrable then by definition they belong to the space $L^1(\mathbb{R}^d)$, and their *Fourier transforms* are well-defined by formula (8.1).

However, if is not only a member of $L^1(\mathbb{R}^d)$, but also $u \in L^2(\mathbb{R}^d)$, we define Fourier transformation a little bit different:

$$\hat{u}(\xi) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-2\pi i \langle \xi, x \rangle} \, u(x) \, dx. \tag{9.1}$$

Then for functions $u, v \in L^2(\mathbb{R}^d)$ the inner product between u and v and the inner product of their Fourier transforms $\hat{u}$ and $\hat{v}$ are related by PLANCHEREL*'s formula*

$$\langle u, v \rangle = \langle \hat{u}, \hat{v} \rangle.$$

Hence *Fourier transformation* $\mathscr{F} : u \mapsto \hat{u}$ constitutes a partial isometry from the space $L^{1,2}(\mathbb{R}^d) := L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$ to the space $L^2(\mathbb{R}^d)$.

It is a crucial fact now that $L^{1,2}(\mathbb{R}^d)$ proves to be dense in $L^2(\mathbb{R}^d)$. Since the mapping $\mathscr{F}$ is linear and continuous on $L^{1,2}(\mathbb{R}^d)$, it is even *uniformly continuous* on that space and can be extended to an *unitary*, that is, a bijective and isometric linear operator $\mathscr{F}_P : L^2(\mathbb{R}^d) \to L^2(\mathbb{R}^d)$, called the FOURIER-PLANCHEREL *operator* or the L^2*-Fourier transformation* in $L^2(\mathbb{R}^d)$.

But in general a function $u \in L^2(\mathbb{R}^d)$ is not an element of $L^1(\mathbb{R}^d)$. Thus it is not possible to define the L^2-Fourier transform $\hat{u}$ of every $u \in L^2(\mathbb{R}^d)$ by means of formula (9.1). However, for $u \in L^2(\mathbb{R}^d)$ and $R > 0$ the function

$$\hat{u}_R(\xi) := \frac{1}{(2\pi)^{d/2}} \int_{|x| \le R} e^{-2\pi i \langle x, \xi \rangle} \, u(x) \, dx$$

is well-defined, and one may obtain the L^2-Fourier transform $\hat{u}$ as the limit of the functions $\hat{u}_R$ for $R \to \infty$ with respect to the 2-norm, that is, $\lim_{R \to \infty} \|\hat{u} - \hat{u}_R\|_2 = 0$.

The Fourier-Plancherel operator can be used to introduce the class of *Bessel potential spaces.* First recall that the mapping $\mathscr{F}_P$ is well-defined in Hilbert space $L^2(\mathbb{R}^d)$, therefore it has also a meaning in the linear subspaces $H^k(\mathbb{R}^d)$ for $k \in \mathbb{N}$. We define for each $k \in \mathbb{N}$ the real-valued mapping

$$p_k : x \in \mathbb{R}^d \mapsto (1 + |x|^{2k})^{1/2} \tag{9.2}$$

and the space

$$\mathcal{H}^k(\mathbb{R}^d) := \{f \in L^2(\mathbb{R}^d) : p_k \hat{f} \in L^2(\mathbb{R}^d)\},$$

which is Hilbertian with inner product $\langle f, g\rangle_{p_k} := \langle p_k\hat{f}, p_k\hat{g}\rangle$ for $f, g \in \mathcal{H}^k(\mathbb{R}^d)$ and derived norm

$$\|f\|_{p_k}^2 := \langle f, f\rangle_{p_k} = \int_{\mathbb{R}^d} (1 + |x|^{2k})\, |\hat{f}|^2\, dx.$$

It turns out that the so-called *Bessel potential space* $\mathcal{H}^k(\mathbb{R}^d)$ is the range of the Fourier-Plancherel operator $\mathscr{F}_P$ when restricted to $H^k(\mathbb{R}^d) \subset L^2(\mathbb{R}^d)$. Hence $\mathscr{F}_P$ is a bijective and norm-preserving mapping from $H^k(\mathbb{R}^d)$ to $\mathcal{H}^k(\mathbb{R}^d)$, i. e.

$$|f|_k = \|\hat{f}\|_{p_k} \qquad \text{for all } f \in H^k(\mathbb{R}^d).$$

In order to extend the definition of the spaces $H^k(\mathbb{R}^d)$ to indices $s \in \mathbb{R}^+$, let p_s be defined as in (9.2) but with $s > 0$ instead of $k \in \mathbb{N}$. We define

$$\mathcal{H}^s(\mathbb{R}^d) := \{\, f \in L^2(\mathbb{R}^d) : p_s\hat{f} \in L^2(\mathbb{R}^d) \,\},$$

then $\mathcal{H}^s(\mathbb{R}^d)$ given this way coincides for $s = k \in \mathbb{N}$ with the classical Sobolev space $H^k(\mathbb{R}^d)$, which motivates to define $H^s(\mathbb{R}^d) := \mathcal{H}^s(\mathbb{R}^d)$ for $s \notin \mathbb{N}$. For $s < 0$, however, we introduce $H^s(\mathbb{R}^d) := H^{-s}(\mathbb{R}^d)^*$.

§10. The Fourier Transform in Distributional Spaces

We finally elaborate that Fourier transformation can be extended to the space $\mathscr{S}'(\mathbb{R}^d)$ of tempered distributions.

First notice that the Fourier transform $\hat{\phi}$ of a Schwartz function $\phi \in \mathscr{S}(\mathbb{R}^d)$ is well defined, since $\mathscr{S}(\mathbb{R}^d)$ is a linear subspace of the convolution algebra $L^1(\mathbb{R}^d, +, *)$. Moreover, Fourier transformation restricted to the Schwartz space $\mathscr{S}(\mathbb{R}^d)$ forms a linear mapping $\mathscr{F} : \mathscr{S}(\mathbb{R}^d) \to \mathscr{S}(\mathbb{R}^d)$, which sends each function $\phi \in \mathscr{S}(\mathbb{R}^d)$ to a further function $\hat{\phi} \in \mathscr{S}(\mathbb{R}^d)$ according to the well-known formula

$$\hat{\phi}(x) = \mathscr{F}(\phi)(x) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{-i<\xi,x>} \phi(\xi)\, d\xi \quad (x \in \mathbb{R}^d). \tag{10.1}$$

The extraordinary meaning of Schwartz functions is founded by the fact that the Fourier transformation has excellent properties on the space $\mathscr{S}(\mathbb{R}^d)$. More precisely, besides the well-known features of Fourier transformation in $L^1(\mathbb{R}^d, +, *)$, the mapping $\mathscr{F}$ is *bijective* on $\mathscr{S}(\mathbb{R}^d)$. Hence the *inverse Fourier transform* exists for all $\phi \in \mathscr{S}(\mathbb{R}^d)$ and is described by the *Fourier inversion formula*

$$\check{\phi}(\xi) = \mathscr{F}^{-1}(\phi)(\xi) := \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\, \phi(x)\, dx \qquad (\xi \in \mathbb{R}^d).$$

We next extend Fourier transformation in $\mathscr{S}(\mathbb{R}^d)$ to the space $\mathscr{S}'(\mathbb{R}^d)$ of tempered distributions by transposition, that is

$$(\mathscr{F}T)\,\phi := T(\mathscr{F}\phi), \qquad \phi \in \mathscr{S}(\mathbb{R}^d),\ T \in \mathscr{S}'(\mathbb{R}^d).$$

This mapping is bijective in the space $\mathscr{S}'(\mathbb{R}^d)$ and even continuous with respect to the weak*-topology on $\mathscr{S}'(\mathbb{R}^d)$. The inverse Fourier transform of a tempered distribution T is represented by the formula

$$\mathscr{F}^{-1}(T)(\phi) := T(\mathscr{F}^{-1}\phi) \qquad (\phi \in \mathscr{S}(\mathbb{R}^d)).$$

For example, the Fourier transform $\mathscr{F}(\delta_0)$ of the Dirac distribution $\delta_0 \in \mathcal{S}'(\mathbb{R})$ is the continuous linear functional

$$\mathscr{F}(\delta_0)(\phi) := \delta_0(\mathscr{F}(\phi)) = \hat{\phi}(0) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \phi(x)\, dx, \quad \phi \in \mathscr{S}(\mathbb{R}),$$

so $\mathscr{F}(\delta_0)$ is nothing but the distribution induced by the constant function $x \mapsto \frac{1}{\sqrt{2\pi}}$.

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VIII.
Linear Operators in Hilbert Space

The presence of the inner product in a Hilbert space can be used to analyze linear operators with respect to this additional feature. For example, a linear operator A with domain of definition $\mathfrak{D}(A)$ acting in a Hilbert space H is called *normal*, if $AA^* = A^*A$ holds, where A^* is the Hilbert adjoint, an operator acting in H as well, which is well-defined whenever A is densely defined.

Important classes of normal operators are the unitary, symmetric and self-adjoint ones. Especially the symmetric and self-adjoint linear operators has been extensively studied and give rise to a very complete theory.

§1. Numerical Range and External Field of Regularity

Besides the spectrum $\sigma(A)$, the *numerical range* $\Theta(A)$ plays an important role in the general theory of a Hilbert space operator A and is defined by $\Theta(A) := \{\langle A, x\rangle : x \in \mathfrak{D}(A), \|x\| = 1\}$. It is a well-known fact, given by the *Toeplitz-Hausdorff theorem*, that $\Theta(A)$ is a convex set within the complex plane. Moreover, let

$$\Delta(A) := \mathbb{C} \setminus \overline{\Theta(A)}$$

be the *external field of regularity* of A. If $\Delta(A) \neq \emptyset$, then it consists of one or two connected components. To be precise, if $\Delta(A)$ has two connected components, then the set $\Theta(A)$ is either a strip or a ray, otherwise it forms a half-strip or a half-ray.

For $\zeta \in \Delta(A)$ we of course have $\langle Ax, x\rangle \neq \zeta$ for $x \in \mathfrak{D}(A)$ and $\|x\| = 1$, and the following chain of inequalities

$$0 < \operatorname{dist}(\zeta, A) \leq |\langle Ax, x\rangle - \zeta| = |\langle (A-\zeta)x, x\rangle| \leq \|(A-\zeta)x\|$$

is valid, where $\operatorname{dist}(\zeta, A)$ denotes the distance between the singleton $\{\zeta\} \subset \Delta(A)$ and the set $\overline{\Theta(A)}$, which in fact is positive, since these sets are closed and have empty intersection. Hence the operator $A - \zeta$ is bounded below, which implies that $A - \zeta$ is injective and has closed range.

In other words, the set $\Delta(A)$ is a subset of the *field of regularity* $\Pi(A) := \{\zeta \in \mathbb{C} : A - \zeta \text{ is bounded below}\}$.

§2. Unitary Operators and Isometries

Let us first consider the class of *unitary* operators. A bijective linear operator $U : H \to H$ acting in a Hilbert space H is called *unitary*, if

$$\langle Ux, Uy\rangle \;=\; \langle x, y\rangle \quad \text{for all } x, y \in H.$$

Another definiton for this notion is given by the condition $U^* = U^{-1}$.

The meaning of unitary operators is motivated by the fact that one can define *unitary equivalence* between operators in a Hilbert space. In particular, two operators A_1 and A_2 are called *unitary equivalent*, if there is an unitary operator U such that

$$A_1 \;=\; U \circ A_2 \circ U^*.$$

Each eigenvalue of a unitary operator U is of modulus 1 and the operator norm $\|U\|$ is equal to 1. The set of unitary operators $\mathscr{U}(H)$ acting in H forms a group, called the *unitary group* of H.

More generally, an *isometry* is a linear mapping $\mathfrak{J} : H_1 \to H_2$ between arbitrary Hilbert spaces H_1 and H_2, which preserves the inner products, that is

$$\langle \mathfrak{J}x, \mathfrak{J}y\rangle \;=\; \langle x, y\rangle \quad (x, y \in H_1).$$

This condition implies that an isometry is always injective. If $H_1 = H_2 = H$, the mapping $\mathfrak{J}$ is called a *partial isometry* (which must not necessary be a surjective mapping). Finally, a partial isometry $\mathfrak{J}$ is an unitary operator if and only if $\mathfrak{J}$ is surjective, so we regard an unitary operator as a special case of an isometry.

§3. Symmetric Linear Operators

Let H be a real or complex Hilbert space. A linear operator acting in H and with domain of definition $\mathfrak{D}(A) \subset H$ is called *symmetric*, if

$$\langle Ax, y\rangle \;=\; \langle x, Ay\rangle \quad \text{for all } x, y \in \mathfrak{D}(A). \tag{3.1}$$

We first consider the symmetry property of operators, which are continuous and everywhere defined. Let $\mathscr{L}(H)$ be the Banach space of *continuous* linear operators $A : H \to H$ furnished with the operator norm. Then, if A is symmetric, we have

$$- \|A\| \leq \langle Ax, x\rangle \leq \|A\|,$$

for $x \in H$ with $\|x\| = 1$, or equivalently,

$$\Theta(A) \subset [-\|A\|, \|A\|].$$

Hence one may introduce in the class of bounded symmetric operators acting in H the partial order

$$A_1 \geq A_2 \text{ if and only if } \langle (A_1 - A_2)\, x, x\rangle \geq 0 \quad (x \in H),$$

which implies that each symmetric linear operator $A \in \mathscr{L}(H)$ can be enclosed by the multiplication operators $A_- := -\|A\|\, 1_H$ and $A_+ := \|A\|\, 1_H$.

On the other hand, the *Hellinger-Toeplitz theorem* quotes that if A is symmetric and $\mathfrak{D}(A)$ is the whole of H, then A must be bounded. In fact, let A be symmetric with $\mathfrak{D}(A) = H$, then for each convergent sequence (x_n) tending to 0 and with (Ax_n) tending to $x \in H$ we have

$$\langle x, x\rangle = \lim_{n\to\infty} \langle Ax_n, x\rangle = \lim_{n\to\infty} \langle x_n, Ax\rangle = \langle 0, Ax\rangle = 0,$$

which implies $x = 0$, so A is closed and even continuous by the closed graph theorem. Hence we may always assume that the domain of definition $\mathfrak{D}(A)$ of a non-continuous symmetric linear operator A is a *proper* subspace of H.

The symmetry of a linear operator with domain of definition dense in H can be characterized by other properties than (3.1):

THEOREM VIII.1. Let A be a densely defined linear operator with domain of definition $\mathfrak{D}(A)$ in Hilbert space H. Then the following statements are equivalent:

1. A is symmetric;
2. $\langle Ax, x\rangle \in \mathbb{R}$ for all $x \in \mathfrak{D}(A)$;
3. $A \prec A^*$.

The last property is of course also valid for symmetric operators not necessarily being densely defined in the entire space.

Since the adjoint A^* of a symmetric operator A forms a closed extension of A, the operator A itself is closable.

Moreover, the closure $\overline{A}$ proves to be symmetric. In fact, to show that $\overline{A}$ is symmetric, let (x_n) and (y_n) be sequences in $\mathfrak{D}(A)$ with $x_n \to x \in \mathfrak{D}(\overline{A})$, $Ax_n \to \overline{A}x$ and $y_n \to y \in \mathfrak{D}(\overline{A})$, $Ay_n \to \overline{A}y$. Since $\langle Ax_n, y_n \rangle = \langle x_n, Ay_n \rangle$ for all $n \in \mathbb{N}$, taking the limit for $n \to \infty$ shows that the closure of A must be symmetric, too.

Recall that a number $\lambda \in \mathbb{C}$ is called an *eigenvalue* of a linear operator A, if there is $x \in \mathfrak{D}(A)$, called the *eigenvector* of A with respect to λ, such that $Ax = \lambda x$. The following properties of eigenvalues and eigenvectors of a symmetric linear operator A are easy to prove:

1. each eigenvalue λ of A is real;
2. eigenvectors x_1, x_2 belonging to different eigenvalues $\lambda_1 \neq \lambda_2$ are *orthogonal* to each other, that is, $\langle x_1, x_2 \rangle = 0$.

§4. Self-Adjoint Operators

If a linear operator A acting in H is symmetric, then every other symmetric linear operator S having the property $A \prec S$ is called a *symmetric extension* of A. Notice that $A \prec S$ always implies $S^* \prec A^*$, which is also true for non-symmetric operators A and S. Furthermore, let $S_1, \ldots, S_k$ be a finite sequence of symmetric extensions of A, where S_{i+1} extends S_i for $i = 1, \ldots, k-1$. Since $S_k \prec S_k^*$ and $S_{i+1}^* \prec S_i^*$ for $i = 1, \ldots, k-1$, we obtain by concatenating all these extensions the chain

$$A \prec S_1 \prec S_2 \prec \ldots \prec S_k \prec S_k^* \prec \ldots \prec S_2^* \prec S_1^* \prec A^*.$$

Thus one may hope to find a *maximal* symmetric extension S_{sa} of A such that

$$A \prec S_{sa} = S_{sa}^* \prec A^*$$

holds, and in this case the operator S_{sa} cannot be extended to a larger symmetric operator. This expectation suggests to introduce the notion of *self-adjointness*: a symmetric linear operator A is called *self-adjoint*, if $A^* \prec A$, or equivalently, if $A = A^*$. For an everywhere defined symmetric and linear operator A in H, however, we have $A^* = A$, so there is no difference between symmetry and self-adjointness in this case, and the HELLINGER-TOEPLITZ theorem claims that A is bounded, which is a consequence of the closed range theorem. In fact, since a self-adjoint operator A is the adjoint of itself, it is closed with closed domain of definition $\mathfrak{D}(A)$, hence it is continuous respectively bounded by the closed range theorem.

To ensure self-adjointess (rsp. essentially self-adjointness) of a symmetric linear operator, the *range condition* serves as the default tool to verify this desired property:

THEOREM VIII.2. Every densely defined, closed and symmetric operator A acting in H is self-adjoint iff

$$\mathfrak{R}(A+i) = \mathfrak{R}(A-i) = H.$$

§5. Essentially Self-Adjointness

The closure of a symmetric linear operator is not necessarily self-adjoint. Hence a closable and symmetric operator A is denoted *essentially self-adjoint*, if $\overline{A}$ proves to be self-adjoint. There is a comparable criterion for essentially self-adjointness available:

THEOREM VIII.3. A densely defined and symmetric linear operator A is essentially self-adjoint if and only if

$$\overline{\mathfrak{R}(A+i)} = \overline{\mathfrak{R}(A-i)} = H.$$

Essentially self-adjointness can also be characterized by means of the adjoint: a densely defined and symmetric linear operator A in H is essentially self-adjoint iff the adjoint A^* is symmetric, too.

§6. The von Neumann Extension Theory

We want to sketch in this paragraph the principles of the *extension theory* for symmetric linear operators. The question is, whether a given symmetric operator owns one or more self-adjoint extensions.

First notice that the closure of a symmetric linear operator is not necessarily self-adjoint. To ensure self-adjointess of a symmetric operator A, the kernels of both operators $A^* \pm i$ are of special interest. In fact, if these are trivial, then $A \pm i$ are surjective and $\pm i$ are both regular values of A.

Based on this observation, F. RIESZ and J. V. NEUMANN have developed an extension theory, which answers the question under which conditions a symmetric and closed linear operator can be extended to a self-adjoint operator and how such an extension can be constructed. Moreover, the possible uniqueness of such an extension if of fundamental interest.

Recall that for a densely defined and closed symmetric operator A the relation $\mathfrak{D}(A) \subset \mathfrak{D}(A^*)$ holds. But more can be said, since the domain $\mathfrak{D}(A^*)$ of the adjoint operator can be represented as sum of three complementary subspaces:

$$\mathfrak{D}(A^*) = \mathfrak{D}(A) \oplus \mathfrak{N}(A^* + i) \oplus \mathfrak{N}(A^* - i). \tag{6.1}$$

Thus the linear operators $A^* - i$ and $A^* + i$ are of special interest. The numbers

$$n_+ := \dim \mathfrak{N}(A^* + i)\,, \qquad n_- := \dim \mathfrak{N}(A^* - i)$$

are called the *defect indices* of A, where the values $n_\pm = \infty$ are admissible, too. One easily derives from (6.1) that a symmetric and closed linear operator A with defect indices (n_+, n_-) can be extended to a self-adjoint operator iff $n_+ = n_-$, and that the operator A itself is self-adjoint iff in addition $n_+ = n_- = 0$ holds. On the other hand, A is called *maximal symmetric*, if at least one of the values n_+ and n_- is zero, which is equivalent to claim that A has no proper symmetric extension.

To extend in case of $n_+ = n_-$ the symmetric linear operator A to a self-adjoint one, it suffices to have a isometric and linear mapping

$$U : \mathfrak{N}(A^* + i) \rightarrow \mathfrak{N}(A^* - i). \tag{6.2}$$

Then, if U is given by (6.2), we define $A_s\, x := Ax$ for $x \in \mathfrak{D}(A)$ and

$$A_s\, z := iz - i\,Uz$$

for $z \in \mathfrak{N}(A^* + i)$, which implies that A_s is also symmetric and hence forms a symmetric extension of A. One can show that $A_s = A^*$, so A_s is even self-adjoint. We emphasize that the set of all self-adjoint extensions is given by the totality of isometric linear operators U as defined above.

§7. Perturbations of Self-Adjoint Operators

This subject deals with operators of the form $A + B$, where A is assumed to be self-adjoint and B to be symmetric with domain of definition $\mathfrak{D}(B)$ being a superset of $\mathfrak{D}(A)$, so $A+B$ is well defined with domain of definition $\mathfrak{D}(A+B)=\mathfrak{D}(A)$. It is our aim to decide under which conditions on the operator B the sum $A+B$ proves to be self-adjoint.

Let H be a complex Hilbert space and the operator $A:\mathfrak{D}(A)\to H$ be self-adjoint. Then the following concept is useful: a symmetric operator $B:\mathfrak{D}(B)\to H$ with $\mathfrak{D}(A)\subset\mathfrak{D}(B)$ is called A-*bounded*, if there are constants $\theta, c\geq 0$ such that

$$\|B\,x\| \;\leq\; \theta\,\|A\,x\| \;+\; c\,\|x\| \qquad \text{for all } x\in\mathfrak{D}(A). \tag{7.1}$$

In order to ensure A-boundedness of B, it is sufficient to ensure (7.1) only for elements x in a core A_0 of A.

The infimum of all numbers θ, for which there is c such that (7.1) holds, is called the A-*bound* of B. Notice that if B is a bounded operator, then it is A-bounded with A-bound $\theta<1$ for any A. In addition, the validness of the estimate

$$\|B\,x\|^2 \;\leq\; \theta^2\,\|A\,x\|^2 \;+\; c^2\,\|x\|^2 \qquad \text{for all } x\in\mathfrak{D}(A).$$

is a sufficient condition for (7.1) to be true.

Let A be self-adjoint in Hilbert space H and B a further symmetric linear operator with domain of definition $\mathfrak{D}(B)\supset\mathfrak{D}(A)$. The KATO-RELLICH *theorem* provides a criterion on self-adjointness of the operator $A+B$, which has been independently found by T. KATO and F. RELLICH:

THEOREM VIII.4. Let the linear operator A be self-adjoint and the linear operator B with $\mathfrak{D}(B)\supset\mathfrak{D}(A)$ be symmetric and A-bounded with $\theta<1$. Then $A+B$ with domain of definition $\mathfrak{D}(A+B)=\mathfrak{D}(A)$ is self-adjoint.

A similar result is obtained for the problem, whether *essentially self-adjointness* of a linear operator A is preserved by a symmetric and A-bounded perturbation B:

THEOREM VIII.5. Let $A:\mathfrak{D}(A)\to H$ be essentially self-adjoint and assume that B is symmetric and A-bounded, that is, $\mathfrak{D}(A)\subset\mathfrak{D}(B)$ and (7.1) holds with bound $\theta<1$. Then the operator $A+B$ is essentially self-adjoint on $\mathfrak{D}(A)$, where $\overline{A+B} \;=\; \overline{A}+\overline{B}$.

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IX.
Spectral Theory of Self-Adjoint Operators

In this section we give an outline on the spectral properties of self-adjoint linear operators. A linear operator A of this type is also suitable to be decomposed into sums of projection operators, where the sum may be mostly countable or be given by an integral representation. Having such a decomposition available, it is possible to define *spectral operators* of the form $f(A)$, where the function $f:\sigma(A)\to\mathbb{C}$ is at least measurable. The mapping $f\mapsto f(A)$ is called a *functional calculus* for A and provides a useful tool to solve evolution equations governed by a *positive* and self-adjoint linear operator A.

§1. Spectral Properties of Normal Operators

Our next topic will be the spectral theory of symmetric and self-adjoint operators. Both kinds of operators can be put together by the notion of *normal* operator, distinguished by the property $A\circ A^*=A^*\circ A$. Symmetric and selfadjoint operators are extremely admissable for the study of their spectral properties.

In order to analyze the spectrum of a symmetric linear operator, we start with a consideration of operators of the form $A-\lambda$, where A is closed, densely defined and symmetric and $\lambda\in\mathbb{C}$ is arbitrary. Basically, let H be a complex Hilbert space.

First remember that a number $\lambda\in\mathbb{C}$ with $\operatorname{Im}\lambda\neq 0$ does not belong to the numerical range $\Theta(A)$ of A, hence λ is contained in the field of external regularity $\Delta(A)$. This means that $A-\lambda$ is bounded below, and since we assume A to be closed, $A-\lambda$ is injective and normally solvable by the closed range theorem, thus upper semi-Fredholm with $\alpha(A-\lambda)=0$. By GOHBERG's punctured neighbourhood theorem, the index $\chi(A-\lambda)$ is locally constant for each fixed $\lambda\notin\mathbb{R}$. In addition, it can be shown that this magnitude is even globally constant in the open upper and lower complex half-plane $\mathbb{H}^+$ and $\mathbb{H}^+$, respectively.

Before proceeding with the study of the range of the operator $A-\lambda$, notice that due to the orthogonality relations the linear subspaces $\mathfrak{R}(A-\lambda)$ and $\mathfrak{N}(A^*-\overline{\lambda})$ are coupled by

$$H = \mathfrak{R}(A-\lambda) \oplus \mathfrak{N}(A^* - \overline{\lambda}).$$

In the following we distinguish between the two cases with respect to surjectivity of $A-\lambda$, or equivalently, to the injectivity of $A^* - \overline{\lambda}$:

1. either $\mathfrak{R}(A-\lambda) = H$, which implies $\mathfrak{N}(A^* - \overline{\lambda}) = \{0\}$, then $A-\lambda$ is invertible, λ is a regular value of A, and $\mathbb{H}^+$ and $\mathbb{H}^-$ belongs to $\rho(A)$, respectively,
2. or $\mathfrak{R}(A-\lambda) \neq H$, then $\mathfrak{N}(A^* - \overline{\lambda}) \neq \{0\}$ and λ belongs to the residual spectrum $\sigma_r(A)$, so $\mathbb{H}^+$ and $\mathbb{H}^-$ are subsets of $\sigma_r(A)$, respectively.

§2. Properties of the Resolvent and the Spectrum

As it has already been indicated, both cases may separately happen for the complex half planes $\mathbb{H}^+$ and $\mathbb{H}^-$, so in conclusion there are four possibilities how the spectrum $\sigma(A)$ of a densely defined and closed symmetric operator A looks like:

1. $\sigma(A) = \mathbb{C}$ in case of $\mathbb{H}^+ \cup \mathbb{H}^- \subset \sigma_r(A)$,
2. $\sigma(A) = \overline{\mathbb{H}^+}$ in case of $\mathbb{H}^+ \subset \sigma_r(A)$ and $\mathbb{H}^- \subset \rho(A)$,
3. $\sigma(A) = \overline{\mathbb{H}^-}$ in case of $\mathbb{H}^- \subset \sigma_r(A)$ and $\mathbb{H}^+ \subset \rho(A)$,
4. $\sigma(A) \subset \mathbb{R}$ in case of $\mathbb{H}^+ \cup \mathbb{H}^- \subset \rho(A)$.

But if the resolvent set $\rho(A)$ contains at least one element $\zeta \in \mathbb{R}$, then only the last case in the previous enumeration can occur. In fact, the property $\sigma(A) \subset \mathbb{R}$ leads us again to the notion of *self-adjointness*, due to the following theorem:

THEOREM IX.1. A closed linear operator A is self-adjoint if and only if $\sigma(A) \subseteq \mathbb{R}$, and in this case we have for $\operatorname{Im}\zeta \neq 0$ the resolvent estimation

$$\|R(A,\zeta)\| \leq \frac{1}{|\operatorname{Im}\zeta|}.$$

Indeed, $\sigma(A)$ being a subset of the real line means that $\mathbb{H}^+$ and $\mathbb{H}^-$ belong to the resolvent set of A, which is true if and only if the range condition is fulfilled, that is, the operators $A \pm i$ are both surjective. The resolvent estimation is a consequence of the inequality $\|R(A,\zeta)\| \leq \delta(\zeta, \sigma(A))^{-1}$, where $\delta(\zeta, \sigma(A)$ denotes the positive distance between $\zeta \in \rho(A)$ and the spectrum of A.

Remember the subsets of the spectrum $\sigma(A)$ of a linear operator A. They have been introduced by the criterion how the perturbed operator $A-\lambda$ fails to be bijective. This approach has led us to the notion of *point spectrum* $\sigma_p(A)$, *continuous spectrum* $\sigma_c(A)$ and *residual spectrum* $\sigma_r(A)$, respectively.

The next result shows that the division of the spectrum for self-adjoint operators is simpler than in the arbitrary case:

THEOREM IX.2. If A is self-adjoint, then the residual spectrum $\sigma_r(A)$ is empty, thus $\sigma(A) = \sigma_p(A) \cup \sigma_c(A)$.

This assertion is proved by the relations $\mathfrak{N}(A) = \mathfrak{R}(A^*)^\perp = \mathfrak{R}(A)^\perp$. By definition, if $\lambda \in \sigma_r(A) \subset \mathbb{R}$, then $A-\lambda$ is injective and consequently $\mathfrak{R}(A-\lambda) = H$.

When dealing with a self-adjoint linear operator A, it is advantageous to breakup the set $\sigma(A)$ into other subsets as for arbitrary linear operators.

First, the *pure point spectrum* $\sigma_{pp}(A)$ is the set

$$\sigma_{pp}(A) := \{\lambda \in \mathbb{C} : \mathfrak{R}(A-\lambda) = \overline{\mathfrak{R}(A-\lambda)} \neq H\}.$$

Second, by *points embedded in continuum* are meant the elements of the set

$$\sigma_{pec}(A) := \{\lambda \in \mathbb{C} : \mathfrak{R}(A-\lambda) \neq \overline{\mathfrak{R}(A-\lambda)} \neq H\}.$$

Both definitions do not require $A-\lambda$ to be injective, thus $\sigma_{pp}(A)$ and $\sigma_{pec}(A)$ possibly contain eigenvalues of A. We join both subsets together and redefine the *point spectrum* $\sigma_p(A) := \sigma_{pp}(A) \sqcup \sigma_{pec}(A)$, that is, $\sigma_p(A)$ for A being self-adjoint consists of those $\lambda \in \mathbb{C}$, for which $\overline{\mathfrak{R}(A-\lambda)} \neq H$. This redefinition leads to the following

THEOREM IX.3. The point spectrum $\sigma_p(A)$ is exactly the set of eigenvalues of A.

Third, the *purely continuous spectrum* is defined to be the set

$$\sigma_{pc}(A) := \{\lambda \in \mathbb{C} : \mathfrak{R}(A-\lambda) \neq \overline{\mathfrak{R}(A-\lambda)} = H\}.$$

We alternatively introduce the *continuous spectrum* according to $\sigma_c(A) := \sigma_{pec}(A) \sqcup \sigma_{pc}(A)$, which of course is nothing but

$$\{\lambda \in \mathbb{C} : \mathfrak{R}(A-\lambda) \neq \overline{\mathfrak{R}(A-\lambda)}\}.$$

In opposite to the originally given definition of $\sigma_c(A)$ (see paragraph VI.2), the set $\sigma_c(A)$ may also contain those elements $\lambda \in \mathbb{C}$, for which $A-\lambda$ is not injective.

For a self-adjoint operator A it is also customary to denote $\sigma_{pp}(A)$ the *discrete spectrum* and its complement relatively to $\sigma(A)$ the *essential spectrum*, i. e.

$$\sigma_{ess}(A) := \sigma(A) \setminus \sigma_{pp}(A),$$

so we have $\sigma_c(A) \subset \sigma_{ess}(A)$.

The *discrete spectrum* of A consists of those values $\lambda \in \mathbb{C}$, for which there is a terminating sequence (e_n) in H consisting of pairwise disjoint elements $e_n \in S_1(0)$. The essential spectrum of A, however, consists of numbers $\lambda \in \mathbb{C}$, for which there is a non-terminating sequence (e_n) in H consisting of pairwise disjoint elements $e_n \in S_1(0)$ such that

$$\lim_{n\to\infty} \|Ae_n - \lambda e_n\| = 0. \tag{2.1}$$

Each sequence having the property (2.1) for some $\lambda \in \sigma(A)$ is called a *Weyl sequence.* In summary, any number $\lambda \in \mathbb{C}$ belongs to the spectrum of a self-adjoint linear operator A, if it owns a (possibly terminating) Weyl sequence.

§3. The Discrete Spectral Theorem

We next consider linear operators k acting in a *separable* Hilbert space H, which are *compact* and *symmetric* at the same time, so k fulfills $\langle kx, y\rangle = \langle x, ky\rangle$ for all $x, y \in H$. We shall call k *compact-symmetric* in this case.

Of course, if k is compact-symmetric, then the properties of compact and and symmetric operators complement, so we may quote that the spectrum $\sigma(k)$ consists of a sequence $\{\lambda_n\}_{n\in\mathbb{N}}$ of *real* eigenvalues of k, which either terminates or converges to $0 \in \mathbb{C}$. Due to the compactness of k, the related eigenspaces E_{λ_n} are finite-dimensional. By the *Gram-Schmidt orthogonalization method*, one can define an orthogonal system $\mathfrak{O}$ of eigenvectors of k such that $\mathfrak{O}$ spans a dense subspace in H and consequently forms an *orthonormal base* in H.

The *Hilbert-Schmidt spectral theorem*, also often called the *discrete spectral theorem*, claims that a compact-symmetric operator k acting in a separable Hilbert space H can be represented as a mostly *countable* sum of linear operators k_{λ_n}, each of them being a simple multiplication operator $k_{\lambda_n} := \lambda_n 1_H$ and acting on a finite-dimensional subspace of H. In other words, the theorem provides an orthogonal decomposition of H into a sequence of k-invariant subspaces:

Theorem IX.4. Let the Hilbert space H be separable. Given a compact-symmetric linear operator $k: H \to H$, there is a null sequence $\{\lambda_n\}_{n\in\mathbb{N}}$ consisting of real eigenvalues and an orthonormal base in H consisting of eigenvectors $\{x_n\}_{n\in\mathbb{N}}$ of k such that for all $x \in H$ the following representation holds:

$$k\,x = \sum_{n=1}^{\infty} \lambda_n \langle x, x_n \rangle\, x_n. \tag{3.1}$$

This suggests to order the set of eigenvalues of A by their moduli. Hence we obtain a sequence $\{\lambda_n\}_{n\in\mathbb{N}}$ of eigenvalues λ_n such that

$$|\lambda_1| \leq |\lambda_1| \leq |\lambda_3| \ldots \to \infty.$$

In the following let A be a compact-symmetric operator acting in a separable Hilbert space H. Then the resolvent operators $R(A,\zeta)$ of A are symmetric-compact, too, and by the discrete spectral theorem IX.4 the set of eigenvectors of each $R(A,\zeta)$, ($\zeta \in \rho(A)$ forms an orthonormal base in H. By the spectral mapping theorem, the eigenvectors of $R(A,\zeta)$ are determined by the eigenvectors of A. In fact, let $\xi_1, \xi_2, \ldots$ be the sequence of eigenvalues of $R(A,\zeta)$. Then $(\lambda-\zeta)^{-1}x$ must be of the form $(\lambda-\zeta)^{-1}x = \xi_n\, x$ for some $n \in \mathbb{N}$, which implies that $\sigma(A)$ is a subset of the set $\{\zeta + \xi_n{}^{-1} : n \in \mathbb{N}\}$.

§4. Projections and Projection-Valued Measures

As the projection theorem in Hilbert space shows, projection mappings are in one-to-one relationship to closed linear subspaces in H. So it is nearby to call two projections π_1, π_2 *orthogonal* to each other, if their related closed subspaces have this property, and in this case the identity $\pi_1 \circ \pi_2 = \pi_2 \circ \pi_1 = \mathsf{0}$ holds, where $\mathsf{0}$ is the projection related with the trivial subspace $\{\mathsf{0}\} \subset H$.

In general, a projection mapping π has the following properties:

1. π is idempotent: $\pi \circ \pi = \pi$.
2. π is linear on H.
3. π is bounded with $\|\pi\| \le 1$, hence $\pi \in \mathscr{L}(H)$.
   If $\mathcal{A}_\pi := \pi(H) = \{0\}$, then $\|\pi\| = 0$, else $\|\pi\| = 1$. The only possible eigenvalues of π are the real numbers 0 and 1.
4. Let $\mathcal{A}_1, \mathcal{A}_2$ be closed subspaces of H with $\mathcal{A}_1 \subseteq \mathcal{A}_2$, then for the associated projections the equation $\pi_{\mathcal{A}_1} \circ \pi_{\mathcal{A}_2} = \pi_{\mathcal{A}_2} \circ \pi_{\mathcal{A}_1} = \pi_{\mathcal{A}_1}$ holds. In this case, the operator $\pi_{\mathcal{A}_2} - \pi_{\mathcal{A}_1}$ is also a projection, and related subspace is given by the set $\mathcal{A}_2 \ominus \mathcal{A}_1 := \mathcal{A}_2 \cap \mathcal{A}_1^{\perp}$.
5. $\pi = \pi^*$, that is, π is symmetric. Conversely, each symmetric and idempotent linear operator is a projection.
6. If $\mathcal{A}$ is a *finite*-dimensional subspace of H, then $\mathcal{A}$ is closed, and the projection $\pi_{\mathcal{A}}$ can be represented by means of a set of base vectors $e_1, \ldots, e_n$ in $\mathcal{A}$, namely

$$\pi_{\mathcal{A}} x = \sum_{i=1}^{n} \langle x, e_i \rangle \, e_i.$$

Finally, let $\mathscr{P}(H)$ be the set of projections in H, then there is a partial order relation $\ge$ on $\mathscr{P}(H)$, defined by means of inclusion between the related closed subspaces:

$$\pi_{\mathcal{A}_1} \ge \pi_{\mathcal{A}_2} \quad :\Longleftrightarrow \quad \mathcal{A}_2 \subseteq \mathcal{A}_1.$$

A *projection-valued measure* is a mapping $P : \mathcal{B}_{\mathbb{R}} \to \mathscr{L}(H)$, where $\mathcal{B}_{\mathbb{R}}$ is the ordinary Borel σ-algebra in $\mathbb{R}$ and P fulfills the following properties:

(P-1) $P(\Omega)$ is a projection operator for each $\Omega \in \mathcal{B}_{\mathbb{R}}$;
(P-2) $P(\mathbb{R}) = 1_H$, the identity operator in H;
(P-3) P is *σ-additiv*, that is, if $\Omega_1, \Omega_2, \ldots$ are measurable sets in the real line and $\Omega := \bigsqcup_{n \in \mathbb{N}} \Omega_n$, then

$$P(\Omega) = \lim_{n \to \infty} \sum_{k=1}^{n} P(\Omega_k),$$

where this kind of convergence has to be understood in the sense of the strong operator topology in the algebra $\mathscr{L}(H)$.

The projection-valued measure P is also called a *spectral measure* or a *resolution of the identity.* Notice that intervals of the form $(-\infty, \lambda]$ are Borel-measurable, so each projection $P((-\infty, \lambda])$ is well-defined. In fact, the one-parameter family of these projections constitutes a projection-valued measure P in H.

§5. Functional Calculus and Spectral Operators

In the sequel it is our aim to introduce a *functional calculus* for measurable functions $f : \mathbb{R} \to \mathbb{C}$ with respect to a projection-valued measure P.

Let $B(\mathbb{R})$ be the linear space of *bounded* measurable functions $f : \mathbb{R} \to \mathbb{C}$ endowed with the norm

$$\|f\|_\infty := \sup_{x \in \mathbb{R}} |f(x)| < \infty,$$

which in fact is a Banach space. Then a function $e \in B(\mathbb{R})$ is called *simple*, if it adopts only finitely many values $e_1, \ldots, e_N$. Moreover, the set

$$S(\mathbb{R}) := \{\, f : \mathbb{R} \to \mathbb{C} : f \text{ is simple} \,\}$$

forms dense subspace of $B(\mathbb{R})$, and the supremum norm of $e \in S(\mathbb{R})$ is given by $\|e\|_\infty = \max\{|e_1|, \ldots, |e_N|\}$.

After these preliminaries, the setup of the spectral integral will be carried out by the following five steps:

1. First of all, each spectral measure P gives rise to a complex and real-valued Borel measure on the real line. For arbitrary $x, y \in H$ let the complex measure $\mu_{x,y}$ be defined by

$$\begin{cases} \mu_{x,y} : \mathcal{B}_\mathbb{R} \to \mathbb{C}, \\ \Omega \mapsto \mu_{x,y}(\Omega) := \langle x, P(\Omega) y \rangle. \end{cases}$$

   Due to the symmetry of $P(\Omega)$, the derived measure $\mu_x(\Omega) := \mu_{x,x}(\Omega)$ for $\Omega \in \mathcal{B}_\mathbb{R}$ is real-valued.

2. We next make sense of the expression $\int_\mathbb{R} f \, dP$ for measurable functions $f : \mathbb{R} \to \mathbb{C}$. Let us start with the case, where f is *simple*, that is, f takes values in the finite set $\{f_1, \ldots, f_N\}$. Define

$$\int_{\mathbb{R}} f\,dP := \sum_{n=1}^{N} f_n P(\chi^{-1}(f_n)),$$

where $\chi^{-1}(f_n)$ is the preimage of f_n in $\mathcal{B}_{\mathbb{R}}$. We further mention that

$$I_{0,P} : f \to \int_{\mathbb{R}} f\,dP$$

is a continuous linear mapping from the normed space $S(\mathbb{R})$ consisting of simple functions to the Banach space $\mathscr{L}(H)$ with operator norm $\|I_{0,P}\| = 1$.

3. We shall continue with the integration of *bounded measurable functions* with respect to P. Due to the B.L.T. theorem, there exists a unique extension I_P from $I_{0,P}$ to the closure $B(\mathbb{R}) = \overline{S(\mathbb{R})}$, where $\|I_P\| = \|I_{0,P}\|$. The mapping I_P is also linear and fulfills

$$\begin{cases} I_P(1_{\mathbb{R}}) = 1_H \\ I_P(fg) = I_P(f) \circ I_P(g) \\ I_P(\overline{f}) = I_P(f)^* \end{cases} \qquad f, g \in B(\mathbb{R}),$$

where $1_{\mathbb{R}}$ is the identity in $\mathbb{R}$, thus I_P even forms a C*-algebra homomorphism between the function algebra $B(\mathbb{R})$ and the operator algebra $\mathscr{L}(H)$.

4. We will also define I_P for an *arbitrary measurable function* $f : \mathbb{R} \to \mathbb{C}$. But this is only possible for those elements $x \in H$ fulfilling

$$\int_{\mathbb{R}} |f|^2\,d\mu_x < \infty. \tag{5.1}$$

The set D_f of elements $x \in H$ satisfying (5.1) forms a linear subspace lying dense in H. For f we introduce an approximating sequence $\{f_n\}_{n\in\mathbb{N}}$ of functions $f_n \in B(\mathbb{R})$ given by

$$f_n := \chi_{\{t\in\mathbb{R}:|f(t)|\le n\}} \cdot f \quad (n \in \mathbb{N}).$$

One can show that f_n is Cauchy in $L^2(\mathbb{R})$, hence $I_P(f_n)$ is Cauchy in H, where I_P is the previously defined operator acting on the space $B(\mathbb{R})$.

5. To the end, for a fixed measurable function f we may define an operator $S(P, f)$, called the *spectral operator* with respect to P,

by virtue of

$$\int_{\mathbb{R}} f\, dP : D_f \mapsto H\,, \qquad x \in D_f \mapsto \Big(\int_{\mathbb{R}} f\, dP\Big)\, x := y\,, \tag{5.2}$$

which enjoys the property

$$\Big(\int_{\mathbb{R}} f\, dP\Big)^* = \int_{\mathbb{R}} \overline{f}\, dP.$$

Especially for $f = 1_{\mathbb{R}}$ the self-adjoint operator $S_P := S(P,1)$ is defined by

$$\mathfrak{D}(S_P) := D_{1_{\mathbb{R}}}, \quad S_P x := \Big(\int_{\mathbb{R}} 1_{\mathbb{R}}\, dP\Big)\, x.$$

§6. The Spectral Theorem for Self-Adjoint Operators

The spectral theorem is definitely the most important assertion with respect to self-adjoint operators. Another way to claim this theorem reads as follows: each self-adjoint linear operator A is a *spectral operator*, whose underlying *projection-valued measure* P can be retrieved from A. In the following we want to make this assertion precise by going on step by step.

In linear algebra the *main axis theorem* quotes that each hermitean matrix is diagonalizable and has purely real-valued eigenvalues. The generalization of this assertion, substituting hermitean matrices by densely defined self-adjoint operators acting in abstract Hilbert space, leads to the so-called *spectral theorem.*

Remember from integration theory that for a given measure space $(X, \mathcal{B}, \mu)$ the linear space $L^2(X)$ of measurable and square-integrable functions $f : X \to \mathbb{C}$ forms a Hilbert space with inner product

$$\langle f, g\rangle := \int_X f\, \overline{g}\, d\mu.$$

If $m : X \to \mathbb{C}$ is a measurable function, then the linear mapping M_m defined by

$$\begin{cases} \mathfrak{D}(M_m) & := \{\, f \in L^2(X) : m \cdot f \in L^2(X)\,\}, \\ (M_m f)(x) & := m(x) \cdot f(x) \;\; \text{for } x \in X \text{ and } f \in \mathfrak{D}(M_m) \end{cases}$$

is called a *multiplication operator* with *generator* m. It turns out that the mapping M_m is symmetric iff m is real-valued, and since the domains of definition of the operators M_m and M_m^* coincide, each multiplication operator with real-valued generator is self-adjoint.

The *spectral theorem* claims that the reverse assertion is also true:

THEOREM IX.5. To each self-adjoint operator $A : \mathfrak{D}(A) \subset H \to H$ there is assigned

1. a measure space $(X, \mathcal{B}, \mu)$,
2. a multiplication operator M_m in $L^2(X)$ with generator function $m : X \to \mathbb{R}$,
3. a bijective and isometric linear operator $U : H \to L^2(X)$

such that $A = U^{-1} \circ M_m \circ U$ holds.

The existence of the operator U gives rise to speak of *unitary equivalence* between the operators A and M_m. If some operators A_1 and A_2 are unitary equivalent, then their spectra $\sigma(A_1)$ and $\sigma(A_2)$ are identical.

In summary, theorem IX.5 implies that an operator A is self-adjoint in Hilbert space H iff it is unitary equivalent to a real-valued multiplication operator M_m on $L^2(X)$, where X is the base set of a suitable chosen measure space $(X, \mathcal{B}, \mu)$.

Using the notion of projection-valued measure, we are now able to quote a further variant of the *spectral theorem for self-adjoint operators*:

THEOREM IX.6. Given a self-adjoint linear operator A, there is a uniquely defined projection-valued measure P such that A is the spectral operator related to P:

$$A = \int_{\mathbb{R}} 1 \, dP.$$

This version of the spectral theorem is extremely useful to define operator-valued mappings $f : \sigma(A) \to f(A)$ by

$$f(A) := \int_{\mathbb{R}} f \, dP,$$

where $f(A)$ will become into a densely defined operator acting in H. Indeed, passing from a self-adjoint linear operator to its family of projection-valued measures and then to the spectral operators defined by (5.2) is nothing but a *functional calculus* for the class of self-adjoint operators. For example, if A is self-adjoint and *positive* at once, that is, $\langle Ax, x\rangle \geq 0$ for all $x \in \mathfrak{D}(A)$, then we may define operators like $|A|$, $\sqrt{A}$, A^n, $\exp(A)$, $\sin(A)$, $\cos(A)$ and others.

It is also possible to formulate the spectral theorem by means of so-called *spectral resolutions* instead of projection-valued measures. Preliminary, to each projection-valued measure P in Hilbert space H there is related the concept of *spectral resolution*, which is nothing but a one-parameter family $\{P_\lambda\}_{\lambda \in \mathbb{R}}$ of linear operators $P_\lambda : H \to H$ such that

(R-1) P_λ is a projection for each $\lambda \in \mathbb{R}$.
(R-2) For each $x \in H$ the mapping $\lambda \mapsto P_\lambda x$ is continuous from the right.
(R-3) $\lambda < \mu$ implies $P_\lambda \leq P_\mu$, that is, $\langle P_\lambda x, x\rangle \leq \langle P_\mu x, x\rangle$ for all $x \in H$.
(R-4) $\lim_{\lambda \to -\infty} P_\lambda\, x = 0$ and $\lim_{\lambda \to \infty} P_\lambda\, x = x$ for all $x \in H$.

Each projection P_λ of a given spectral resolution is related to a closed linear subspace $\mathcal{A}_\lambda \subset H$ and the difference $P_\mu - P_\lambda$ for $\mu \geq \lambda$ also forms a projection related to the closed subspace $\mathcal{A}_\mu \ominus \mathcal{A}_\lambda$.

Now, using spectral resolutions instead of projection-valued measures, the spectral theorem claims that each self-adjoint operator A owns a spectral resolution $\{P_\lambda\}_{\lambda \in \mathbb{R}}$ such that

$$\mathfrak{D}(A) = \{\, x \in H : \int_{\mathbb{R}} \lambda^2 \, d\langle P_\lambda\, x, x\rangle < \infty \,\},$$

$$\langle A\, x, y\rangle = \int_{\mathbb{R}} \lambda \, d\langle P_\lambda\, x, y\rangle \quad \text{for } x \in \mathfrak{D}(A) \text{ and } y \in H.$$

The latter formula is often written down in the symbolic form

$$Ax = \int_{\mathbb{R}} \lambda \, dP_\lambda x.$$

To the end of this paragraph we want to investigate the relationship between the shape of the spectrum and the spectral resolution of a self-adjoint operator.

In fact, the following theorem shows that the resolvent set and the different kinds of spectra of a self-adjoint linear operator A are determined by its spectral resolution:

THEOREM IX.7. Let A be self-adjoint in H and $\mathcal{A}_\mu$ be the closed subspace in H associated with the projection P_μ for some $\mu \in \mathbb{R}$. Then

(a) $\mu \in \mathbb{R}$ belongs to the *point spectrum* $\sigma_p(A)$ iff

$$0 < \dim\,(\mathcal{A}_{\mu+0} \ominus \mathcal{A}_\mu) < \infty.$$

(b) $\mu \in \mathbb{R}$ belongs to the *continuous spectrum* $\sigma_c(A)$ iff for each $\epsilon > 0$ the subspace $\mathcal{A}_{\mu+\epsilon} \ominus \mathcal{A}_{\mu-\epsilon}$ is infinite-dimensional.

(c) $\mu \in \mathbb{R}$ is a *regular value* of A iff there is an open interval $(\alpha, \beta) \subset \mathbb{R}$ such that $\mu \in (\alpha, \beta)$ and P_μ is constant in (α, β).

Especially the resolvents $R(A, \zeta)$ for $\zeta \in \rho(A)$ of a self-adjoint linear operator A can be represented by means of the spectral resolution:

$$R(A,\zeta)\,x \;=\; \int_{-\infty}^{\infty} \frac{1}{\lambda - \zeta}\, dP_\lambda x, \quad (x \in H,\ \mathrm{Im}\,\zeta \neq 0).$$

On the other hand, STONE*'s formula* provides a constructive method to describe the spectral resolution of A in an interval $[a, b]$ by means of the resolvents $R(A, \zeta)$. Let $x \in H$ be arbitrary, then

$$\frac{1}{2}\{P[a,b] + P(a,b)\}\,x = \lim_{\epsilon \downarrow 0} \frac{1}{2\pi i} \int_a^b \{R(A, t+i\epsilon)\,x - R(A, t-i\epsilon)\,x\}\,dt.$$

§7. Positive Self-Adjoint Operators and Cauchy Problems

In the Hilbert space setting the analysis of linear operator-differential equations can be done wth the aid of spectral theory, where the differential equation $u' + Au = 0$ is governed by a positive self-adjoint operator A. The following existence and uniqueness theorem purely relies on the spectral theorem and the functional calculus for unbounded self-adjoint operators.

1. Remember that a linear operator A acting in a Hilbert space H is called *positive*, if it is symmetric and accretive. In this case any solution $x : [0, \tau) \to H$ of the homogeneous Cauchy problem

$$x'' + Ax \;=\; 0, \quad x(0) \;=\; x_0 \tag{7.1}$$

satisfies the estimate

$$\|x(t)\| \leq \|x(0)\| \qquad \text{for all } t > 0\,, \tag{7.2}$$

which is a consequence of the inequality $\|x(t)\|^2 = -2\,\langle Ax, x\rangle \leq 0$ and therefore causes the solution of the homogeneous Cauchy problem to be unique.

To verify the existence of a solution for (7.1), let $(P_\lambda)_{\lambda\in\mathbb{R}}$ be the spectral resolution of the self-adjoint operator A given by the spectral theorem. We notice that $P_\lambda = 0$ for $\lambda < 0$, since A is positive. Then the functional calculus applied to the continuous function $t \mapsto e^{-t}$ gives rise to the spectral operator

$$L_t(x) := e^{-tA}x = \int_0^\infty e^{-\lambda t}\, dP_\lambda\, x \qquad (t \geq 0,\, x \in H).$$

We apparently have $L_0 = 1_X$, and (7.2) implies $\|L_t\| \leq 1$ for $t > 0$.

In conclusion, the existence of the solution of (7.1) is ensured by the following

THEOREM IX.8. The function $x : t \mapsto L_t(x_0)$ is continuous in $[0,\infty)$, continuously differentiable in $(0,\infty)$ and forms a solution of problem (7.1).

2. The functional calculus for self-adjoint linear operators is also helpful to solve further classes of abstract initial-value problems. By the *abstract Schrödinger equation* we mean the linear operator-differential equation

$$x'(t) + iA\,x(t) = 0\,, \tag{7.3}$$

where A is a self-adjoint operator in Hilbert space H. The operator iA is called *skew-adjoint* in this case. Each solution x of (7.3) satisfies the *energy equation*

$$\|x(t)\| = \|x(0)\| \tag{7.4}$$

for $t \in \mathbb{R}$, which immediately follows from

$$\frac{d}{dt}\,\|x(t)\|^2 = \langle -iAx(t), x(t)\rangle + \langle x(t), -iAx(t)\rangle = 0.$$

If an initial value $x(0) = x_0$ is given, then (7.4) implies that the solution of the initial-value problem for (7.3) is unique.

A solution of (7.3) is also provided by semigroup theory and especially by STONE's theorem (see paragraph X.5), since the operator iA is the infinitesimal generator of a unitary group of bounded operators. To construct a solution for problem (7.3) by the functional calculus method for unbounded self-adjoint operators, we define the spectral operator

$$L_t(x) := e^{-itA}x = \int_{-\infty}^{\infty} e^{-i\lambda t} dP_\lambda\, x \qquad (t > 0),$$

where P_λ again is the spectral resolution of A having support in the whole real line. Then the family of operators L_t yields the solution of the abstract Schrödinger problem:

THEOREM IX.9. The mapping $t \mapsto L_t(x_0)$ is the unique solution of the Cauchy problem for (7.3), where for each $t > 0$ the operator L_t is unitary.

3. The functional calculus for self-adjoint linear operators may also be applied to the second-order homogeneous Cauchy problem

$$x'' + Ax = 0, \quad x(0) = x_0, \quad x'(0) = x_1, \tag{7.5}$$

since the square root operator $\sqrt{A}$ is well-defined, due to the fact that A is positive.

Each solution x of problem (7.5) satisfies the energy conservation law

$$\|x'(t)\|^2 + \|\sqrt{A}\, x(t)\|^2 = \|x'(0)\|^2 + \|\sqrt{A}\, x(0)\|^2 \quad (0 < t < \tau),$$

which implies that the problem owns mostly one solution.

Let (P_λ) again be the spectral resolution of A. We define a function $x : [0, \tau) \to H$ by means of the spectral operator L_t induced by the functions $t \mapsto \cos(\sqrt{t})$ and $t \mapsto \sin(\sqrt{t})/\sqrt{t}$. In fact, the operator $x \mapsto L_t x$ is built as the sum of two single spectral operators for each $0 \le t < \tau$, namely

$$x(t) := L_t(x_0, x_1) := \int_0^\infty \cos\sqrt{\lambda}t\, dP_\lambda\, x_0 + \int_0^\infty \frac{\sin\sqrt{\lambda}t}{\sqrt{\lambda}}\, dP_\lambda\, x_1.$$

Having defined the family of linear operators $\{L_t(x_0, x_1)\}$ this way, we obtain the following existence and uniqueness theorem for our second order problem:

THEOREM IX.10. The mapping $t \mapsto L_t(x_0, x_1)$ is continuous in $[0, \tau)$, twice continuously differentiable in $(0, \tau)$ and is the unique solution of problem (7.5).

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X.
Semigroups of Bounded Linear Operators

Our aim in this part is to sketch the basic notions of semigroup theory, which forms one of the most important topics within functional analysis with respect to applications to ordinary and partial differential equations.

The notion of *semigroup* has been introduced by J. A. Séguier in 1904. Later some applications to partial differential equations were detected, but the first systematic and comprehensive treatement of semigroup theory was published in the pioneering monograph [4] by E. Hille and R. Phillips. Nowadays semigroup theory forms a powerful method in solving linear evolution problems with a multiplicity of applications to partial differential equations governed by linear-elliptic operators.

§1. Strongly Continuous Semigroups

Given a Banach space X, a C_0*-semigroup* or a *strongly continuous semigroup of bounded linear operators* is an operator-valued mapping

$$S : [0,\infty) \to \mathscr{L}(X),$$

shortly denoted $\{S(t)\}_{t\geq 0}$ and distinguishes by the properties

(G-1) $S(t+s) = S(t) \circ S(s)$ for all $t, s \geq 0$ (the *semigroup property*);
(G-2) $\lim_{t\downarrow 0} S(t)\, x = x$ for all $x \in X$.

It follows from (G-2), which is nothing but strong continuity from the right-hand side in $t=0$, that $S(0)$ is the identity mapping in X. The conditions (G-1) and (G-2) together imply continuity from the right of the semigroup even in every $t > 0$. In case of $X = \mathbb{R}$, semigroups share both properties with the family $\{\mathcal{E}(t)\}_{t\geq 0}$ of exponential functions $\mathcal{E}(t) : \lambda \in \mathbb{R} \mapsto e^{\lambda t}$, so $\{\mathcal{E}(t)\}_{t\geq 0}$ serves as an elementary example of a strongly continuous semigroup.

Notice that the totality of C_0-semigroups may be classified as follows: for each such semigroup $\{S(t)\}_{t\geq 0}$ there are constants $M \geq 0$ and $\omega \in \mathbb{R}$ such that

$$\|S(t)\| \leq Me^{\omega t} \quad \text{for } t > 0. \tag{1.1}$$

By the *orbit map* or *trajectory* η_x of a C_0-semigroup $\{S(t)\}_{t\geq 0}$ and associated with an initial element $x \in X$ we mean the vector-valued mapping $\eta_x : [0, \infty) \to X$ given by $\eta_x(t) := S(t)\,x$. Due to condition (G-1), strong continuity of $\{S(t)\}_{t\geq 0}$ implies continuity of the function η_x. But η_x is even differentiable under a very mild assumption: if $\{S(t)\}_{t\geq 0}$ is a C_0-semigroup, then for each $x \in X$ the mapping η_x is differentiable in $(0, \infty)$ if and only if η_x is right differentiable in $t = 0$.

§2. Infinitesimal Generators of Semigroups

If $\{S(t)\}_{t\geq 0}$ is a strongly continuous semigroup, its *infinitesimal generator* (briefly denoted the *generator* of the semigroup) is defined as a mapping A in X by means of the graph

$$\Gamma_A := \{\, (x, y) \in X \times X :\ y = \lim_{h \downarrow 0} \frac{1}{h}(S(h)x - x) \text{ exists } \}.$$

Hence $\mathfrak{D}(A)$ of A consists of those elements $x \in X$, for which the orbit map η_x is differentiable from the right in $t = 0$, that is,

$$\mathfrak{D}(A) = \{x \in X : \eta_x \text{ is differentiable}\}.$$

It is easy to show that A is linear, so it constitutes a linear operator acting in X with domain of definition $\mathfrak{D}(A) \subset X$.

The meaning of strongly continuous semigroups is confirmed by the solution theory for homogeneous Cauchy problems governed by linear operators. Indeed, the *main theorem of semigroup theory* reads as follows:

THEOREM X.1. Let A be a densely defined and closed linear operator acting in a Banach space X. Then the following assertions are equivalent to each other:

(a) The operator $-A$ generates a strongly continuous semigroup of type (ω, M).
(b) The resolvent set of A fulfills $(-\infty, \omega) \subset \rho(A)$ and the resolvent function $R(A, \zeta)$ satisfies

$$\|R(A, \zeta)^n\| \leq M(\omega - \zeta)^{-n} \quad (n \in \mathbb{N}, \zeta < \omega). \tag{2.1}$$

The equivalence of these assertions is the content of the celebrated FELLER-MIYADERA-HILLE-YOSIDA *theorem.* However, since the number of conditions on the operator A is infinite, this theorem has limited worth with respect to applications. But the original theorem for generators of *contraction semigroups* due to HILLE and YOSIDA in 1948 quotes that in this case only the first inequality ($n = 1$, with $\omega = 0$ and $M = 1$) of (2.1) has to be verified:

THEOREM X.2. The linear operator $-A$ acting in a Banach space X is the infinitesimal generator of a C_0-semigroup of *contractions* if and only if

(a) A is densely defined and closed,
(b) the interval $(-\infty, 0)$ belongs to the resolvent set $\rho(A)$,
(c) $\|\zeta\, R(A,\zeta)\| \leq 1$ for all $\zeta < 0$.

If $-A$ owns the properties mentioned in theorem X.2, then the semigroup $\{S(t)\}_{t \geq 0}$ generated by $-A$ may be constructed by

$$S(t)\, x := \lim_{n\to\infty} (1_X + \frac{t}{n} A)^{-n}\, x \text{ for each } t > 0 \text{ and } x \in X.$$

§3. Accretive Operators and Contractive Semigroups

In the following let H be a complex Hilbert space. Then a linear operator A acting in H and with domain of definition $\mathfrak{D}(A) \subset H$ is called *accretive*, if

$$\operatorname{Re} \langle A\, x, x\rangle \geq 0 \quad \text{for all } x \in \mathfrak{D}(A), \tag{3.1}$$

or equivalently, if $\Theta(A) \subset \{\zeta \in \mathbb{C} : \operatorname{Re}\zeta \geq 0\}$. Otherwise, a linear operator A is said to be *dissipative*, if $-A$ is accretive.

We first notice that if A is accretive (dissipative), then ζA is so for each $\zeta > 0$. Furthermore, since

$$\|A\, x + \zeta x\|^2 - \|\zeta x\|^2 \geq 2\zeta \operatorname{Re} \langle A\, x, x\rangle \quad \text{for all } x \in \mathfrak{D}(A),$$

the operator A is accretive if and only if

$$\|(A + \zeta)x\| \geq \|\zeta x\| \quad \text{for all } \zeta \geq 0 \text{ and } x \in \mathfrak{D}(A) \tag{3.2}$$

or eqivalently

$$\|(A+\zeta)x\| \geq \operatorname{Re}\zeta \cdot \|x\| \qquad \text{for all } \operatorname{Re}\zeta \geq 0 \text{ and } x \in \mathfrak{D}(A). \quad (3.3)$$

These two properties can be used to define accretivity and dissipativity also for linear operators in Banach spaces, where no inner product is present.

Inequality (3.1) quotes that for $\zeta > 0$ the operator $A + \zeta$ is bounded below and thus injective. In addition, if A is closed, then the bounded below theorem implies that A has closed range, and by the *closed graph theorem* the inverse operator

$$(A+\zeta)^{-1} : \mathfrak{R}(A+\zeta) \to \mathfrak{D}(A)$$

is bounded with norm $\|(A+\zeta)^{-1}\| \leq \zeta^{-1}$, so it is even a contraction for $\zeta \geq 1$.

Let us now assume that for the accretive operator A there is $\zeta > 0$ such that $A+\zeta$ is surjective. A linear operator with this property is said to fulfill the *range condition.* In this case there is an open interval I containing ζ such that $A+\zeta$ is surjective for all $\zeta \in I$. Finally, it turns out that $A+\zeta$ is surjective for every $\zeta > 0$.

An accretive linear operator A is called *maximal accretive* or shortly *m-accretive* if the range condition is fulfilled. The following enumeration contains a couple of properties of the m-accretive operator A:

(a) A is densely defined in H;
(b) A is closed;
(c) $\mathbb{C}^- := \{\zeta \in \mathbb{C} : \operatorname{Re}\zeta < 0\} \subset \rho(A)$;
(d) $A+\zeta$ is either surjective for all $\zeta \in \mathbb{C}^+$ or *not* surjective for all $\zeta \in \mathbb{C}^+$.

The main theorem of this paragraph characterizes m-accretive linear operators by other conditions:

THEOREM X.3. If A is accretive, the following assertions are equivalent to each other:
(a) A is m-accretive;
(b) A is accretive and does not have a proper accretive extension;
(c) $A + i\alpha$ is m-accretive for all $\alpha \in \mathbb{R}$;
(d) $A + \alpha$ is m-accretive for all $\alpha > 0$;
(e) $\mathbb{C}^- \subset \rho(A)$ and $\|R(A,\zeta)\| \leq (\operatorname{Re}\zeta)^{-1}$ for all $\zeta \in \mathbb{C}^-$;
(f) $(-\infty, 0) \subset \rho(A)$ and $\sup_{t<0} \|t\, R(A,t)\| \leq 1$;

(g) A is densely defined and closed, and the adjoint A^* of A is m-accretive, too.

Besides, if A is densely defined in H and m-accretive, then $\mathfrak{N}(A) = \mathfrak{N}(A^*)$ and consequently $\mathfrak{N}(A) \subset \mathfrak{D}(A) \cap \mathfrak{D}(A^*)$.

We shall denote a linear operator A *coercive*, if $\mathrm{Re}\langle Ax, x\rangle \geq \theta \, \|x\|^2$ for some $\theta > 0$ and all $x \in \mathfrak{D}(A)$. It should be clear that every coercive operator is accretive.

In the following we want to characterize the spectral properties (b) and (c) in theorem X.2 for a linear operator A by equivalent conditions.

The usage of m-accretivity is illustrated by the next theorem. By means of this concept, the celebrated *Lumer-Phillips theorem* classifies those linear operators, which are infinitesimal generators of contractive C_0-semigroups:

THEOREM X.4. Let A be a linear operator in Hilbert space. Then $-A$ generates a contractive C_0-semigroup iff A is m-accretive.

The proof of this theorem relies on the classical Hille-Yosida theorem for C_0-semigroups of type $(0, 1)$. Indeed, A is m-accretive, then A is densely defined, closed and $-A$ fullfills the estimate (**??**), hence $-A$ is the generator of a contractive C_0-semigroup.

Reversely, if $-A$ is the infinitesimal generator of a C_0-semigroup of contractions, then we have $(0, \infty) \subset \rho(A)$ and therefore $\mathfrak{R}(A - \zeta) = H$ for all $\zeta > 0$. In addition, if $x \in \mathfrak{D}(A)$, then

$$|\langle S(t)x, x\rangle| \;\leq\; \|S(t)x\| \, \|x\| \;\leq\; \|x\|^2$$

and

$$\mathrm{Re}\langle S(t)x - x, x\rangle \;=\; \mathrm{Re}\langle S(t)x, x\rangle \;-\; \|x\|^2 \;\leq\; 0.$$

Dividing this line by $t > 0$ and letting $t \downarrow 0$ yields $\mathrm{Re}\langle Ax, x\rangle \leq 0$, which shows that $-A$ is accretive, hence in summary the operator A even proves to be m-accretive.

§4. Sectorial Operators and Analytic Semigroups

Regarding the definition of sectorial operator, we shall follow close to [3] and denote a linear operator A in a Banach space X and with domain of definition $\mathfrak{D}(A)$ (not necessarily dense in X) to be *sectorial* of angle $\psi \in [0, \pi)$ and vertex $\omega \in \mathbb{R}$, if the following conditions are true:

(S-1)' $\sigma(A) \subset \overline{\Sigma_{\psi,\omega}}$, where $\Sigma_{\psi,\omega} := \{\zeta \in \mathbb{C} : \zeta \neq \omega, arg(\zeta - \omega) < \psi\}$ is the sector with vertex ω and semi-angle ψ. This implies $\mathbb{C} \setminus \overline{\Sigma_{\psi,\omega}} \subset \rho(A)$;

(S-2)' $\sup\{\, \|\zeta R(A,\zeta)\| : \zeta \in \mathbb{C} \setminus \overline{\Sigma_{\phi,\omega}}\,\} < \infty$ for all $\psi < \phi < \pi$.

The operator A is called *sectorial*, if it is so for some pair $(\psi, \omega) \in [0, \pi) \times \mathbb{R}$. Notice that if A is sectorial with respect to the angle ψ, then there is $0 < \epsilon < \psi$ such that A is also sectorial with respect to the smaller angle $\psi - \epsilon$. The infimum ψ_0 of all these angles is called the *angle of sectoriality* of A.

The following theorem, whose proof can be found in [5], p. 43, provides useful criterions to ensure sectorialness of A:

THEOREM X.5. Let A be a linear operator in X with domain of definition $\mathfrak{D}(A) \subset X$ having the following properties:

(a) the resolvent set $\rho(A)$ contains $\mathbb{C}_\omega := \{\zeta \in \mathbb{C} : \operatorname{Re}\zeta \leq \omega\}$;

(b) there is $M > 0$ such that $\|\zeta\, R(A,\zeta)\| \leq M$ for all $\zeta \in \mathbb{C}_\omega$.

Then A is sectorial.

A further theorem shows that in the Hilbert space setting the notation of sectorial operator is close to KATO's concept of *m-sectorial operator*:

THEOREM X.6. Let H be a Hilbert space and A be a closed linear operator in H with numerical range $\Theta(A) \subset \overline{\Sigma_{\psi,0}}$ for $\psi \leq \pi/2$ and $A + 1_H$ being surjective. Then A is sectorial of angle ϕ.

This theorem especially implies that a self-adjoint and positive linear operator in Hilbert space is sectorial of angle 0.

In the following we collect further features of a sectorial linear operator A acting in ordinary Banach spaces:

1. $x \in \overline{\mathfrak{D}(A)}$ if and only if $\lim_{t\to\infty} t(t + A)^{-1}x = x$;
2. $x \in \overline{\mathfrak{R}(A)}$ if and only if $\lim_{t\to 0} t(t + A)^{-1}x = 0$;
3. If X is reflexive, then A is densely defined and $X = \mathfrak{N}(A) \oplus \overline{\mathfrak{R}(A)}$.

Semigroups of bounded linear operators are usually parametrized on the half line $[0, \infty)$, but it is also customary to study possible extensions to a symmetric sector

$$\Sigma_\theta := \{z = re^{i\alpha} \in \mathbb{C} : r > 0\,, |\alpha| < \theta\}$$

of angle 2θ for some $\theta \in (0, \frac{\pi}{2}]$.

A semigroup $\{S(t)\}_{t\geq 0}$ is called *analytic*, if there is an angle $\theta \in (0, \frac{\pi}{2})$ and an analytic extension $S' : \Sigma_\theta \to \mathscr{L}(X)$ of S, which is *locally bounded*, that is,

$$\sup\{\,\|S'(z)\| : z \in \Sigma_\theta,\ \|z\| \leq 1\} < \infty.$$

Next we want to sketch a functional calculus for sectorial operators A, which especially enables us to define bounded linear operators of the form e^{-tA}.

If X is a Banach space and A is a sectorial operator acting in X, then for each $t > 0$ it is possible to introduce the bounded linear operator $e^{-tA} \in \mathscr{L}(X)$ by means of the contour integral

$$e^{-tA}x := \frac{1}{2\pi i}\int_\gamma e^{-t\zeta}\, R(A,\zeta)x\, d\zeta \quad (x \in X), \tag{4.1}$$

which generalizes the well-known RIESZ-DUNFORD *formula*. Here γ constitutes a closed curve in the complex plane that drifts within the sector $S_{\phi,\omega}$ and includes the spectrum of the operator $-tA$. The family of operators $\{S(t)\}_{t\geq 0}$ defined by (4.1) forms a semigroup. In addition, if A is densely defined in X, then the semigroup is even strongly continuous.

Due to the property of A being sectorial, the domain of the parameter t is not restricted to the interval $(0,\infty)$, but can be extended to a certain sector $\Sigma_\theta \subset \mathbb{C}$, in which the semigroup property $S(t+s) = S(t)\,S(s)$ for $s, t \in \Sigma_\theta$ holds. Thus we obtain the following

THEOREM X.7. If A is a sectorial operator, then the semigroup $\{S(t)\}_{t\geq 0}$ of bounded linear operators generated by $-A$ is analytic.

We finally turn to semigroups being analytic and contractive at the same time. Let $\{S(t)\}_{t\geq 0}$ be an analytic semigroup. If in addition $\|S(\zeta)\| \leq 1$ for all $\zeta \in \Sigma_\theta$, then it is called a *sectorially contractive analytic* C_0-semigroup of angle θ.

The generation theorem for this subclass of analytic semigroups reads as follows:

THEOREM X.8. Let A be a densely defined operator on a Banach space X and let $\theta \in (0, \frac{\pi}{2}]$. The following assertions are equivalent:

(a) The operator $-A$ generates a sectorially contractive analytic C_0-semigroup of angle θ.

(b) $\rho(A) \subset \mathbb{C} \setminus \Sigma_\theta$ and $\|\zeta(A+\zeta)^{-1}\| \leq 1$ for $\zeta \in \Sigma_\theta$.

§5. Skew-Adjoint Operators and Unitary Groups

Recall that a bijective linear operator U in Hilbert space is called *unitary*, if $\mathfrak{D}(U) = H$ and $U^{-1} = U^*$. The spectrum $\sigma(U)$ of such an operator U forms a subset of the unit circle in the complex plane.

A semigroup $\{U(t)\}_{t\geq 0}$ of bounded linear operators acting in a Hilbert space H is called *unitary*, if each operator $U(t)$ is unitary. Furthermore, let $U_{-t} := U_t$ for $t > 0$, then a *group* $(U_t)_{t\in\mathbb{R}}$ of unitary operators is defined, which fulfills the group property

$$U_{s+t} = U_s \circ U_t \qquad (t, s \in \mathbb{R}).$$

Reversely, STONE's theorem claims that the infinitesimal generator of a unitary group is skew-adjoint:

THEOREM X.9. Each unitary group $(U_t)_{t\in\mathbb{R}}$ in a Hilbert space is infinitesimally generated by a linear operator of type iA, where A is self-adjoint.

It should be mentioned that if a self-adjoint linear operator A is given, then the unitary group $(U_t)_{t\in\mathbb{R}}$ can be constructed by means of the spectral decomposition of A and the functional calculus for spectral operators with respect to the function $t \mapsto e^{it}$, that is, we have for each $t > 0$

$$U_t x := \int_{\mathbb{R}} e^{i\lambda t}\, dP_\lambda\, x \quad (x \in H).$$

§6. Well-Posedness of Second-Order Cauchy Problems

Similar to the first-order Cauchy problem, a function x is called a *classical solution* of the second-order homogeneous Cauchy problem

$$x'' + Ax = 0, \quad x(0) = x_0 \tag{6.1}$$

if $x \in C([0,\tau),\mathfrak{D}(A)) \cap C^2((0,\tau),X)$ and satisfies both initial conditions as well as the differential equation in each point $t \in (0,\tau)$.

To apply results from semigroup theory to the second order Cauchy problem, let $U := (x, x')$. We suppose that A is a *positive* and *self-adjoint* linear operator in a Hilbert space H. Then the differential equation $x''(t) + A\,x(t) = 0$ in (6.1) can be written as first order problem

$$\begin{cases} U'(t) = B\,U(t) \\ U(0) = (x_0, x_1), \end{cases} \tag{6.2}$$

where the operator B is defined by

$$B := \begin{pmatrix} 0 & 1 \\ \sqrt{A} & 0 \end{pmatrix}. \tag{6.3}$$

The main idea in proving existence and uniqueness of a solution is to show that the operator B defined on a suitable chosen domain $\mathfrak{D}(B)$ is the infinitesimal generator of a semigroup of contractions. The set $\mathfrak{D}(B)$ is defined according to

$$\mathfrak{D}(B) := \mathfrak{D}(A) \times \mathfrak{D}(\sqrt{A}) \subset \mathfrak{D}(\sqrt{A})_{\|\cdot\|_A} \times H,$$

where $\mathfrak{D}(\sqrt{A})_{\|\cdot\|_A}$ is the Hilbert space $\mathfrak{D}(\sqrt{A})$ furnished with the graph norm.

The following theorem is for example proved in [2], p. 111:

THEOREM X.10. Let the linear operator A be positive self-adjoint and the operator B be defined by (6.3). Then B is the infinitesimal generator of a C_0-semigroup of contractions, whose orbit η determined by $\eta(0) = U(0)$ solves (6.2).

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XI.
Coercive Sesquilinear Forms

Our aim in this part is to present the general theory of coercive sesquilinear forms and its applications to stationary and evolutional linear equations. Sesquilinear forms are mostly efficient when studying abstract problems in the Hilbert space setting.
A further goal is to indicate the correspondence of sesquilinear forms and linear operators. In fact, each sesquilinear form gives rise to a couple of linear operators, and reversely, by taking the inner products $\langle Ax, y\rangle$, every linear operator A defines a sesquilinear form on its domain of definition $\mathfrak{D}(A)$.

The totality of applications of the presented theory to the boundary and inital-boundary value problems of linear partial differential equations are often denoted as *form methods* or as *Hilbert space methods.*

§1. Generalities on Sesquilinear Forms

By a *sesquilinear form* defined in complex linear space V is meant a mapping $\mathfrak{a}\colon V\times V\to\mathbb{C}$, which is linear in the first and conjugate linear in the second argument, that is, for all $x, y, z\in V$ and $\lambda, \mu\in\mathbb{C}$ the computation rules

$$\left\{\begin{array}{rcl} \mathfrak{a}(x+y,z) & = & \mathfrak{a}(x,z) \ + \ \mathfrak{a}(y,z) \\ \mathfrak{a}(x,y+z) & = & \mathfrak{a}(x,y) \ + \ \mathfrak{a}(x,z) \\ \mathfrak{a}(\lambda x,\mu y) & = & \lambda\overline{\mu}\,\mathfrak{a}(x,y) \end{array}\right.$$

are valid. If the linear space V is real, however, then $\mathfrak{a}$ is assumed to be linear also in the second argument, so it becomes into a *bilinear form* in this case.

The mapping $x\in V\mapsto \mathfrak{a}[x] := \mathfrak{a}(x,x)\in\mathbb{C}$ is called the *quadratic form* of $\mathfrak{a}$. Notice that all values $\mathfrak{a}(x,y)$ can be recovered from the values $\mathfrak{a}[x]$ and $\mathfrak{a}[y]$ by certain polarization formulas.

A sesquilinear form $\mathfrak{a}$ may also be regarded as a linear mapping $\mathcal{A}_{\mathfrak{a}}\colon V\to V_a^{\#}$, where $V_a^{\#}$ is the (algebraic) conjugate space of V, that is, the linear space of conjugate linear functionals $\phi\colon V\to\mathbb{C}$. More precisely, given a sesquilinear form $\mathfrak{a}$, each element $x\in V$ gives rise to a conjugate linear functional $\phi_x\colon V\to\mathbb{C}$, defined by

$\phi_x(y) := \mathfrak{a}(x,y)$ for $y \in V$. Let $V_a^{\#}$ again be the linear space of conjugate linear functionals on V. Then the mapping

$$\mathcal{A}_\mathfrak{a} : \begin{cases} \mathcal{A}_\mathfrak{a} : V \to V^{\#}, \\ \mathcal{A}_\mathfrak{a}\, x := \phi_x \quad (x \in V) \end{cases}$$

proves to be linear and is said to be the *form operator* or the *representation operator* associated with $\mathfrak{a}$.

Reversely, for any linear operator $\mathcal{A} \colon V \to V_a^{\#}$, there is a sesquilinear form $\mathfrak{a} \colon V \times V \to \mathbb{K}$ such that $\mathfrak{a}(x,y) = (\mathcal{A}\,x)y$ for $x, y \in V$. Hence sesquilinear forms acting on a complex linear space V appear in a one-to-one correspondence to linear mappings acting between the spaces V and $V_a^{\#}$. This correspondence remains valid, if V is a *real* linear space, $\mathfrak{a}$ is a *bilinear* form and $\mathcal{A}$ is the associated mapping from V to the adjoint space $V^{\#}$.

From now on we assume that V is a *normed* linear space with underlying field $\mathbb{C}$. Then any sesquilinear form $\mathfrak{a} \colon V \times V \to \mathbb{C}$ is called *bounded*, if there is $\Theta > 0$ such that

$$|\mathfrak{a}(x,y)| \le \Theta\, \|x\| \cdot \|y\| \qquad (x, y \in V).$$

Due to the *uniform boundedness principle*, this property is equivalent to say that $\mathfrak{a}$ is continuous on the space $V \times V$.

The prototypical example of a continuous form is the inner product in a complex Hilbert space H, where boundedness is ensured by the CAUCHY-SCHWARZ inequality $|\langle x, y\rangle| \le \|x\| \cdot \|y\|$.

Moreover, for each continuous sesquilinear form $\mathfrak{a}$ on H there is a bounded linear operator $B_\mathfrak{a} : H \to H$ such that

$$\mathfrak{a}(x,y) = \langle B_\mathfrak{a} x, y\rangle \quad \text{for all } x, y \in H, \tag{1.1}$$

which is a representation for the form $\mathfrak{a}$. In fact, if $x \in H$ is fixed, then the mapping $\mathfrak{a}(x,\cdot) : H \to \mathbb{C}$ causes a continuous antilinear functional $\phi_x \in H_a^*$, and by RIESZ's representation theorem there is $\tilde{x} \in H$ such that $\phi_x(y) = \langle \tilde{x}, y\rangle$ for all $y \in H$. This suggests to define $B_\mathfrak{a} x := \tilde{x}$ for every $x \in H$. If (1.1) holds for some operator $B \in \mathscr{L}(H)$, then the form $\mathfrak{a}$ is said to be represented by B.

The following theorem characterizes continuity of sesquilinear forms by equivalent assertions:

THEOREM XI.1. Let $\mathfrak{a}: H \times H \to \mathbb{C}$ be a sesquilinear form. Then:
(a) $\mathfrak{a}$ is continuous;
(b) $\mathfrak{a}$ is sequentially continuous in (x, y) for every $x, y \in H$;
(c) $\mathfrak{a}$ is sequentially continuous in $(0, 0) \in H \times H$;
(d) there is $\Theta \geq 0$ such that $|\mathfrak{a}(x, y)| \leq \Theta \|x\| \cdot \|y\|$ for $x, y \in H$;
(e) The representation operator $A_{\mathfrak{a}}$ is continuous.

§2. Coercive Forms and Variational Equations

Any continuous sesquilinear form $\mathfrak{a}: H \times H \to \mathbb{C}$ is called *coercive* if there is $\theta > 0$ such that

$$\operatorname{Re} \mathfrak{a}(x, x) \geq \theta \|x\|^2 \qquad \text{for all } x \in H. \tag{2.1}$$

The estimate (2.1) especially implies the validness of

$$\|B_{\mathfrak{a}} x\| \, \|x\| \;\geq\; |\langle B_{\mathfrak{a}} x, x\rangle| = |\mathfrak{a}(x, x)| \;\geq\; \operatorname{Re} \mathfrak{a}(x, x) \;\geq\; \theta \|x\|^2$$

for all $x \in H$. Hence the representation operator $B_{\mathfrak{a}}$ is bounded below with lower bound θ, and we may conclude by the fundamental *bounded-below theorem* that it is injective and the range $\mathfrak{R}(B_{\mathfrak{a}})$ forms a closed linear subspace in H.

Coercivity of a sesquilinear form $\mathfrak{a}$ is of great interest with respect to the solvability of the so-called *variational problem*, which may be formulated as follows: given a continuous and conjugate linear functional $\phi \in V_a^*$, it is to find $x \in V$ such that

$$\mathfrak{a}(x, y) \;=\; \phi(y) \qquad \text{for all } y \in V. \tag{2.2}$$

The LAX-MILGRAM *theorem* is of outstanding meaning in applied functional analysis and ensures well-posedness of the variational problem for any right-hand side $\phi \in V_a^*$:

THEOREM XI.2. If $\mathfrak{a}: V \times V \to \mathbb{C}$ is bounded and coercive, then for each $\phi \in V_a^*$ the equation (2.2) owns exactly one solution $x_\phi \in V$. In addition, $\| x_\phi \| \leq \theta^{-1} \|\phi\|$ holds.

In other words, if $\mathfrak{a}$ is bounded and coercive, then the form operator $\mathcal{A}_{\mathfrak{a}}: V \to V_a^*$ is bjective. The bounded inverse theorem implies that the inverse operator ${\mathcal{A}_{\mathfrak{a}}}^{-1}$ is bounded, too, so the solution x_ϕ of (2.2) depends continuously on ϕ.

There is also an *operator version* of theorem XI.2. Let $A: V \to V$ be linear and bounded. We shall denote A *coercive* if

$$\operatorname{Re}\langle Ax, x\rangle > \theta \, \|x\|^2$$

holds for some $\theta > 0$ and all $x \in V$. It should be clear that the mapping $(x, y) \mapsto \langle Ax, y\rangle$ defines a coercive form on V. Then the Lax-Milgram theorem implies invertibility of A and boundedness of A^{-1}, where $\|A^{-1}\| \leq \theta^{-1}$ holds.

§3. Triples of Hilbert Spaces

It is often indicated to study abstract linear problems in the setting of so-called *pairs* (V, H) of Hilbert spaces, where the space V is continuously and densely embedded into a second space H. Let us assume that V, H are Hilbert spaces and

(T-1) there is a linear mapping $\iota_1 : V \to H$ such that $\overline{\mathfrak{R}(\iota_1)} = H$;

(T-2) there is $c > 0$ such that $\|\iota_1 v\|_H \leq c \, \|v\|_V$ for all $v \in V$.

By means of the embedding ι_1, one can also think of V as a dense linear subspace in the space H.

Each Hilbert pair (V, H) with embedding $\iota : V \to H$ gives rise to introduce the *conjugate pair* (H_a^*, V_a^*), where the conjugate space H_a^* can be seen as continuously and densely embedded into the conjugate space V_a^* by a further embedding $\iota_2 : H_a^* \to V_a^*$. Then, by identifying the spaces H and H_a^* via the Riesz map, combining both Hilbert pairs leads to the triple

$$V \overset{\iota_1}{\hookrightarrow} H \overset{\iota_2}{\hookrightarrow} V_a^*, \tag{3.1}$$

sometimes called a *Gelfand triple*, an *evolution triple* or a *rigged Hilbert space*. The center space H is often denoted the *pivot space* of (3.1).

Gelfand triples own a remarkable property. Let $(\cdot\,, \cdot)$ and $\langle \cdot\,, \cdot \rangle$ be the inner products in V and H, respectively, then for fixed chosen $x \in H$ the action of the antilinear linear functional $f_x := \iota_2 \, x \in V_a^*$ to any element $v \in V$ is represented by the inner product of x and v in H, i. e. $f_x(v) = \langle x, \iota_1 \, v\rangle$ holds for all $v \in V$.

§4. Linear Operators Associated with a H-Elliptic Form

The presented theory of coercive sesquilinear forms in a Hilbert space V given by § 2 is often not sufficent for applications. Hence it is quite natural to examine sesquilinear forms also with respect to pairs (V, H) of Hilbert spaces. In applications it is often the case that sesquilinear forms are merely continuous but not coercive. In the setting of a Hilbert pair (V, H), however, we try to make a form $\mathfrak{a}$ coercive by adding an additional term, which is just a multiplicity of the inner product in H.

Indeed, if $\mathfrak{a}: V \times V \to \mathbb{C}$ is continuous and the slightly modified form

$$\mathfrak{a}_\lambda(u,v) := \mathfrak{a}(u,v) + \lambda\langle \iota_1 u, \iota_1 v\rangle \quad (u, v \in V)$$

is coercive for some $\lambda \in \mathbb{R}$, that is, there is $\theta > 0$ such that

$$\operatorname{Re} \mathfrak{a}(u,u) + \lambda \|\iota_1 u\|_H^2 \geq \theta \|u\|_V^2 \quad (u \in V), \tag{4.1}$$

then $\mathfrak{a}$ is called *quasi-coercive*, *H-elliptic* or simply a *Gårding form.* Clearly, if $\mathfrak{a}_{\lambda_0}$ is coercive for some $\lambda_0 \in \mathbb{R}$, then $\mathfrak{a}_\lambda$ is so for $\lambda > \lambda_0$.

Quasi-coercive sesquilinear forms are often used to define densely defined and closed linear operators acting in the pivot space H of a given Gelfand triple.

Let $\mathfrak{a}$ be a sesquilinear form defined on a Hilbert space V, which is the left hand side of a Gelfand triple (V, H, V_a^*) with continuous embeddings $\iota_1 : V \to H$ and $\iota_2 : H \to V_a^*$. It is our aim to introduce a linear operator $A_\mathfrak{a}$ with domain space H and range space H. This will be done by restricting the form operator $\mathcal{A}_\mathfrak{a} : V \to V_a^*$ associated with $\mathfrak{a}$ to those elements $u \in V$, for which $\mathcal{A}_\mathfrak{a} u \in H$. Indeed, the domain $\mathfrak{D}(A_\mathfrak{a})$ is defined to be the subspace of those elements $u \in V$, for which $\mathcal{A}_\mathfrak{a} u \in \iota_2(H) \subset V_a^*$ holds, briefly denoted

$$\mathfrak{D}(A_\mathfrak{a}) := \{ u \in V : \mathcal{A}_\mathfrak{a} u \in H \}.$$

Clearly, the set $\mathfrak{D}(A_\mathfrak{a})$ can be regarded as a linear subspace of H by means of the embedding ι_1. Then for $u \in \mathfrak{D}(A)$ there exists $y \in H$ such that $\iota_2 y = \mathcal{A}_\mathfrak{a} u$. We set $x := \iota_1 u$ and $A_\mathfrak{a} x := y$ for $x \in \mathfrak{D}(A_\mathfrak{a})$ and denote $A_\mathfrak{a}$ the *operator in H associated with* $(V, \mathfrak{a})$, or shortly, the *variational operator* derived from $\mathfrak{a}$. Another phrase is to say that the operator $A_\mathfrak{a}$ is the *part of* $\mathcal{A}_\mathfrak{a}$ *in H*.

The following theorem collects a couple of nice properties of the operator $A_{\mathfrak{a}}$ associated with a continuous and quasi-coercive form $\mathfrak{a}$. Besides the LAX-MILGRAM theorem, is is our second important theorem:

THEOREM XI.3. Let $(V, H, V_{\mathfrak{a}}^*)$ be a triple of Hilbert spaces and the sesquilinear form $\mathfrak{a}$ be continuous and quasi-coercive on V with parameter $\lambda_0 \in \mathbb{R}$. Then the following assertions are true:
(a) $A_{\mathfrak{a}}$ is closed.
(b) $A_{\mathfrak{a}}$ is densely defined in H.
(c) If the embedding of V into H is *compact*, then the operator $A_{\mathfrak{a}} + \lambda$ has a compact inverse $(A_{\mathfrak{a}} + \lambda)^{-1}$ for each $\lambda > \lambda_0$.
(d) The operator $A_{\mathfrak{a}} + \lambda$ is m-accretive.
(e) The operator $A_{\mathfrak{a}} + \lambda$ is sectorial.

It is often helpful to consider the *adjoint form* $\mathfrak{a}^*$ of a given sesquilinear form $\mathfrak{a}$ defined by

$$\mathfrak{a}^*(u, v) := \overline{\mathfrak{a}(v, u)} \qquad (u, v \in V).$$

Notice that if $\mathfrak{a}$ is continuous, then $\mathfrak{a}^*$ is so. Moreover, the form $\mathfrak{a}$ is called *symmetric*, if $\mathfrak{a} = \mathfrak{a}^*$ holds[7].

With respect to the variational operators $A_{\mathfrak{a}}$ and $A_{\mathfrak{a}^*}$, the following theorem is valid:

THEOREM XI.4. If $\mathfrak{a}^*$ is the adjoint form of $\mathfrak{a}$, then ${A_{\mathfrak{a}}}^* = A_{\mathfrak{a}^*}$. Furthermore, if $\mathfrak{a}$ is symmetric, then $A_{\mathfrak{a}}$ is self-adjoint.

Given a Hilbert pair (V, H), the theorem XI.4 is very helpful to construct self-adjoint linear operators in H by starting with symmetric forms in V. This approach can especially be performed, if the form is the inner product in V. The operator A associated with the inner product $\langle \cdot , \cdot \rangle_V$, called the *generating operator* of (V, H), is coercive and even self-adjoint in H by theorem XI.4.

In the following we derive the generating operator of a pair (V, H) of Hilbert spaces in a straightforward way. First of all, each fixed $y \in H$ gives rise to a linear functional on V according to

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[7] A well-known example of a symmetric form is the inner product $\langle \cdot , \cdot \rangle_H$ in a Hilbert space H, which differs from arbitrary symmetric sesquilinear forms only by positive definiteness.

$$f_y(v) := \langle \iota v, y\rangle_H, \qquad (v \in V).$$

The mapping f_y is also bounded, i. e. $f_y \in V_a^*$. Then, by the Riesz representation theorem, there is $v_y \in V$ such that

$$\langle v, v_y\rangle = f_y(v) \quad \text{for all } v \in V.$$

We therefore obtain a mapping $B : H \to V$, by $By := v_y$ for $y \in H$, which proves to be linear and continuous, and the inner product in H is obtained from that in V by

$$\langle \iota x, y\rangle_H = \langle x, By\rangle_V \qquad (x, y \in H).$$

But the operator B is also symmetric and strictly positive. These properties imply the existence of the inverse operator

$$B^{-1} : \mathfrak{D}(B^{-1}) = \mathfrak{R}(B) \subset V \to H,$$

which indeed is not necessarily bounded, but uniquely defined by the pair (V, H).

The operator B^{-1} is symmetric and strictly positive, too, hence there exists the square root operator $B^{-1/2}$ having the property

$$\langle B^{-1/2}x, B^{-1/2}y\rangle_H = \langle x, y\rangle_V \qquad (x, y \in H). \tag{4.2}$$

It turns out that $\mathfrak{D}(B^{-1/2}) = V$, which can be shown by contradiction. Especially for $y = x$ in (4.2) we obtain the equality

$$\|B^{-1/2}x\|_H = \|x\|_V \qquad (x \in V).$$

§5. Sectorial Forms and m-Sectorial Operators

In this paragraph we introduce the notations of *closed sectorial form* and *m-sectorial operator* and show the relationships between them.

Let H be a complex Hilbert space and $\mathfrak{a} : \mathfrak{D}(\mathfrak{a}) \times \mathfrak{D}(\mathfrak{a}) \to \mathbb{C}$ be a sesquilinear form, but defined on a linear subspace $\mathfrak{D}(\mathfrak{a})$ of H. Like for linear operators, the *numerical range* of $\mathfrak{a}$ is defined by

$$\Theta(\mathfrak{a}) := \{\, \mathfrak{a}(x, x) : x \in \mathfrak{D}(\mathfrak{a}),\ \|x\| = 1\,\}.$$

We then denote the form $\mathfrak{a}$ to be *sectorial*, if $\Theta(\mathfrak{a})$ is contained in a sector $\Sigma_{\psi,\omega}$, that is,

$$\Theta(\mathfrak{a}) \subset \Sigma_{\psi,\omega} := \{\zeta \in \mathbb{C} : \zeta \neq \omega, arg(\zeta - \omega) < \psi\}, \tag{5.1}$$

where $\omega \in \mathbb{R}$ is the vertex and $0 \leq \psi < \pi/2$ the semi-angle of the sector. Notice that sectors are symmetric with respect to the real line. Especially for $\psi = 0$ let $\Sigma_{0,\omega} := (\omega, \infty)$, and for $\psi = \pi/2$ we have $\Sigma_{\pi/2,0} = \mathbb{C}_+$. We also say that A is (ψ, ω)-sectorial, if (5.1) holds.

Now, if $\mathfrak{a}$ is symmetric and $\mathfrak{a} = \mathfrak{a}^*$, respectively, then $\Theta(\mathfrak{a}) \subset \mathbb{R}$ and $\omega \leq \inf \Theta(\mathfrak{a})$, so in this case we denote $\mathfrak{a}$ to be *half-bounded* or *bounded below* and call ω the lower bound of $\mathfrak{a}$. Moreover, if $\mathfrak{a}$ is half-bounded and (ψ, ω)-sectorial, then the estimates

$$\begin{cases} \operatorname{Re} \mathfrak{a}(x,x) \geq \omega \, \|x\|^2 \\ \operatorname{Im} \mathfrak{a}(x,x) \leq \tan \psi \cdot (\operatorname{Re} \mathfrak{a}(x,x) - \omega \, \|x\|^2) \end{cases}$$

hold for all $x \in \mathfrak{D}(\mathfrak{a})$.

Let $\operatorname{Re} \mathfrak{a} := (\mathfrak{a} + \mathfrak{a}^*)/2$ be the *real part* of $\mathfrak{a}$. Then for every $\alpha > 0$ the slightly modified form $[\cdot,\cdot]_\alpha := \operatorname{Re} \mathfrak{a} - \omega + \alpha$ constitutes an inner product on the linear space $\mathfrak{D}(\mathfrak{a})$. It turns out that the norms $|\cdot|_\alpha$ derived from these inner products $[\cdot,\cdot]_\alpha$ are equivalent to each other. We especially take $\alpha = 1$, pay attention to the inner product

$$[x,y]_1 = \operatorname{Re} \mathfrak{a}(x,y) - \omega\langle x,y\rangle + \langle x,y\rangle, \qquad \text{for all } x,y \in \mathfrak{D}(\mathfrak{a})$$

and distinguish between the two cases with respect to completeness of the norm

$$|x|_1 := [x,x]_1^{1/2} \qquad (x \in \mathfrak{D}(\mathfrak{a})).$$

In both cases it is our goal to obtain a Hilbert space $H_\mathfrak{a}$ with inner product $[x,y]_1$ and norm $|x|_1$ for $x, y \in H_\mathfrak{a}$, together with a linear embedding $\iota : H_\mathfrak{a} \to H$, which should be continuous if possible, so $(H_\mathfrak{a}, H)$ hopefully becomes into a pair of Hilbert spaces.

1. The normed linear space $(\mathfrak{D}(\mathfrak{a}), |\cdot|_1)$ is *complete.*
   We denote the form $\mathfrak{a}$ *closed* in this case, and $\mathfrak{D}(\mathfrak{a})$ becomes into a Hilbert space in its own right, hence we may define $H_\mathfrak{a} := \mathfrak{D}(\mathfrak{a})$ in this case. Notice that the norm $|\cdot|_1$ on $H_\mathfrak{a}$ is stronger than the norm $\|\cdot\|$ in H restricted to $H_\mathfrak{a}$, so the natural inclusion mapping $\iota : H_\mathfrak{a} \to H$ proves to be continuous.
2. The normed linear space $(\mathfrak{D}(\mathfrak{a}), |\cdot|_1)$ is *not* complete rsp. $\mathfrak{a}$ is not closed.

We define the Hilbert space $H_\mathfrak{a}$ as the completion of $\mathfrak{D}(\mathfrak{a})$ with respect to the norm $|\cdot|_1$. Moreover, Cauchy sequences in $H_\mathfrak{a}$ are also Cauchy sequences in H, hence for every $x \in H_\mathfrak{a}$ there is $x' \in H$ such that $x' = \iota x$ for a linear map $\iota : H_\mathfrak{a} \to H$.

We want to extend the form $\mathfrak{a}$ from $\mathfrak{D}(\mathfrak{a})$ to $H_\mathfrak{a}$ as well, and this will be done by defining

$$\overline{\mathfrak{a}}(x, y) := \lim_{n\to\infty} \mathfrak{a}(x_n, y_n)$$

for $x, y \in H_\mathfrak{a}$ and arbitrary sequences $\{x_n\}_{n\in\mathbb{N}}, \{y_n\}_{n\in\mathbb{N}} \subset \mathfrak{D}(\mathfrak{a})$ with $x_n \to x$ and $y_n \to y$. It turns out that the definition of $\overline{\mathfrak{a}}(x, y)$ is independent of the sequences $\{x_n\}_{n\in\mathbb{N}}$ and $\{y_n\}_{n\in\mathbb{N}}$, so this value is in fact well-defined for $x, y \in H_\mathfrak{a}$.

In summary, we obtain in both cases a Hilbert space $H_\mathfrak{a}$ with inner product $[\cdot,\cdot]_1$ and norm $|\cdot|_1$ together with a linear embedding $\iota : H_\mathfrak{a} \to H$, but which in the second case is not necessarily continuous. Regarding the second case, the form $\mathfrak{a}$ is said to be *closable*, if the mapping ι is continuous. This definition especially means that every closed form is closable. If $\mathfrak{a}$ is closable, then the final result in both cases will be that there exists a closed form, namely $\mathfrak{a}$ itself in the first and $\overline{\mathfrak{a}}$ in the second case, together with a continuous embedding of $H_\mathfrak{a}$ into H.

Let us continue with the concept of *sectorialness* of a closed linear operator in a complex Banach or Hilbert space. This class of operators has been introduced and first studied by T. Kato. Let again $\Sigma_{\psi,\omega}$ be an open sector with vertex $\omega \in \mathbb{R}$ and semi-angle $0 < \psi \leq \pi$. In his book [2], Kato denotes a linear operator A in a Hilbert space to be *sectorial*, if the following conditions are fulfilled:

(S-1) A is quasi-accretive, i. e. $\langle (A+\omega)x, x\rangle \geq 0$ for some $\omega \in \mathbb{R}$ and $x \in \mathfrak{D}(A)$,

(S-2) the numerical range $\Theta(A)$ is a subset of the sector $\Sigma_{\psi,\omega}$.

Furthermore, A is called *m-sectorial*, if it is at once sectorial and m-quasi-accretive, that is, $A + \omega$ is m-accretive for some $\omega \in \mathbb{R}$.

We finally want to elaborate the correspondence between closed sectorial forms and m-sectorial operators. It should be clear that if A is a m-sectorial operator in H with domain of definition $\mathfrak{D}(A)$, then the sesquilinear form

$$\mathfrak{a}_A : \begin{cases} \mathfrak{D}(\mathfrak{a}_A) := \mathfrak{D}(A), \\ \mathfrak{a}_A(x,y) := \langle A\,x, y\rangle \text{ for } x, y \in \mathfrak{D}(\mathfrak{a}_A) \end{cases}$$

is sectorial and closed.

Reversely, let $\mathfrak{a}_0$ be a closable sectorial form in H with domain $\mathfrak{D}(\mathfrak{a})$. We consider the closed completion $\mathfrak{a}$ on the space $H_\mathfrak{a}$. A densely defined linear subspace $D \subset \mathfrak{D}(\mathfrak{a})$ is denoted a *core* of $\mathfrak{a}$, or equivalently, the set D is a core of $\mathfrak{a}$, if $\mathfrak{a}$ is the closure of the restriction of $\mathfrak{a}$ to D.

KATO*'s first representation theorem* claims that a densely defined and closed sectorial form gives rise to a m-sectorial linear operator:

THEOREM XI.5. Let $\mathfrak{a}$ be a densely defined and closed sectorial form in H with vertex ω, semi-angle ψ and domain of definition $H_\mathfrak{a}$, and let ι be the continuous embedding from $H_\mathfrak{a}$ into H. Then there is a m-sectorial operator $A_\mathfrak{a}$ in H, whose domain of definition $\mathfrak{D}(A_\mathfrak{a})$ is defined as follows: Any element $x \in H$ belongs to $\mathfrak{D}(A_\mathfrak{a})$, if $x = \iota x_0$ for some $x_0 \in H_\mathfrak{a}$ and $\mathfrak{a}(x_0, z) = \langle y, \iota z\rangle$ holds for some $y \in H$ and all $z \in H_\mathfrak{a}$, which suggests to define $A_\mathfrak{a}\, x := y$.

The operator $A_\mathfrak{a}$ owns a couple of properties:

(a) $\mathfrak{D}(A_\mathfrak{a})$ is dense in H and the preimage $\iota^{-1}\mathfrak{D}(A_\mathfrak{a})$ is dense in $H_\mathfrak{a}$.
(b) If $x \in H_\mathfrak{a}$, $y \in H$ and $\mathfrak{a}(x,y) = \langle \iota x, y\rangle$ for all x in a core of $\mathfrak{a}$, then $\iota x \in \mathfrak{D}(A_\mathfrak{a})$ and $y = A(\iota x)$.
(c) $\Theta(A_\mathfrak{a})$ is a dense subset of $\Theta(\mathfrak{a})$.
(d) The adjoint form $\mathfrak{a}^*$ is also closed and sectorial with vertex ω and semi-angle ψ.
(e) The adjoint $A_\mathfrak{a}{}^*$ of $A_\mathfrak{a}$ is the m-sectorial operator associated with the adjoint form $\mathfrak{a}^*$.

Thus m-sectorial linear operators are in one-to-one correspondence with closed sectorial forms.

§6. The Friedrichs Extension Theorem

We want to apply the results of the last paragraph to the following class of operators: a *symmetric* linear operator A is called *half-bounded*, if there is $\theta \in \mathbb{R}$ such that

$$\langle A\,x, x\rangle \geq \theta\,\|x\|^2 \qquad \text{for all } x \in \mathfrak{D}(A),$$

or equivalently, if $\Theta(A) \subset [\theta, \infty)$ for some $\theta \in \mathbb{R}$. The operator A is called *positive* (*strictly positive*), if $\theta = 0$ ($\theta > 0$) can be chosen. In case of $\theta > 0$, the Cauchy-Schwarz inequality implies that A is also bounded below, hence A is surely injective. In addition, if A is closed, then the range $\mathfrak{R}(A)$ is closed in H, which follows from the bounded-below theorem, but $\mathfrak{R}(A)$ may be a proper subspace of H.

The main result in our topic is given by the fact that each densely defined and half-bounded linear operator A acting in a Hilbert space owns at least one *self-adjoint extension.*

First notice that if A is half-bounded with $\theta \in \mathbb{R}$, then it is $(0, \theta)$-sectorial, and the sesquilinear form $\mathfrak{a}(x, y) := \langle Ax, y \rangle$ is sectorial and densely defined. But $\mathfrak{a}$ is also closable, so we may switch to the sesquilinear form $\overline{\mathfrak{a}}$ defined on the Hilbert space $H_{\mathfrak{a}}$, called the *energetic space* of the operator A. Then by theorem XI.5 there is a m-sectorial and even self-adjoint linear operator $A_{\mathfrak{a}}$, that forms an extension of the half-bounded operator A, called the *Friedrichs extension* of A.

The domain of definition $\mathfrak{D}(A_{\mathfrak{a}})$ of the operator $A_{\mathfrak{a}}$ coincides with the linear subspace $H_{\mathfrak{a}} \cap \mathfrak{D}(A^*)$. Moreover, the operator $A_{\mathfrak{a}}$ owns a useful property: if one adds to the half-bounded operator A the multiplication operator $x \in H \mapsto \theta x$ with $\theta \in \mathbb{R}$, then the Friedrichs extensions of A and $A + \theta$ are related to each other by $(A + \theta)_{\mathfrak{a}} = A_{\mathfrak{a}} + \theta$.

§7. Variational Evolution Equations

Having the concept of Hilbert-Sobolev space available, we want to review the variational approach to first and second order evolution equations, which has been developed by J.-L. Lions in the midth of the last century. It can be seen as the straightforward extension of the Lax-Milgram theory from stationary equations to evolutional ones.

Let $\tau > 0$ and denote λ the Lebesgue-Borel measure on $[0, \tau)$. By an *abstract variational problem of parabolic type* in a real Hilbert space triple (V, H, V^*) we mean a λ-nearly everywhere in $(0, \tau)$ valid linear operator-differential equation

$$\begin{cases} u'(t) + \mathcal{A}_{\mathfrak{a}}(t)\, u(t) = f(t), \\ u(0) = u_0, \end{cases} \tag{7.1}$$

where $u_0 \in H$, $f : [0, \tau) \to V^*$ and $\mathcal{A}_{\mathfrak{a}\, t \in [0,\tau)}$ is a family of linear operators $\mathcal{A}_{\mathfrak{a}} : V \to V^*$.

Let $\mathfrak{a}(t)$ be the form associated with the operator $\mathcal{A}_{\mathfrak{a}}(t)$. Then we call $u \in H^1(0, \tau; V, V^*)$ a *weak solution* of (7.1), if $u(0) = u_0$ and

$$\langle u'(t), v(t)\rangle + \mathfrak{a}(t, u(t), v(t)) = f(t)(v)$$

for all $v \in V$ and λ-nearly everywhere $t \in [0, \tau)$.

Regarding the existence and uniqueness of a *weak solution* for the initial value problem (7.1), the following theorem is of fundamental meaning:

THEOREM XI.6. Let $\{\mathfrak{a}(t)\}_{t \in [0,\tau)}$ fulfill the following conditions:

(a) the mapping $t \in [0, \tau) \mapsto \mathfrak{a}(t, u, v)$ is Lebesgue-measurable for every fixed pair $(u, v) \in V \times V$,

(b) $\mathfrak{a}$ is *uniformly* bounded in $[0, \tau)$, that is, there is a constant $M \geq 0$ such that

$$|\mathfrak{a}(t, u, v)| \leq M \cdot \|u\|_V \|v\|_V \qquad \text{for all } u, v \in V,\ t \in [0, \tau),$$

(c) $\mathfrak{a}(t)$ is quasi-coercive for each $t \in [0, \tau)$.

Then for each $u_0 \in H$ and each $f \in L^2(0, \tau; V^*)$ there is exactly one solution $u \in H^1(0, \tau; V, V^*)$ of (7.1), which continuously depends on the data u_0 and f.

The proof of this theorem has been published in [3] as *Théorème 1.1*, page 46. Both terms u' and $\mathcal{A}_{\mathfrak{a}}(t)u$ in (7.1) belong to the same space $L^2(0, \tau; V^*)$ as f, which gives rise to say that the problem has *maximal regularity* in V^*.

The concept of weak solution makes also sense for linear evolution equations of second order.

An *abstract hyperbolic variational problem* with respect to the Hilbert space triple (V, H, V^*) and a family of bilinear forms $\mathfrak{a}(t) : [0, \tau] \times V \times V \to \mathbb{R}$ is given by

$$\begin{cases} \langle u''(t), v(t)\rangle + \mathfrak{a}(t, u(t), v(t)) = \langle f(t), v\rangle \quad (v \in V, \\ u(0) = u_0 \in V, \\ u'(0) = u_1 \in H, \end{cases} \tag{7.2}$$

where $\langle \cdot, \cdot \rangle$ is the inner product in H.

The next theorem shows that problem 7.2 is *weakly solvable* under a couple of conditions on the involved bilinear forms:

THEOREM XI.7. Let $\{\mathfrak{a}(t)\}_{t\in[0,\tau)}$ be a family of bilinear forms on V such that

(a) $t \mapsto \mathfrak{a}(t)$ is *uniformly* bounded in $[0,\tau)$,
(b) $\mathfrak{a}(t)$ is H-elliptic for each $t\in[0,\tau)$,
(c) $\mathfrak{a}(t)$ satisfies the symmetry condition $\mathfrak{a}(t,u,v)=\mathfrak{a}(t,v,u)$ for all $u,v\in V,\ t\in[0,\tau)$,
(d) the mapping $t\mapsto \mathfrak{a}(t,u,v)$ is continuously differentiable in $[0,\tau)$ for each fixed $(u,v)\in V\times V$,
(e) there is $M_1 \geq 0$ such that

$$|\frac{d}{dt}\mathfrak{a}(t,u,v)| \leq M_1 \cdot \|u\|_V \, \|v\|_V \qquad \text{for all } t\in[0,\tau).$$

Then the initial value problem (7.2) has a unique weak solution $u\in H^1(0,\tau;\,V,\,V^*)$ for all $u_0\in V$, $u_1\in H$ and $f\in L^2(0,\tau;H)$.

The symmetry condition (c) in XI.7 is essential to prove this theorem. Moreover, the solution map

$$R: L^2(0,\tau;H)\times V\times H\to L^2(0,\tau;H)$$

given by $(f,u_0,u_1)\mapsto u$ is linear in each argument and continuous, thus problem (7.2) proves to be well-posed.

References

[1] W. Arendt, A.F.M. ter Elst: *From Forms to Semigroups.* Operator Theory: Advances and Applications, Vol. 221, pp. 47-69.

[2] T. Kato: *Perturbation Theory for Linear Operators*, Springer Verlag, Grundlehren der mathematischen Wissenschaften, Band 132, 1995.

[3] J.-L. Lions: *Équations Différentielles Opérationelles et Problèmes aux Limites.* Springer, Grundlehren der mathematischen Wissenschaften, Band 111, 1961.

XII.
Applications to Differential Operators

This final section of this text deals with applications of the presented theory to linear differential operators and some standard problems related with them.

Remember first the notation of *multi-index*, which is nothing but a d-tuple $\alpha = (\alpha_1, \ldots, \alpha_d)$ with values $\alpha_i \in \mathbb{N}_0$ and with modulus $|\alpha| := \alpha_1 + \ldots + \alpha_d$. Multi-indices are used to shorten multiple partial derivatives $\partial^\alpha f := \partial_1^{\alpha_1} \partial_2^{\alpha_2} \ldots \partial_d^{\alpha_d} f$ of a sufficiently often differentiable and complex-valued function f, but also to denote the multiple powers $\xi^\alpha := \xi_1^{\alpha_1} \cdot \xi_2^{\alpha_2} \cdot \ldots \cdot \xi_d^{\alpha_d}$ of a vector $\xi = (\xi_1, \xi_2, \ldots, \xi_d) \in \mathbb{C}^d$.

By the usage of multi-indices, a *linear partial differential operator* $\mathcal{A}$ of even order $2m$ is defined by

$$\mathcal{A}(x, \cdot) := \sum_{|\alpha| \leq 2m} c_\alpha(x)\, \partial^\alpha \tag{0.1}$$

for $x \in \overline{G}$, the closure of a *domain* G, that is, an open and connected subset of an Euclidean space $\mathbb{R}^d$. This domain may be bounded or unbounded and even the whole of $\mathbb{R}^d$.

We shall call $\mathcal{A}$ *essentially real*, if the coefficients c_α of highest order $|\alpha| = 2m$ are real-valued. Moreover, the expression $\mathcal{A}$ is said to be an *operator with constant coefficients*, if all functions c_α are constant in $\overline{G}$.

§1. Free Space and Boundary Value Problems

In case of $G = \mathbb{R}^d$ respectively $\partial G = \emptyset$, where $\partial G := \overline{G} \setminus \mathring{G}$ is the boundary of G, we denote the linear partial differential equation $\mathcal{A}u = f$ with $\mathcal{A}$ given by (0.1) a *free space problem.* On the other hand, partial differential equations considered in domains with non-empty boundary are usually furnished with *boundary conditions*, which fix possible solutions and their normal derivatives up to a certain order in the boundary ∂G. Thus, when having in mind linear equations of the form $\mathcal{A}u = f$ with imposed boundary conditions, we shall speak of *linear boundary value problems.*

The most general form of a boundary value problem posed in a domain $G \subsetneq \mathbb{R}^d$ is of the form

$$\begin{cases} \mathcal{A}(x,u) \;=\; f(x) & (x \in G), \\ \mathcal{B}_k(x,u) \;=\; g_k(x) & (x \in \partial G \text{ and } k = 0,\ldots,m-1), \end{cases} \tag{1.1}$$

where $\mathcal{A}$ is a linear differential operator of order $2m$ and $\mathcal{B}_0,\ldots,\mathcal{B}_{m-1}$ are *boundary operators* of order m_k, each of them defined by

$$\mathcal{B}_k(x,u) \;=\; \sum_{|\alpha|\le m_k} b_{k,\alpha}(x)\,\partial^\alpha u(x).$$

To introduce the boundary operators $\mathcal{B}_k$, it is sufficent that the real or complex-valued functions $b_{k,\alpha}$ are defined in a neighbourhood of the boundary ∂G only. Then each operator $\mathcal{B}_k$ is a linear differential operator in its own right of order $m_k \le 2m-1$, which maps the function $u : \overline{G} \to \mathbb{C}$ to a function $\mathcal{B}_k u : \partial G \to \mathbb{C}$ belonging to a so-called *boundary space* $B_k(\partial G)$.

The collection $\{B_k\}_{k=0}^{m-1}$ of boundary operators in (1.1) is said to be *homogeneous* if $\mathcal{B}_k(x,\cdot) = 0$ for $x \in \partial G$ and $k = 0,\ldots,m-1$. Moreover, it is denoted *normal* if

(B-1) $m_i \ne m_j$ for $i \ne j$,

(B-2) for all $\xi \ne 0$ being normally positioned in $x \in \partial G$ we have

$$\sum_{|\alpha| = m_k} b_{k,\alpha}(x)\,\xi^\alpha \ne 0 \qquad (k = 0,\ldots,m-1).$$

Finally, our collection of boundary operators is called a *Dirichlet system*, if it is normal and the values m_k form a permutation of the set $\{0,\ldots,m-1\}$, so one may reorder the operators $\mathcal{B}_k$ to obtain $m_k = k$ $(0 \le k \le m-1)$. The most regarded set of boundary conditions in applications is the *homogeneous Dirichlet system* defined by

$$\mathcal{B}_k(x,u) \;:=\; (\partial_\nu^k u)(x) \;=\; 0 \tag{1.2}$$

for $x \in \partial G$ and $k = 0,\ldots,m-1$, where $(\partial_\nu^k u)(x)$ is the k-th derivative of a function u along the outer normal vector ν in $x \in \partial G$.

If the set of boundary conditions in (1.1) is a Dirichlet system, then we denote a solution $u : \overline{G} \to \mathbb{C}$ of this problem *classical*, if

(a) $u \in C^{2m}(G) \cap C^{m-1}(\overline{G})$,

(b) u fullfills $\mathcal{A}(x,u) = f(x)$ in each $x \in G$ and

(c) the $m-1$ equations $\mathcal{B}_k(x,u) = g_k(x)$ hold in every $x \in \partial G$.

Especially to each *second-order* boundary value problem there is assigned exactly one boundary operator $\mathcal{B}_0$, hence problem (1.1) becomes into

$$\begin{cases} \mathcal{A}(x,u) = f(x) & (x \in G), \\ \mathcal{B}_0(x,u) = g_0(x) & (x \in \partial G). \end{cases} \tag{1.3}$$

Boundary value problems of the form (1.3) and governed by a linear differential operator $\mathcal{A}$ of second order are obtained by various boundary operators $\mathcal{B}_0$. We denote

(BC-1) $\mathcal{B}_0(x,u) = u(x)$ for $x \in \partial G$ a *Dirichlet condition*,

(BC-2) $\mathcal{B}_0(x,u) = u_\nu(x)$ for $x \in \partial G$ a *Neumann condition*,

(BC-3) $\mathcal{B}_0(x,u) = \alpha(x)u(x) + \beta(x)u_\nu(x) = 1$ for $x \in \partial G$ a *Robin condition*, where the functions α, β must fulfill

$$\alpha(x)^2 + \beta(x)^2 \neq 0 \qquad (x \in \partial G).$$

Due to physical reasons, one often has to split the boundary set ∂G into disjoint subsets $\Gamma_1, \Gamma_2, \Gamma_3$, on which the conditions (BC-1), (BC-2) and (BC-3) are prescribed. We then shall speak of *mixed* boundary value problems for the second-order operator $\mathcal{A}$.

§2. Operators of Elliptic, Parabolic and Hyperbolic Type

The concept of *ellipticity* has been shown to be helpful in order to classify linear partial differential expressions by a simple algebraic criterion. Let $\mathcal{A}$ be a linear differential operator of order $2m$ be given by (0.1) and let

$$p(x,\xi) := \sum_{|\alpha|=2m} c_\alpha(x)\, \xi^\alpha \qquad \text{for } \xi = (\xi_1, \ldots, \xi_d) \in \mathbb{R}^d$$

be the *characteristic function* or the *main symbol* of $\mathcal{A}$, which is built up by the summands of highest order in (0.1).

(E-1) The expression $\mathcal{A}$ is called *elliptic* in $x \in \overline{G}$, if

$$|p(x,\xi)| \neq 0 \qquad \text{for all } \xi \neq 0,$$

and $\mathcal{A}$ is said to be *uniformly elliptic* in G (in $\overline{G}$), if it is elliptic in *each* point $x \in G$ (in $x \in \overline{G}$). We then call a boundary value problem $\mathcal{A}u = f$ *elliptic*, if $\mathcal{A}$ is so.

(E-2) $\mathcal{A}$ is called *properly elliptic* in $\overline{G}$, if it is elliptic and if for each fixed $x \in \partial G$ and all linear independent vectors $\xi, \eta \in \mathbb{R}^d$ the main symbol $P_{\mathcal{A}}(x, \xi + \tau\eta)$ has, with respect to the variable τ, exactly m roots with positive imaginary part and exactly m roots with negative imaginary part. It can be shown that if $\mathcal{A}$ is *essentially real* or $G \subset \mathbb{R}^d$ with $d \geq 3$, then any elliptic differential operator is also properly elliptic.

(E-3) The concept of ellipticity is often not good enough to obtain satisfying results. Hence the expression $\mathcal{A}$ of order $2m$ is called *strongly elliptic* in $x \in \overline{G}$, if there is $\theta_x > 0$ such that

$$\operatorname{Re}(-1)^m\, p(x,\xi) \;\geq\; \theta_x \cdot |\xi|^{2m} \qquad \text{for all } \xi \neq 0.$$

(E-4) $\mathcal{A}$ is called *uniformly strongly elliptic* in $\overline{G}$, if there is $\theta > 0$ such that

$$\operatorname{Re}(-1)^m\, p(x,\xi) \;\geq\; \theta \cdot |\xi|^{2m} \qquad \text{for all } \xi \neq 0 \text{ and } x \in \overline{G}.$$

The most well-known linear partial differential operator of second order is the *Laplacian*

$$\Delta = \partial^2/\partial {x_1}^2 + \ldots + \partial^2/\partial {x_d}^2,$$

whose negative version $-\Delta$ constitutes the prototype of what is meant by an uniformly strongly elliptic operator. One easily computes $\operatorname{Re} -p_{-\Delta}(x,\xi) = |\xi|^2$ for $\xi \in \mathbb{R}^d$, hence the expression $-\Delta$ often serves as a first example when studying linear-elliptic operators.

Besides the class of linear-elliptic operators there are two further important classes of linear partial differential operators. They are called to be of *parabolic* and *hyperbolic* type. In order to introduce these concepts, we consider differential expressions on cylindrical subsets $I \times G$, where $I = (0,\tau) \subset \mathbb{R}$ and $G \subset \mathbb{R}^d$ is a domain. Then a *parabolic* differential operator $\mathcal{A}$ is one of the form

$$\mathcal{A}(t,x,u) \;=\; \frac{\partial u}{\partial t}(x) \;+\; \mathcal{A}_1(x,u),$$

where $\mathcal{A}_1$ is *uniformly elliptic* in G, and $\mathcal{A}$ is of *hyperbolic* type, if it is of the form

$$\mathcal{A}(t,x,u) \;=\; \frac{\partial^2 u}{\partial t^2}(x) \;+\; \mathcal{A}_1(x,u),$$

where again $\mathcal{A}_1$ is uniformly elliptic in G. Simple examples of parabolic and hyperbolic operators are the *heat operator* $\partial_t - \Delta$ and the *wave operator* $\partial_{tt} - \Delta$, respectively. With these operators we associate the linear evolution equations

$$\frac{\partial u}{\partial t} + \mathcal{A}(\cdot, u) = f \quad \text{and} \quad \frac{\partial^2 u}{\partial t^2} + \mathcal{A}(\cdot, u) = f,$$

both of them called *initial value problems*, when considered with the prescribed start value $u(0) = u_0$ (and in addition $\dot{u}(0) = u_1$ in the hyperbolic case).

When considering parabolic or hyperbolic equations, we speak of so-called *initial-boundary value problems*, if for every time $t \in (0, T]$ the solution fulfills a certain set of boundary conditions in every $x \in \partial G$, where we especially have in mind the homogeneous Dirichlet boundary conditions (1.2). However, in case of $G = \mathbb{R}^d$, we obtain free space problems or pure initial value problems for the differential operators of parabolic or hyperbolic type.

§3. Functional Analytic Methods for Differential Operators

It is the aim of this part to apply all the presented functional analytic tools to the just enumerated problems. Let $\mathcal{A}$ be a linear partial differential operator of order $k \in \mathbb{N}$ (not necessarily even) and given by

$$\mathcal{A}(x, u) := \sum_{|\alpha| \le k} c_\alpha(x)\, \partial^\alpha u. \tag{3.1}$$

Then by a *realization* of $\mathcal{A}$ in a Banach space $\mathbb{E}(G)$ of functions $u : G \to \mathbb{C}$ is meant any linear operator

$$A : \mathfrak{D}(A) \subset \mathbb{E}(G) \to \mathbb{E}(G)$$

with domain of definition $\mathfrak{D}(A)$. The Banach space $\mathbb{E}(G)$ should contain the space $C_c^\infty(G)$ of test functions[8] as a dense linear subspace.

[8] The space $C_c^\infty(G)$ consists of indefinitely differentiable functions having *compact* support

$$\operatorname{supp}(\phi) := \overline{\{\, x \in G : \phi(x) \neq 0 \,\}},$$

usually denoted the *space of test functions*. This space plays an outstanding role in analysis, since it lies dense in a multiplicity of other functional spaces.

We also require for each $u \in \mathfrak{D}(A)$ that first $\mathcal{A}\,u$ is well-defined by $Au = \mathcal{A}\,u$ and second $\mathcal{A}\,u \in \mathbb{E}(G)$ holds. Clearly, our first condition means that the set $\mathfrak{D}(A)$ may contain only functions $u : G \to \mathbb{C}$ being sufficiently often differentiable in some sense. But $\mathfrak{D}(A)$ should also include the space $C_c^\infty(G)$ of test functions, so $\mathfrak{D}(A)$ lies dense in $\mathbb{E}(G)$ as well and A appears as a densely defined operator in $\mathbb{E}$.

In this section we want to point out the following facts:

- for a linear partial differential expression $\mathcal{A}$ there can be defined a couple of densely defined and closable linear operators A acting either in the Hölder spaces $C^{k,\gamma}(G)$ ($k \in \mathbb{N}_0$, $0 < \gamma \leq 1$) or in the Lebesgue spaces $L^p(G)$ ($1 \leq p < \infty$);
- using elliptic estimates, the realization of a strongly elliptic differentiable operator acting in Sobolev spaces under homogeneous Dirichlet boundary conditions is closed;
- the study on realizations of differential operators can be supported by that on their adjoints, where those are determined by so-called *adjointness theorems*;
- by starting with quasi-coercive sesquilinear forms, created by strongly elliptic differential expressions, we can obtain realizations in $L^2(G)$, which prove to be densely defined and closed;
- weak solutions of initial-boundary value problems exist and are unique, supposed the governing sesquilinear form is the Dirichlet form of a strongly elliptic differential operator;
- the realizations of a regularly elliptic expression $\mathcal{A}$ in $L^2(G)$ are Fredholm, so the alternative theorem may be applied to the equation $\mathcal{A}\,u = f$;
- strongly elliptic expressions give rise to m-accretive and sectorial realizations in $L^2(G)$, so the related parabolic and hyperbolic initial value problem are well-posed;
- formally self-adjoint and half-bounded differential expressions lead to self-adjoint realizations in $L^2(G)$;

This enumeration is of course only a small selection within the totality of applications of functional analytic methods to linear partial differential operators and related equations, but may give a good impression of all the capabilities.

§4. Realizations in Hölder and Lebesgue Spaces

We basically want to distinguish between two classes of functional spaces, in which realizations of a linear differential operator $\mathcal{A}$ are usually studied:

1. The notation of classical derivative $\partial^\alpha u$, applied to a function $u : G \to \mathbb{C}$, leads us to the usage of the spaces $C^\gamma(G)$ of *Hölder-constinuous* functions, which constitute Banach spaces when endowed with a suitable norm.

   A realization of a differential operator $\mathcal{A}$ of order $2m$ acting in Hölder space $C^\gamma(G)$ is nothing but a mapping

   $$A : \mathfrak{D}(A) \subset C^\gamma(G) \to C^\gamma(G),$$

   where $\mathfrak{D}(A)$ forms a linear subspace of $C^{2m,\gamma}(G)$ restricted by boundary conditions, for example by a Dirichlet system.

   There is an extensive theory on the realizations of linear partial differential operators acting in Hölder spaces, which provide nice results especially for operators of second order. We should mention J. SCHAUDER as the outstanding pioneer in the investigation of this subject. But SCHAUDER uses classical methods for the setup of his theory, coming from potential theory and complex analysis, so we do not want to go into the details (a good reference for this subject is [2]).

2. Instead of classical derivatives, the modern theory of partial differential equations is based on the concept of *weak derivative*, since this approach leads to larger spaces of differentiable functions than the already mentioned Hölder spaces, usually denoted as *Lebesgue-Sobolev spaces.*

   Recall that for any domain $G \subset \mathbb{R}^d$ the Hilbert space $L^2(G)$ is a complete normed space consisting of measurable functions $f : G \to \mathbb{C}$ such that

   $$\int_G |f|^2 \, dx < \infty,$$

   where the inner product in $L^2(G)$ is defined by

   $$\langle f, g \rangle := \int_G f \cdot \overline{g} \, dx \quad \text{for } f, g \in L^2(G).$$

This space is a specific member in the scale of the so-called *Lebesgue spaces* $L^p(G)$ $(1 \le p < \infty)$, whose norms are defined by

$$|f|_p := \Big(\int_G |f|^p \, dx\Big)^{\frac{1}{p}} < \infty$$

for equivalence classes of measurable functions $f \in L^p(G)$.

Now a realization of a linear differential operator $\mathcal{A}$ of order k in Lebesgue space $L^p(G)$ is any linear mapping

$$A : \mathfrak{D}(A) \subset L^p(G) \to L^p(G),$$

for which $\mathfrak{D}(A)$ forms a linear subspace of $W^{k,p}(G)$, for example restricted by homogeneous Dirichlet boundary conditions.

§5. The Closeness of Strongly Elliptic Dirichlet Operators

One often has to deal with the realization A_D of a strongly elliptic differential operator $\mathcal{A}$ in $L^p(G)$ having order $2m$ and domain of definition

$$\mathfrak{D}(A_D) := \{u \in W_0^{m,p}(G) : \mathcal{A}u \in L^p(G)\},$$

so the functions in $\mathfrak{D}(A_D)$ satisfy all the m boundary conditions of the homogeneous Dirichlet system (1.2). By this reason it is reasonable to denote the realization A_D a *Dirichlet operator.*

We turn to the question of closeness of the realization A_D. Let the domain G and the expression $\mathcal{A}$ fulfill the following conditions:

(I) G is bounded in $\mathbb{R}^d$,

(I') G satisfies the m-th *extension property*, that is, the linear space $R_0(G)$ consisting of those functions $\phi \in W^{m,p}(G)$, which are restrictions $\phi|_G$ of functions $\phi \in C_c^\infty(\mathbb{R}^d)$, is dense in $W^{m,p}(G)$;

(II) $\mathcal{A}$ is uniformly strongly elliptic in G;

(III) the coefficient functions $a_{\alpha\beta}$ are essentially bounded in G, that is, $a_{\alpha\beta} \in L^\infty(G)$;

(III') the coefficient functions $a_{\alpha\beta}$ of highest order $|\alpha| + |\beta| = 2m$ are uniformly continuous in the closure $\overline{G}$ of G.

Then, by showing the validness of the *elliptic estimate*

$$|u|_{2m} \le c \cdot \{\, |u|_p + |A_D u|_p \,\}$$

for $u \in \mathfrak{D}(A_D)$, where $|u|_p = \|u\|$ is the norm of u in $L^p(G)$, theorem II.2 implies that the operator A_D is closed. Moreover, A_D is injective and the range of A_D is a closed subspace in $L^p(G)$.

§6. Formally Adjoint Differential Expressions

As we have already mentioned, it is often helpful to combine the study of a linear operator with that of its formally adjoint. Supposed the linear partial differential operator $\mathcal{A}$ of order k has coefficients $c_\alpha \in C^{|\alpha|}(G)$. We then find for $u \in C_c^\infty(G)$ and $\psi \in C_c^\infty(G)$ by repeated partial integration

$$\int_G \sum_{|\alpha|\le k} (c_\alpha\, \partial^\alpha u) \cdot \overline{\psi}\, dx \;=\; \int_G u \cdot \sum_{|\alpha|\le k} (-1)^{|\alpha|}\, \overline{\partial^\alpha(\overline{c_\alpha}\, \psi)}\, dx.$$

Hence there is a *formally adjoint* differential expression

$$\mathcal{A}^\dagger(x,u) \;:=\; \sum_{|\alpha|\le k} (-1)^{|\alpha|} \partial^\alpha(\overline{c_\alpha(x)}\, u(x)) \tag{6.1}$$

such that the identity

$$\int_G \mathcal{A}\,\phi\,\overline{\psi}\, dx \;=\; \int_G \phi\, \overline{\mathcal{A}^\dagger\,\psi}\, dx$$

holds for $\phi, \psi \in C_c^\infty(G)$, briefly $\langle \mathcal{A}\,\phi, \psi\rangle = \langle \phi, \mathcal{A}^\dagger\,\psi\rangle$ by means of the inner product in $C_c^\infty(G)$. Using LEIBNIZ's rule to compute the terms $\partial^\alpha(\overline{c_\alpha}\, u)$ in (6.1), there are functions $c'_\alpha : G \to \mathbb{C}$ such that

$$\mathcal{A}^\dagger(x,u) \;=\; \sum_{|\alpha|\le k} c'_\alpha(x)\, \partial^\alpha u(x).$$

Thus $\mathcal{A}^\dagger$ is uniquely determined by $\mathcal{A}$ and constitutes a differential expression of type (3.1), too. It may happen that $\mathcal{A}^\dagger = \mathcal{A}$ holds, and in this case we shall call the expression $\mathcal{A}$ *formally self-adjoint.*

A useful criterion to ensure closability of a linear operator A acting in Hilbert space $L^2(G)$ and induced by a differential expression $\mathcal{A}$ of order k is given by the following theorem (see also [5], p. 78), whose proof uses the presence of the formally adjoint expression $\mathcal{A}^\dagger$:

THEOREM XII.1. Let the expression $\mathcal{A}$ of order n be given by (3.1) with coefficients $c_\alpha \in C^k(G)$ for $|\alpha| \le k$ and the linear operator

$$A: \begin{cases} \mathfrak{D}(A) := \{u \in C^k(G) \cap L^2(G) : \mathcal{A}u \in L^2(G)\}, \\ Au := \mathcal{A}u \;\text{ for } u \in \mathfrak{D}(A) \end{cases}$$

be a realization of $\mathcal{A}$ in $L^2(G)$. Then A is closable.

Note that the domain of definition $\mathfrak{D}(A)$ of the operator A considered in theorem XII.1 is not restricted to any boundary conditions. It should also be mentioned that rather the closability of the differential operator A with domain of definition $\mathfrak{D}(A)$ like in theorem XII.1 is ensured, it is not an easy task to give an exact description for the domain of definition of its closure. But if the coefficients of $\mathcal{A}$ are constant, then the set $\mathfrak{D}(\overline{A})$ can be determined and coincides with the Hilbert-Sobolev space $H^k(G)$.

The closure $A_{min} := \overline{A_0}$ of a closable linear differential operator A_0 with domain of definition $\mathfrak{D}(A_0) = C_c^\infty(G)$ is denoted the *minimal realization* or the *minimal operator* of $\mathcal{A}$. The *maximal operator* A_{max} of $\mathcal{A}$, however, is by definition the adjoint of the minimal operator of $\mathcal{A}^\dagger$ with domain of definition $\mathfrak{D}({A^\dagger}_0) = C_c^\infty(G)$, that is, $A_{max} := ({A^\dagger}_{min})^*$. One always has $A_{min} \prec A_{max}$, and often any operator A fulfilling

$$A_{min} \prec A \prec A_{max}$$

is called a *realization* of $\mathcal{A}$, in opposite to our selected definition of this concept. Using this definition, we also want to consider the linear operators A_{min} and A_{max} as realizations of $\mathcal{A}$, too.

To the end, the question arises whether any realization of the formally adjoint expression $\mathcal{A}^\dagger$ is in relationship with the adjoint of some realization of $\mathcal{A}$. This problem is examined in [1] (section 3: Adjoints) even in the context of L^p-spaces. BROWDER's result for Hilbert spaces $L^2(G)$ and the Dirichlet realization A_D of a regularly elliptic differential expression $\mathcal{A}$ sounds as follows (corresponding to theorem 5, see p. 57):

THEOREM XII.2. Let the $2m$-order expression $\mathcal{A}$ be strongly elliptic with sufficiently smooth coefficients and defined on a smooth domain $G \subset \mathbb{R}^d$ such that the formal adjoint $\mathcal{A}^\dagger$ exists and its coefficients are uniformly bounded in G.

If A and $A^\dagger$ are the realizations of $\mathcal{A}$ and $\mathcal{A}^\dagger$ in $L^2(G)$ with homogeneous Dirichlet boundary conditions and domains of definition

$$\mathfrak{D}(A) \;=\; \mathfrak{D}(A^\dagger) \;:=\; H^{2m}(G) \cap H_0^m(G),$$

then $A^\dagger$ is the Hilbert adjoint of A, that is, $A^* = A^\dagger$.

A similar result holds in the setting of the spaces $L^p(G)$ for $1 < p < \infty$. As it is quoted in [3], lemma 7.3.4, we have $A_p{}^* = A^\dagger{}_q$, where $q = p/(p-1)$ respectively $\frac{1}{p} + \frac{1}{q} = 1$ and both operators

$$\begin{cases} \mathfrak{D}(A_p) \;:=\; W^{2m,p}(G) \cap W_0^{m,p}(G), \\ A_p u \;:=\; \mathcal{A} u \text{ for } u \in \mathfrak{D}(A_p), \end{cases}$$

$$\begin{cases} \mathfrak{D}(A^\dagger{}_q) \;:=\; W^{2m,q}(G) \cap W_0^{m,q}(G), \\ A^\dagger{}_q u \;:=\; \mathcal{A}^\dagger u \text{ for } u \in \mathfrak{D}(A^\dagger{}_q). \end{cases}$$

are densely defined and closed.

§7. Gårding's Inequality

We next want to illustrate the usage of abstract sesquilinear forms for the analysis of linear equations of the form $\mathcal{A} u = f$, where $\mathcal{A}$ is a linear partial differential operator of order $2m$ and u, f are real or complex-valued mappings defined on a domain $G \subset \mathbb{R}^d$.

Preliminary, a linear differential expression $\mathcal{A}$ of order $2m$ is said to be in *divergence form*, if there are functions $a_{\alpha\beta} \in C^{|\alpha|}(G)$ for $|\alpha|, |\beta| \leq m$ such that $\mathcal{A}$ can be written as

$$\mathcal{A} \;:=\; \sum_{|\alpha| \leq m, |\beta| \leq m} (-1)^{|\alpha|}\, \partial^\alpha (\, a_{\alpha\beta}(x)\, D^\beta). \tag{7.1}$$

In opposite to this definition let us call an arbitary differential expression given by (3.1) to be *in non-divergence form.*

It is a matter of fact that every expression $\mathcal{A}$ in divergence form can be transformed into an expression $\mathcal{A}'$ in *non-divergence form.* Reversely, a non-divergence form expression $\mathcal{A}'$ given by (3.1) can be transformed into an expression $\mathcal{A}$ in divergence form (7.1), if $c_\alpha \in C^{2m-|\alpha|}(G)$ for $|\alpha| \leq 2m$, that is, if the coefficients are sufficiently many differentiable in G. Comparing the expressions (3.1) and (7.1), it turns out that $\mathcal{A}'$ is strongly elliptic in G if and only if $\mathcal{A}$ is so.

To show the purpose of a linear differential operator being in divergence form, let the space $C_c^\infty(G)$ of test functions be furnished with the inner product

$$\langle \phi, \psi \rangle := \int_G \phi \overline{\psi} \, dx \quad \text{for all } \phi, \psi \in C_c^\infty(G).$$

Now, applying $\mathcal{A}$ given in divergence form to $\phi \in C_c^\infty(G)$, then multiplying $\mathcal{A}\phi$ with $\overline{\psi} \in C_c^\infty(G)$ and finally performing m-times partial integration leads to

$$B_{\mathcal{A}}(\phi, \psi) := \langle \mathcal{A}\phi, \psi \rangle = \sum_{|\alpha| \leq m, |\beta| \leq m} \int_G a_{\alpha\beta} \, \partial^\alpha \phi \, D^\beta \overline{\psi} \, dx,$$

since all the boundary terms vanish. The mapping

$$B_{\mathcal{A}} := C_c^\infty(G) \times C_c^\infty(G) \to \mathbb{C}$$

is a sesquilinear form, called the *Dirichlet form* of $\mathcal{A}$ in G. If the coefficent functions $a_{\alpha\beta}$ of $\mathcal{A}$ are measurable and essentially bounded, then the mapping $B_{\mathcal{A}}$ is continuous and thus can continuously be extended to the product space $H_0^m(G) \times H_0^m(G)$.

We are mainly interested in a $2m$-order linear differential operator $\mathcal{A}$ given in divergence form, which together with the underlying domain G fulfills the already mentioned conditions (I), (I'), (II), (III) and (III'). GÅRDING*'s inequality* can be seen as a bridge from the theory of linear-elliptic differential operators to abstract functional analysis:

THEOREM XII.3. Let the domain G fulfill (I), (I') and let the linear differential expression $\mathcal{A}$ of order $2m$ fulfill (II), (III), (III').
Then there is $\lambda_0 \in \mathbb{R}$ and $a_0 > 0$ such that for all $u \in H_0^m(G)$

$$B_{\mathcal{A}}(u, u) + \lambda_0 \|u\|^2 = \langle (\mathcal{A} + \lambda_0) u, u \rangle \geq a_0 \, |u|_m{}^2. \tag{7.2}$$

In other words, theorem XII.3 quotes that the Dirichlet form associated with the operator $\mathcal{A} + \lambda_0$ is coercive, hence the Lax-Milgram theorem may be applied. It should be clear that this assertion also holds for all $\lambda > \lambda_0$.

The most important class of linear partial differential operators in divergence form with respect to applications is that of *second order*, that is, our operator $\mathcal{A}$ is of the form

$$\mathcal{A}(x,\cdot) = -\sum_{i=1}^{d} \partial_i \,(\sum_{j=1}^{d} a_{ij}(x)) + \sum_{j=1}^{d} (b_j(x)\,\partial_j) + c(x), \tag{7.3}$$

which is nothing but (7.1) for $m=1$. The Dirichlet form $B_{\mathcal{A}}$ belonging to $\mathcal{A}$ is the sesquilinear form

$$B_{\mathcal{A}}(u,v) \;=\; \int_G (\sum_{i,j=1}^{d} a_{ij}\,\partial_i u\,\partial_j \overline{v} \;+\; \sum_{j=1}^{d} b_j\,\partial_j u\,\overline{v} \;+\; c\,u\,\overline{v})\,dx.$$

A short computation shows that if the function c is large enough in comparison with the sum of partial derivatives of the functions b_j, then $B_{\mathcal{A}}$ is coercive. More precisely, $B_{\mathcal{A}}$ proves to be coercive if

$$c(x) \;\geq\; \frac{1}{2}\sum_{k=1}^{d} \frac{\partial b_k}{\partial x_k}(x) \tag{7.4}$$

holds in all points $x \in G$.

On the other hand, if the matrix $A(x) = (a_{ij}(x))_{i,j=1}^{d}$ of second-order coefficients in (7.3) fails to be positive definite for all $x \in G'$, where $G' \subset G$ has positive measure, then $B_{\mathcal{A}}$ cannot be coercive on $H_0^1(G)$. Also, if $A(x)$ is positive definite in all $x \in G$ and condition (7.4) does not hold. then the functions b_j disturb the coercivity of $B_{\mathcal{A}}$. But after all, the form $B_{\mathcal{A}}$ proves at least to be quasi-coercive, which is nothing but the assertion of GÅRDING's inequality.

GÅRDING's inequality remains even true for the Dirichlet forms of second-order differential operators, if the basic conditions (I)' and (III') are dropped:

THEOREM XII.4. Let $G \subset \mathbb{R}^d$ fulfill (I) and the differential operator $\mathcal{A}$ of second order and given in divergence form fulfill (II) and (III). Then there is $\lambda_0 \in \mathbb{R}$ and $a_0 > 0$ such that

$$B_{\mathcal{A}}(u,u) \;+\; \lambda_0\,\|u\|^2 \;=\; \langle\,(\mathcal{A}{+}\lambda_0)\,u,u\rangle \;\geq\; a_0\,|u|_1{}^2$$

for all $u \in H_0^1(G)$, where a_0 is the uniform lower bound of the eigenvalues of the matrices $A(x)$ with $x \in G$.

§8. Weak Solutions of Elliptic Boundary Value Problems

The variational approach to stationary and evolutional boundary value problems provides the concept of weak solution, and in this section we want to apply the abstract theory boundary value problems governed by strictly elliptic differential operators by quoting the related existence and uniqueness assertions.

We basically assume that

(DP-1) $\mathcal{A}$ is a the differential operator of order $2m$, strongly elliptic in a *bounded* domain $G \subset \mathbb{R}^d$ and given in divergence form;

(DP-2) the coefficients $a_{\alpha\beta}$ belong to $L^\infty(G)$ and the top-order coefficients are uniformly continuous in $\overline{G}$.

It is our first aim to solve the *homogeneous Dirichlet problem*

$$\begin{cases} (\mathcal{A}\,u)(x) \;=\; f(x) & (x \in G), \\ u(x) = \partial u/\partial\nu(x) = \ldots = \partial u^{m-1}/\partial\nu^{m-1}(x) = 0 & (x \in \partial G). \end{cases}$$

Let $V = H_0^m(G)$ and $H = L^2(G)$, hence the pair (V, H) is a Hilbert triple. By GÅRDING's inequality, the Dirichlet form

$$B_{\mathcal{A}}(u,v) \;:=\; \langle \mathcal{A}\,u, v\rangle \qquad (u, v \in V)$$

is quasi-coercive with some $\lambda_0 \in \mathbb{R}$. Then, by means of the Lax-Milgram theorem, we obtain the following *well-posedness theorem*:

THEOREM XII.5. For $\lambda \geq \lambda_0$, the variational equation

$$B_{\mathcal{A}}(u,v) \;+\; \lambda\langle u, v\rangle \;=\; f(v) \quad (v \in H_0^m(G))$$

owns for $f \in H^{-m}(G) \;:=\; H_0^m(G)^*$ a unique solution $u \in H_0^m(G)$ depending continuously on the right-hand side f and called the *weak solution* of the homogeneous Dirichlet problem.

§9. The Fredholm Property of Elliptic Operators

This statement is properly not correct, since it only proves to be true for properly elliptic differential operators defined in a *bounded* domain $G \subset \mathbb{R}^d$ and endowed with certain boundary conditions, for example those given by a homogeneous Dirichlet system.

We first want to present RELLICH*'s embedding theorem*, which holds for Sobolev spaces whose underlying domain is *bounded*. It can be seen as a key theorem in the analysis of linear-elliptic boundary problems on such domains:

THEOREM XII.6. If $G \subset \mathbb{R}^d$ is bounded and owns smooth boundary ∂G, then the continuous embeddings $\iota_k : H_0^k(G) \hookrightarrow L^2(G)$ are compact for all $k \in \mathbb{N}$.

The basic approach to investigate the Fredholm property of a properly elliptic operator $\mathcal{A}$ in divergence form and of order $2m$ consists of two small changes for the Dirichlet realization A_D having domain of definition $\mathfrak{D}(A) = H^{2m}(G) \cap H_0^m(G)$: let us consider the linear operator A_D as

(F-1) a mapping $A_D : H_0^m(G) \to L^2(G)$, which is continuous due to the essential boundedness of the coefficients $a_{\alpha,\beta}$ in (7.1);

(F-2) the sum $A_1 + A_2$ of two further linear operators, where we shall assume that $A_1 : H_0^m(G) \to L^2(G)$ is the main part having weak derivatives of order $2m$ only, whereas $A_2 : H_0^m(G) \to L^2(G)$ is the part of A_D consisting of weak derivatives with order less than $2m$.

Let λ_0 be the smallest parameter for the Dirichlet form $B_{A+\lambda_0}$ to be coercive. Since

$$A_D = (A_1 + \lambda_0) + (A_2 - \lambda_0)$$

and $A_2 - \lambda_0$ may be seen as a compact perturbation of $A_1 + \lambda_0$, which maps $H^m(G)$ one-to-one onto $L^2(G)$ and consequently is Fredholm with vanishing index, the operator A_D is Fredholm again with index $\chi(A_D) = 0$.

§10. Spectral Theory of Strongly Elliptic Operators

Let us consider again a strongly elliptic differential operator $\mathcal{A}$ given in divergence form and its Dirichlet realization A_D in a bounded domain $G \subset \mathbb{R}^d$. By Gårding's inequality, the Dirichlet form $B_{\mathcal{A}}$ is quasi-coercive with respect to the parameter λ_0, hence $A_D + \lambda$ proves to be bijective for all $\operatorname{Re}\lambda > \lambda_0$. Furthermore, the homogeneous Dirichlet problem in $L^2(G)$

$$A_D\, u + \lambda u = f \qquad (f \in L^2(G)) \tag{10.1}$$

is well-posed, which is a direct consequence of theorem XII.5.

In the special case, where $f \in L^2(G)$, the solutions u of (10.1) belong to the Hilbert-Sobolev space $H^{2m}(G)$ and are called *strong*. It should be clear that each strong solution of (10.1) is also a weak solution. But in order to show that these solutions u are even classical, that is, $u \in C^{2m}(G) \cap C^m(\overline{G})$, methods of *regularity theory* for elliptic problems have to be applied.

Let us turn to the spectrum of A_D. Since $-\Delta_D + \lambda$ forms a bijective and closed operator between the spaces $H^{2m}(G) \cap H_0^m(G)$ and $L^2(G)$ for each $\lambda \in C$ with $\operatorname{Re}\lambda \geq \lambda_0$, we surely have

$$\{\zeta \in \mathbb{C} : \operatorname{Re}\zeta \geq \lambda_0\} \subset \rho(A_D) \text{ and } \sigma(A_D) \subset \{\zeta \in \mathbb{C} : \operatorname{Re}\zeta < \lambda_0\}.$$

Furthermore, if G is bounded, then A_D is an operator with pure point spectrum.

We can also make assertions on the accretivity and sectorialness of the operator A_D. As theorem (XI.3) (d), (e) shows, $A_D + \lambda_0$ is the generator of a semigroup of bounded linear operators that is strongly continuous and analytic at the same time. Hence the initial value problems for the heat and wave equation governed by the operator $A_D + \lambda_0$ is well-posed. By perturbation theory of semigroups, this is also true for both problems governed by the operator A_D itself.

Let us finally consider the heat equation for the negative Laplacian $-\Delta_D$ in the entire space $\mathbb{R}^d$, which of course is a free space problem. Theorem X.7 ensures that the realization $-\Delta_D$ in Hilbert space $L^2(\mathbb{R}^d)$ with domain of definition

$$\mathfrak{D}(-\Delta_D) = H^2(\mathbb{R}^d)$$

is the infinitesimal generator of a sectorially contractive analytic semigroup $\{W(t)\}_{t\geq 0}$, denoted the *Gauss-Weierstrass semigroup*. Hence there is a uniquely defined solution $u \in C^1(I \times \mathbb{R}^d, \mathbb{R})$ for the homogeneous Cauchy problem

$$\begin{cases} u_t - \Delta u = 0, \\ u(x,0) = u_0(x) \text{ for } x \in \mathbb{R}^d \end{cases}$$

with initial mapping $u_0 \in H_0^2(\mathbb{R}^d)$. Then one may represent the semigroup $\{W(t)\}_{t\geq 0}$ explicitly by means of the *heat kernel* function Φ_d, that is, for all $t > 0$ and $x \in \mathbb{R}^d$

$$(W(t)\,u_0)\,(x) = \frac{1}{(4\pi\,t)^{d/2}} \int_{\mathbb{R}^d} e^{-\frac{|x-y|^2}{4t}}\,u_0(y)\,dy = (\Phi_d(t) * u_0)\,(x).$$

§11. Self-Adjoint Extensions of Half-Bounded Operators

Remember the definition of the differential expression $\mathcal{A}^\dagger$. We then denote a linear differential expression $\mathcal{A}$ *formally self-adjoint* in case of $\mathcal{A}^\dagger = \mathcal{A}$. If $\mathcal{A}$ has constant coefficients, then $\mathcal{A}$ proves to be formally self-adjoint iff $c_\alpha \in \mathbb{R}$ for $|\alpha|$ even and $c_\alpha \in i\,\mathbb{R}$ for $|\alpha|$ odd.

In the following let us collect a couple of formally self-adjoint linear differential expressions and describe some self-adjoint realizations of them:

(a) *Sturm-Liouville operators.* Our first example of a formally self-adjoint linear differential operator (given in divergence form) is the *Sturm-Liouville operator*

$$\mathcal{A}_{sl}(u,x) := -(p(x)\,u'(x))' + q(x)\,u(x), \qquad x \in (a,b)$$

with $p(x) > 0$ and $q(x) \geq 0$ for $x \in [a,b]$, $-\infty < a < b < \infty$. This operator determines the elongations of a vibrating string being fixed at the endpoints of $[a,b]$. Then the corresponding Dirichlet form reads as

$$B_{\mathcal{A}_{sl}}(u,v) = \int_a^b (p\,u'\,\overline{v}' + q\,u\,\overline{v})\,dx,$$

which shows that the expression $\mathcal{A}_{sl}$ is symmetric and half-bounded on the domain of definition $C_c^\infty(G)$. The energetic space $H_{\mathcal{A}_{sl}}$ is the Hilbert-Sobolev space $\overset{\circ}{H}{}^1_0(a,b)$ containing the restrictions of *absolutely continuous functions* $u : [a,b] \to \mathbb{R}$ with boundary conditions $u(a) = u(b) = 0$ to the interval (a,b).

The self-adjoint extension $A_{sl,F}$ of the Sturm-Liouville operator in $L^2(a,b)$ has domain of definition

$$\mathfrak{D}(A_{sl,F}) = H^2(a,b) \cap \overset{\circ}{H}{}^1_0(a,b).$$

The operator $A_{sl,F}$ has pure point spectrum, since the interval (a,b) is assumed to be bounded. Especially in case of $p(x) = 1$ and $q(x) = 0$ for $x \in [a,b]$, the eigenvalues and eigenfunctions of the operator $\mathcal{A}_{sl,F} = -\,d^2/dx^2$ can exactly be determined. In fact, the eigenvalue distribution of $\mathcal{A}_{sl,F}$ is given by

$$\lambda_n = \frac{n^2\pi^2}{(b-a)^2} \qquad (n \in \mathbb{N}),$$

so the asymptotical behaviour of the eigenvalues is $\lambda_n \sim n^2$. Each eigenvalue λ_n owns a one-dimensional eigenspace, and the eigenfunction u_n of λ_n is given by

$$u_n(x) := \sin\left(\frac{n\pi x}{b-a}\right) \quad \text{for } x \in [a,b].$$

Finally, the sequence of the eigenfunctions u_n constitutes after suitable renormations an orthonormal base in $L^2(a,b)$.

(b) *(Negative) Laplacian in various domains $G \subset \mathbb{R}^d$.* In order to apply the theory of self-adjoint operators to the investigation of the negative Laplacian, we start with the realization $A_{-\Delta,0}$ of the expression $-\Delta$ in the space $C_c^\infty(G)$ of test functions.

Using GREEN's second formula, and due to the boundary condition $\phi(x) = 0$ for $x \in \partial G$ and $\phi \in C_c^\infty(G)$, we compute for $\phi, \psi \in C_c^\infty(G)$

$$\int_G -\Delta\phi\,\overline{\psi}\,dx = \sum_{i=1}^n \int_G \frac{\partial^2\phi}{\partial x_i^2}\,\overline{\psi}\,dx$$

$$= \sum_{i=1}^n \int_G u\,\overline{\frac{\partial^2\phi}{\partial x_i^2}}\,dx = \int_G \phi\,\overline{\Delta\psi}\,dx,$$

which implies symmetry of the operator $A_{-\Delta,0}$. Moreover, we obtain for $\phi \in C_c^\infty(G)$ the inequality

$$\langle -\Delta\phi, \phi\rangle = \sum_{i=1}^n \int_G \frac{\partial^2\phi}{\partial x_i^2}\,\overline{\phi}\,dx = \sum_{i=1}^n \int_G \left|\frac{\partial\phi}{\partial x_i}\right|^2 dx \geq 0,$$

hence $A_{-\Delta,0}$ is half-bounded with lower bound 0 and fulfills the prerequisites to apply the Friedrichs extension method. Thus we may quote that the operator $A_{-\Delta,0}$ owns at least one self-adjoint extension, and this assertion is true actually for arbitrary domains $G \subset \mathbb{R}^d$.

In the following enumeration we consider various self-adjoint realizations of the negative Laplacian in a domain G, all of them being extensions of the operator $A_{-\Delta,0}$:

(i) *(Negative) Laplacian in* $\mathbb{R}^d$. The analysis of the operator $-\Delta$ in the entire Euclidean space $\mathbb{R}^d$ is easy, since the differential operator proves to be unitarily equivalent to the multiplication operator A_m generated by the function

$$m:(x_1,\ldots,x_n)\mapsto(x_1^2+\ldots+x_n^2) \;=\; |x|^2 \quad (x\in\mathbb{R}^d).$$

In fact, the operator A_m is real-valued and self-adjoint, which is a consequence of the *Fourier-Plancherel transformation*[9]. This implies that the operator A_{sa} with domain of definition $\mathfrak{D}(A_{sa}) := H^2(\mathbb{R}^d)$ must be self-adjoint, too. The spectrum of A_{sa} is that of the operator A_m, that is, it is purely continuous and $\sigma_c(A_{sa})=[0,\infty)$. Moreover, since A_{sa} is the closure of the operator $A_{-\Delta,0}$, the latter proves to be essentially self-adjoint.

(ii) *Dirichlet-Laplacian.* Let the domain $G\subset\mathbb{R}^d$ be *bounded* and consider $-\Delta$ with domain of definition $\mathfrak{D}(-\Delta) := C_c^\infty(G)$. The related Dirichlet form $B_{-\Delta}$ is given by

$$B_{-\Delta}(\phi,\psi) \;:=\; \int_G \nabla\phi\,\nabla\psi\,dx \qquad (\phi,\psi\in C_c^\infty(G)),$$

which can continuously be extended to the sesquilinear form $\overline{B_{-\Delta}}$ with domain of definition $\mathfrak{D}(\overline{B_{-\Delta}}) := H_0^1(G)\times H_0^1(G)$.

Due to theorem XI.3, the operator A_D associated with the form $B_{-\Delta}$ and defined by

$$\left\{\begin{array}{lcl} \mathfrak{D}(A_D) &:=& \{u\in H_0^1(G):\Delta u\in L^2(G)\} \\ A_D u &:=& -\Delta u \;\text{ for } u\in\mathfrak{D}(A_D) \end{array}\right.$$

[9] We introduce the *Fourier-Plancherel transformation* in $L^2(\mathbb{R}^d)$ by

$$\hat{u}(\xi) \;:=\; \frac{1}{(2\pi)^{d/2}}\int_{\mathbb{R}^d} e^{-2\pi i\langle\xi,x\rangle}\,u(x)\,dx$$

for all $u\in C_c^\infty(\mathbb{R}^d)$, which forms a partial isometry $\mathscr{F}:u\mapsto\hat{u}$ from the space $C_c^\infty(\mathbb{R}^d)$ to the space $L^2(\mathbb{R}^d)$. Then for functions $u,v\in L^2(\mathbb{R}^d)$ the inner product between u and v and the inner product of their Fourier-Plancherel transforms $\hat{u}$ and $\hat{v}$ are related by PLANCHEREL*'s formula* $\langle u,v\rangle=\langle\hat{u},\hat{v}\rangle$. Since the mapping $\mathscr{F}$ is linear and continuous on $C_c^\infty(\mathbb{R}^d)$, which is a dense subspace of $L^2(\mathbb{R}^d)$, it can be extended to an *unitary*, that is, a bijective and isometric linear operator $\mathscr{F}_P: L^2(\mathbb{R}^d)\to L^2(\mathbb{R}^d)$, called the *Fourier-Plancherel operator* or the L^2*-Fourier transformation* in $L^2(\mathbb{R}^d)$.

is self-adjoint, called the *negative Laplacian with Dirichlet boundary conditions* or briefly the *Dirichlet-Laplacian.*

We obviously have $H_0^1(G) \cap H^2(G) \subset \mathfrak{D}(A_D)$, and one can prove that if G is bounded and owns a C^2-boundary or $G = \mathbb{R}^n_+$, then the reversed inclusion also holds, hence $\mathfrak{D}(A_D) = H_0^1(G) \cap H^2(G)$. This equality is even true in the case, where G is bounded and convex.

Furthermore, if G has the 1-extension property, then the space $H_0^1(G)$ is compactly embedded into the space $L^2(G)$ and A_D is an operator with pure point spectrum. In this case it is an important property of the Dirichlet-Laplacian that the number 0 does *not* belong to the spectrum $\sigma(A_D)$, so A_D is invertible and the operator A_D^{-1} proves to be bounded.

(iii) *Neumann-Laplacian.* Let $G \subset \mathbb{R}^d$ be a bounded domain with C^1-boundary ∂G and let the realization A_N of the negative Laplacian in $L^2(G)$ be defined by

$$\begin{cases} \mathfrak{D}(A_N) := \{u \in H^2(G) : u_v(x) = 0,\ x \in \partial G\} \\ A_N u := -\Delta u \ \text{ for } u \in \mathfrak{D}(A_N). \end{cases}$$

The realization A_N is said to be the *(negative) Laplacian with Neumann boundary condition* or briefly the *Neumann-Laplacian.* It owns $\lambda = 0$ as the minimal eigenvalue, and all constant functions in $\mathfrak{D}(A_N)$ are eigenfunctions of $\lambda = 0$.

(c) *Schrödinger operators.* By the *Schrödinger operator* is meant the linear partial differential expression given by

$$\mathcal{H}_q(u, x) := (-\Delta u)(x) + q(x),$$

defined in the entire Euclidean space $\mathbb{R}^d$. In the special case, where $q = 0$, self-adjointness of the operator $-\Delta$ acting in $L^2(\mathbb{R}^d)$ and with domain of definition $H^2(\mathbb{R}^d)$ is well-known.

The so-called *potential function* $q : \mathbb{R}^d \to \mathbb{R}$ gives rise to a multiplication operator M_q acting in $L^2(\mathbb{R}^d)$, that is,

$$\begin{cases} \mathfrak{D}(M_q) := \{u \in L^2(\mathbb{R}^d) : q \cdot u \in L^2(\mathbb{R}^d)\}, \\ M_q u := q \cdot u \ \text{ for } u \in \mathfrak{D}(M_q), \end{cases}$$

so the operator $\mathcal{H}_q$ can be seen as a perturbation of the negative Laplacian by the operator M_q.

By the time-dependent *Schrödinger equation* is meant the initial value problem

$$i\,\dot{u} \;=\; -\Delta u \,+\, M_q u$$

with prescribed initial value $u(0) = u_0 \in H^2(\mathbb{R}^d)$, where it is supposed that $\mathcal{H}_q = -\Delta + M_q$ is a self-adjoint operator in $L^2(\mathbb{R}^d)$ with domain of definition $\mathfrak{D}(\mathcal{H}_q) = H^2(\mathbb{R}^d)$. The existence and uniqueness of a solution for this problem is ensured by the functional calculus for self-adjoint operators using the spectral resolution of the operator $\mathcal{H}_q$ and the function $f : \lambda \in \mathbb{R} \mapsto e^{it\lambda}$. The solution u of the problem is then provided by a group of unitary operators $\{U(t)\}_{t\in\mathbb{R}}$, which implies $u(t) = U(t)u_0$ for every $t \in \mathbb{R}$ (see also section IV.8 for the treatment of the abstract Schrödinger equation).

The study of *time-independent solutions* of the Schrödinger equation leads to the eigenvalue problem

$$\mathcal{H}_q u \;:=\; -\Delta u + M_q u \;=\; \lambda u, \tag{11.1}$$

which means to find those real numbers λ, for which there are non-trivial solutions of (11.1). The discrete spectrum $\sigma_d(\mathcal{H}_q)$ of the operator $\mathcal{H}_q$ is in fact of great interest, since the eigenfunctions of eigenvalues $\lambda \in \sigma_d(\mathcal{H}_q)$ describe the stationary states of a quantum mechanical system.

The presence of a potential function q can fairly change the spectrum of a Schrödinger operator like the example of the *harmonic oscillator* shows: the spectrum of the negative Laplacian is the connected set $[0, \infty)$, but the spectrum of the operator $-\Delta + M_q$ with $q : x \mapsto \frac{1}{2}x^2$ consists of discrete points only.

It is a further and interesting problem in quantum mechanics to determine lower and upper bounds for the discrete spectrum of the operator $\mathcal{H}_q$. In the following we want to present three assertions, which contain information on the bounds of the set $\sigma_d(\mathcal{H}_q)$, namely the *Birman-Schwinger bound*, the *Lieb-Cwikel-Rozenblum bound* and Kato*'s theorem*:

1. The *Birman-Schwinger bound.* This assertion presents an upper bound for the set $\sigma_d(\mathcal{H}_q)$, which is valid for certain potentials q in case of $d = 3$. Let $q \in L^\infty(\mathbb{R}^3)$ and

$$B_q := \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \frac{q(x)\, q(y)}{|x-y|^2}\, dx\, dy < +\infty.$$

Then the total number $N(\mathcal{H}_q)$ of eigenvalues of the operator $\mathcal{H}_q$ can be estimated according to

$$N(\mathcal{H}_q) \le \frac{1}{16\pi^2} B_q.$$

2. The *Lieb-Cwikel-Rozenblum theorem* quotes an estimate on the number $N(\mathcal{H}_q)$ of negative eigenvalues of the operator $\mathcal{H}_q$: if $d > 3$ and $q \in L^\infty_{loc}(\mathbb{R}^d)$, then

$$N(\mathcal{H}_q) \le c_d \int_{\mathbb{R}^d} \min\{q(x), 0\}^{n/2}\, dx,$$

where c_d is a constant only depending on the dimension d.
3. Our last result is often denoted as *Kato's theorem.* It does not make assertions on bounds for the discrete spectrum, but quotes the absence of positive eigenvalues under mild assumptions on the function q: if $q \in L^\infty_{loc}(\mathbb{R}^d)$ and

$$\lim_{|x|\to\infty} |x|\, q(x) = 0,$$

then the operator $\mathcal{H}_q$ does not own any positive eigenvalue.

§12. Eigenvalue Problems in Bounded Domains

We now turn to the eigenvalue equation $\mathcal{A}u - \lambda u = f$ of a strongly elliptic differential operator $\mathcal{A}$ of second order in a *bounded* domain $G \subset \mathbb{R}^d$ with homogeneous Dirichlet boundary conditions $u(x) = 0$ for $x \in \partial G$.

Sine the realization A of $\mathcal{A}$ in Hilbert space $L^2(G)$ is an operator with pure point spectrum, we may conclude that the eigenvalue problem is well-posed for every $\lambda \in \mathbb{R}$ up to countable many isolated points (λ_k) with $|\lambda_k| \to \infty$ for $k \to \infty$, and for these points the space of solutions of the homogeneous equation $\mathcal{A}u = \lambda_k u$ is finite-dimensional.

The mentioned facts will be illustrated for the negative Laplacian $-\Delta$ in bounded domains and under homogeneous Dirichlet boundary conditions. Unfortunately, the eigenvalues and eigenfunctions can exactly be determined only in the case, where the shape of G is of elementary type:

1. If G is a finite interval $(a,b) \subset \mathbb{R}$, then the homogeneous boundary conditions are $u(a)=u(b)=0$. In this case the eigenvalue distribution is given by
$$\lambda_n = \frac{n^2\pi^2}{(b-a)^2} \qquad (n \in \mathbb{N}),$$
and the related eigenfunctions are the mappings
$$u_n(x) := \sin\left(\frac{n\pi x}{b-a}\right) \text{ for } x \in (a,b)$$
forming an orthonormal base in $L^2(a,b)$. Thus, the smallest eigenvalue is the number $\pi^2/(b-a)^2$, and the asymptotical behaviour of the eigenvalues is $\lambda_n \sim n^2$. Notice that for each eigenvalue λ_n the eigenspace of $-\Delta - \lambda_n$ is 1-dimensional.
   The Sturm-Liouville operator $\mathcal{L}\,u = -(pu')' + qu$ with domain of definition $H^2(a,b) \cap H_0^1(a,b)$ also forms an operator with pure point spectrum, supposed (a,b) is a bounded subset of the real line. Especially for $p(x)=1$ and $q(x)=0$ for $x \in (a,b)$ we obtain the negative Laplacian $-d^2/dx^2$.

2. In case of $G=(0,a)\times(0,b)$ we obtain similar results. In fact, the separation approach $u(x,y)=u_1(x)\cdot u_2(y)$ for the eigenvalue problem $-\Delta u=\lambda u$ leads to separated differential equations for u_1 and u_2 with solution functions
$$u_{1,n}(x) = \sin\left(\frac{n\pi x}{a}\right), \qquad u_{2,k}(x) = \sin\left(\frac{k\pi y}{b}\right),$$
eigenvalues $\lambda_{n,k} = \frac{n^2\pi^2}{a^2}+\frac{k^2\pi^2}{b^2}$ and related eigenfunctions $u_{1,n}\cdot u_{2,k}$.

3. Finally, the eigenvalue problem for the $2d$-circle reads as
$$G = \{\,(r,\theta) : 0 \le r < 1, 0 \le \theta \le 2\pi\,\}.$$
It is appropriate to transform the eigenvalue equation into an equation with respect to polar coordinates (r,θ)
$$v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta} + \lambda v = 0\,,$$

and boundary conditions $v(1,\theta)=0$ for $0\le\theta\le 2\pi$. The separation approach $v(r,\theta)=R(r)\cdot\Theta(\theta)$ leads to encoupled differential equations for R and Θ:

$$\begin{cases} r^2R''(r) + rR'(r) + (\lambda r^2-\sigma)R(r) = 0\,, \\ \Theta''(\theta) + \sigma\,\Theta(\theta) = 0\,, \end{cases}$$

where the second one is solved by $\Theta_n(\theta)=a\cos(n\theta)+b\sin(n\theta)$ for $a,b\in\mathbb{R}$. The index $n:=\sqrt{\sigma}$ can only be of integer type, due to the 2π-periodicity of the function Θ. Moreover, the solutions $R_n(r)$ of the first differential equation are the *Bessel functions*

$$J_n(\sqrt{\lambda}r) := \frac{\lambda^n r^n}{2^n}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!(n+k)!}\,\frac{\lambda^k r^{2k}}{4^k}.$$

The eigenvalues of this problem coincide with the squares of the infinitely many zero points $\mu_{n,1},\mu_{n,2},\ldots$ of the functions J_n $(n\in\mathbb{N}_0)$ having the eigenfunctions

$$v_{n,m}(r,\theta) = J_n(\mu_{n,m}\cdot r)\cdot\{\,a\cos(n\theta) + b\sin(n\theta)\,\}.$$

§13. Linear Differential Operators in Distributional Spaces

We next study the action of linear partial differential operators in distributional spaces. First recall that if $\psi\in C^\infty(G)$ and $F\in\mathscr{D}'(G)$, then ψF defined by

$$(\psi F)\phi := F(\psi\phi), \qquad \phi\in C_c^\infty(G)$$

also belongs to $\mathscr{D}'(G)$. This implies that the linear differential operators $c_\alpha\,\partial^\alpha$ with $c_\alpha\in C^\infty(G)$ are acting on the space $\mathscr{D}'(G)$, since for $F\in\mathscr{D}'(G)$ the mapping $c_\alpha\partial^\alpha F$ is a distribution again. By taking linear combinations of such operators, we conclude that the action of a linear partial differential operator

$$\mathcal{A}(x,u) := \sum_{|\alpha|\le 2m} c_\alpha(x)\,\partial^\alpha u(x)$$

having smooth coefficients and acting on the space $\mathscr{D}'(G)$ is well-defined, which of course is especially true in case of $G=\mathbb{R}^d$.

In the following we introduce the concept of *fundamental solution* for a linear partial differential operators $\mathcal{A}$ having smooth coefficients c_α. It is nothing but a distribution $\Phi_{\mathcal{A}} \in \mathscr{D}'(\mathbb{R}^d)$ that forms a solution of the differential equation $\mathcal{A}\Phi_{\mathcal{A}} = \delta_0$, where δ_0 is the Dirac distribution with respect to the origin $x = 0$.

It is useful to know a fundamental solution of $\mathcal{A}$ as the following computation shows: let $F \in \mathscr{D}'(\mathbb{R}^d)$ be any Schwartz distribution and consider the inhomogeneous differential equation $\mathcal{A}\,U = F$. If convolution between $\Phi_{\mathcal{A}}$ and F exists, then the equalities

$$\mathcal{A}(\Phi_{\mathcal{A}} * F) \;=\; (\mathcal{A}\Phi_{\mathcal{A}}) * F \;=\; \delta_0 * F \;=\; F$$

are valid, so the convolutional product $\Phi_{\mathcal{A}} * F$ solves the distributional differential equation $\mathcal{A}\,U = F$. Notice that there is a distributional solution U of this equation especially for all distributions F having compact support.

The fundamental solution of a linear differential operator need not be unique, which can be showed by the following example: let $U \in \mathscr{D}'(\mathbb{R}^d)$ be a solution of the homogeneous equation $\mathcal{A}\,U = 0$, then the distribution $\Phi_{\mathcal{A}} + U$ is a fundamental solution of $\mathcal{A}$, too.

Regarding the existence of fundamental solutions for linear differential operators, the MALGRANGE-EHRENPREIS *theorem* has been shown to be a milestone in PDE theory:

THEOREM XII.7. Each linear differential operator $\mathcal{A} \neq 0$ with constant coefficients has a fundamental solution in the distributional space $\mathscr{D}'(\mathbb{R}^d)$.

It should be mentioned that there are counterexamples which demonstrate that this theorem is not valid for differential operators whose coefficients are non-constant.

In the following we describe the fundamental solutions of three linear differential operators, namely the *Laplace operator*, the *heat operator* and the *wave operator*.

1. The fundamental solution Φ^d_Δ of the Laplacian Δ depends on the number of spatial coordinates. Since Δ is invariant with respect to translations and rotations, it is nearby to conjecture $\Phi^d_\Delta(x) = \phi_d(|x|)$. Indeed, the functions

$$\Phi_\Delta(x) = \begin{cases} \frac{1}{2}|x| & (d=1) \\ -\frac{1}{2\pi}\ln|x| & (d=2) \\ \frac{1}{(d-2)\,\omega_d}|x|^{2-d} & (d\geq 3) \end{cases}$$

are solutions of the differential equations $\Delta\Phi_\Delta^d = \delta_0$ for $d \in \mathbb{N}$, where ω_d is the Lebesgue-Borel measure of the d-dimensional unit sphere. Each other fundamental solution of the Laplacian is of the form $\Phi_\Delta^d + P_1$, where P_1 is any polynomial of degree less than two.

2. The fundamental solution of the *heat operator* $\mathcal{A} := \partial_t - \Delta u$ in $\mathbb{R}^d$ is given by the so-called *heat kernel* function

$$\Phi_d(x,t) = \begin{cases} \frac{1}{(4\pi t)^{d/2}} \cdot e^{-\frac{|x|^2}{4t}}, & x \in \mathbb{R}^d,\ t > 0, \\ 0, & x \in \mathbb{R}^d,\ t \leq 0. \end{cases}$$

Each other fundamental solution of the heat operator adopts the form $\Phi_d + c$ $(c \in \mathbb{C})$, so the kernel of the heat operator purely consists of constant distributions.

3. When investigating the fundamental solutions of the hyperbolic *wave operator* $\square := \partial_{tt} - \Delta u$, one has to distinguish between even and odd spatial dimensions.

Let H be the *Heaviside function*, defined by $H(t) := 0$ for $t < 0$ and $H(t) := 1$ for $t \geq 0$. Then we obtain for $d \leq 3$

$$\Phi_\square^d(x,t) = \begin{cases} \frac{1}{2}H(t-|x|) & (d=1) \\ \frac{1}{2\pi}\,(t^2-|x|^2)^{-1/2}\,H(t-|x|) & (d=2) \\ \frac{1}{2\pi}\,\delta(t^2-|x|^2)\,H(t) & (d=3). \end{cases}$$

It is a notable fact that all the higher fundamental solutions of the wave operator can be derived from the functions $\Phi_\square^1$ and $\Phi_\square^2$. Using the variable $r := |x|$, the functions $\Phi_\square^{d+2}$ $(d \in \mathbb{N})$ are recursively given by

$$\Phi_\square^{d+2}(t,r) = \frac{-1}{2\pi r} \cdot \frac{\partial}{\partial_r}\Phi_\square^d(t,r), \quad d \geq 1.$$

To determine the support of a fundamental solution $\Phi_{\square}^d$, we first define for fixed $x_0 \in \mathbb{R}^d$ the *light cone*

$$C^+(x_0) \;:=\; \{(t,x) : t>0, |x-x_0|<t\}$$

and obtain the support of the fundamental solution according to

$$\operatorname{supp}(\Phi_{\square}^d) \;=\; \begin{cases} \partial C^+(0), & d \text{ odd or } d \geq 3, \\ \overline{C^+(0)}, & d \text{ even or } d=1. \end{cases}$$

A linear differential operator $\mathcal{A}$ is called *hypoelliptic*, if with respect to the linear differential equation $\mathcal{A}u = f$ the condition $f \in C^\infty(G)$ implies $u \in C^\infty(G)$. Restricting to differential operators with constant coefficients, hypoellipticity can be characterized by means of a feature of the fundamental solution $\Phi_{\mathcal{A}_c}$: $\mathcal{A}_c$ is hypoelliptic if and only if $\Phi_{\mathcal{A}_c} = F_f$, where F_f is a regular distribution with generating function f being indefinitely differentiable in every $x \neq 0$.

We next introduce the concept of *singular support* $\Sigma(F)$ for $F \in \mathscr{D}'(\mathbb{R}^d)$: it is the smallest subset $S \in \mathbb{R}^d$ such that F remains regular on $\mathbb{R}^d \backslash S$. Then hypoellipticity of linear differential operators with constant coefficients are characterized as follows:

THEOREM XII.8. A linear partial differential operator $\mathcal{A}_c$ with constant coefficients is hypoelliptic if and only if $\Sigma(\mathcal{A}_c F) = \Sigma(F)$ for all $F \in \mathscr{D}'(\mathbb{R}^d)$.

Knowing the fundamental solutions of the elliptic operator Δ and the parabolic heat operator $\partial_t - \Delta$, it turns out that both operators are hypoelliptic. But theorem XII.8 implies that the hyperbolic wave operator $\square = \partial_{tt} - \Delta$ is not hypoelliptic.

§14. Free Space Problems

To show the benefit of knowing the fundamental solution of a linear partial differential operator with constant coefficients, we consider free space problems ($G = \mathbb{R}^d$) related to the Laplacian, the heat and the wave operator.

1. The *Poisson equation* in the entire space $\mathbb{R}^d$ is given by

$$\Delta u \;=\; -4\pi f, \quad f \in C(\mathbb{R}^d). \tag{14.1}$$

A function $u:\mathbb{R}^d \to \mathbb{R}$ is called a *classical solution* of (14.1), if it is twice continuously differentiable and fulfills the differential equation in each $x \in \mathbb{R}^d$. However, $U \in \mathscr{D}'(\mathbb{R}^d)$ is said to be a *distributional solution* of the Poisson equation, if

$$\Delta U = -4\pi F, \quad F \in \mathscr{D}'(\mathbb{R}^d) \tag{14.2}$$

holds in the sense of distributions.

Let Φ_Δ be the fundamental solution of the Laplacian. If the convolutional product $\Phi_\Delta * F$ exists, which is especially true for each regular distribution F_f with generating function $f \in C_c(\mathbb{R}^d)$, then the solution of (14.2) is obtained by convolution, that is, $U = \Phi_\Delta * F$. For the physically interesting case $d = 3$ and for $f \in C_c(\mathbb{R}^3)$ we get a classical solution u_f, which is denoted the *Newton potential* for the function f:

$$u_f(x) = \int_{\mathbb{R}^3} \frac{f(y)}{|x-y|}\, dy, \quad x \in \mathbb{R}^3.$$

2. By the *heat equation* equation in $\mathbb{R}^d$ is meant the inital value problem

$$\begin{cases} u_t(t,x) - \Delta u(t,x) = f(t,x) & (x \in \mathbb{R}^d,\ t > 0) \\ u(0,x) = 0, & (x \in \mathbb{R}^d). \end{cases} \tag{14.3}$$

We first define the concept of distributional solution for problem (14.3): let $U_0 \in \mathscr{D}'(\mathbb{R}^d)$ and $F \in \mathscr{D}'(\mathbb{R}^{d+1})$ are Schwartz distributions satisfying $\operatorname{supp}(F) \subset [0,\infty) \times \mathbb{R}^d$, then $U \in \mathscr{D}'(\mathbb{R}^{d+1})$ is called a *distributional solution*, if $\operatorname{supp}(U) \subset [0,\infty) \times \mathbb{R}^d$ and

$$U_t - \Delta U = F + U_0 \otimes \delta_0, \tag{14.4}$$

where $\delta_0 \in \mathscr{D}'(\mathbb{R})$ is the Dirac distribution. Here the initial condition of the problem has been transformed into an inhomogeneous part of the differential equation.

The existence of a solution for (14.4) is ensured, if $U_0 \in \mathcal{E}'(\mathbb{R}^d)$ and $F \in \mathcal{E}'(\mathbb{R}^{d+1})$. In this case the solution $U \in \mathscr{D}'(\mathbb{R}^d)$ is given by convolution of the fundamental equation Φ_d with the inhomogeneous parts of (14.4),

$$U = \Phi_d * (F + U_0 \otimes \delta_0) = \Phi_d * F + \Phi_d * U_0.$$

Given a classical solution u of (14.3), the related Schwartz distribution F_u is a solution in the distributional sense. Conversely, let F_u be regular and solve (14.4), then u forms a classical solution, if some regularity conditions are valid:

a. U, U_0 and F are regular distributions with associated functions $u, u_0, f \in L^1_{loc}(G)$,
b. u_0 is continuous on $\mathbb{R}^d$ and f is continuous on $[0,\infty) \times \mathbb{R}^d$,
c. u is twice continuously differentiable on $[0,\infty) \times \mathbb{R}^d$.

3. The initial-value problem for the *wave equation* in $\mathbb{R}^d$ is given by

$$\begin{cases} u_{tt}(t,x) - \Delta u(t,x) \;=\; f(t,x) & (x \in \mathbb{R}^d,\, t>0) \\ u(0,x) = u_0(x) & (x \in \mathbb{R}^d) \\ u_t(0,x) = u_1(x) & (x \in \mathbb{R}^d. \end{cases} \tag{14.5}$$

As for the heat equation, we define for (14.5) the concept of distributional solution, where again both initial value conditions are transformed into inhomogeneous parts of the equation by means of tensor products.

Let $U_0 \in \mathscr{D}'(\mathbb{R}^d)$, $U_1 \in \mathscr{D}'(\mathbb{R}^d)$ and $F \in \mathscr{D}'(\mathbb{R}^{d+1})$ be distributions, then U is called a distributional solution of (14.5), if

$$U_{tt} - \Delta U \;=\; F + U_0 \otimes \delta_0' \;+\; U_1 \otimes \delta_0\,, \tag{14.6}$$

where δ_0 is the Dirac distribution for $t=0$ and δ_0' is the distributional derivative of δ_0 with respect to the variable t.

Since the support of the function $U_0 \otimes \delta_0' + U_1 \otimes \delta_0$ belongs to the set $\mathbb{R}^{d+1}$, there is to each distribution $F \in \mathscr{D}'(\mathbb{R}^{d+1})$ with support

$$\operatorname{supp}(F) \;=\; \{\,(x,t) \in \mathbb{R}^d \times \mathbb{R} : t \geq 0\,\}$$

a further distribution $U_F := \Phi^d_\square * (F + U_0 \otimes \delta_0' + U_1 \otimes \delta_0) \in \mathscr{D}'(\mathbb{R}^{d+1})$, which in fact solves the distributional differential equation (14.6).

Each classical solution of (14.5) is also a distributional solution. However, the reverse statement only holds under the conditions

a. U, U_0, U_1 and F are regular and generated by ordinary functions $u, u_0, u_1, f \in L^1_{loc}(G)$,
b. $u_0 \in C^1(\mathbb{R}^d)$ and $u_1 \in C(\mathbb{R}^d)$,
c. $f \in C([0,\infty) \times \mathbb{R}^d)$ and

d. $u \in C^2((0,\infty)\times\mathbb{R}^d)\cap C([0,\infty)\times\mathbb{R}^d)$.

§15. Fourier Transform Methods

The usability of Fourier transformation mainly relies on the fact that elementary differential operators ∂_k, applied to Schwartz functions ϕ, appear as multiplication operators to the Fourier transform $\mathscr{F}(\phi)$ and via verce, that is, we always have

$$\partial_k\,\phi(x) \overset{\mathscr{F}}{\longmapsto} i\,\xi_k\cdot\hat{\phi}(\xi)\,, \qquad i\,x_k\phi(x) \overset{\mathscr{F}}{\longmapsto} \partial_k\,\hat{\phi}(\xi)\,,$$

for $k=1,\ldots,d$, as one can verify by partial integration. For any linear differential operator $\mathcal{A}_c$ with constant coefficients c_α and symbol $S_{\mathcal{A}_c}$ we accordingly obtain

$$(\mathcal{A}_c\,\phi)(x) \overset{\mathscr{F}}{\longmapsto} S_{\mathcal{A}_c}(i\xi)\cdot\hat{\phi}(\xi).$$

This is nothing but the fact that all the differential operators $\mathcal{A}_c$ are Fourier multipliers. They operate on functions $\phi\in\mathscr{S}(\mathbb{R}^d)$ like multiplications with the symbols $S_{\mathcal{A}_c}$ on the Fourier transform $\mathscr{F}\phi$. Hence a linear differential operator $\mathcal{A}_c$ with constant coefficients is conjugated to its symbol function $S_{\mathcal{A}_c}$ in the sense of

$$\mathcal{A}_c\phi \;=\; (\mathscr{F}^{-1}\circ S_{\mathcal{A}_c}\circ\mathscr{F})\,\phi \;=\; \mathscr{F}^{-1}(S_{\mathcal{A}_c}\hat{\phi}) \quad \text{for all } \phi\in\mathscr{S}(\mathbb{R}^d).$$

We want incident how Fourier transformation can be used to solve linear differential equations in the whole space $\mathbb{R}^d$ and governed by the Laplacian. The basic idea is to transform the differential equation either into a purely algebraic equation or into an ordinary differential equation. Once these are solved, inverse Fourier transformation is applied to get the solution of the original problem.

Two examples for these methods will be presented. In the first one, the bijectivity of Fourier transformation on the Schwartz space is used to solve a specific differential equation, while in the second one the simplified computation of convolution products after performing Fourier transformation is helpful to find fundamental solutions of the underlying differential operators.

1. The equation $\mathcal{A}_c\,u=f$ with $f\in\mathscr{S}(\mathbb{R}^d)$ and governed by a differential operator $\mathcal{A}_c$ having constant coefficents can be converted by applying Fourier transformation into the algebraic equation

$$S_{\mathcal{A}_c}(i\xi)\,\hat{u}(\xi) \;=\; \hat{f}(\xi) \qquad (\xi \in \mathbb{R}^d). \tag{15.1}$$

Now, if the symbol function $S_{\mathcal{A}_c}$ of $\mathcal{A}_c$ owns the property

$$S_{\mathcal{A}_c}(i\xi) \neq 0 \quad \text{for all } \xi = (\xi_1, \dots, \xi_d) \in \mathbb{R}^d, \tag{15.2}$$

equation (15.1) can be resolved for $\hat{u}(\xi)$, and the solution u of $\mathcal{A}_c\, u = f$ is obtained by applying inverse Fourier transformation:

$$\hat{u}(\xi) = \hat{f}(\xi) \,/\, S_{\mathcal{A}_c}(i\xi)$$

and

$$u(x) \;=\; \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{i\xi x} \cdot S_{\mathcal{A}_c}^{-1}(i\xi) \cdot \hat{f}(\xi)\, d\xi.$$

2. If (15.2) holds, the Fourier transform method can also be applied to solve the distributional differential equation $\mathcal{A}_c\, U = F$ with $F \in \mathscr{S}'(\mathbb{R}^d)$. Since $\mathcal{A}_c$ is bijective on $\mathscr{S}(\mathbb{R}^d)$ and

$$P_{\mathcal{A}_c}(i\xi) \;=\; P_{\mathcal{A}_c^*}(-\xi) \quad \text{for all } \xi \in \mathbb{R}^d$$

holds, the formally adjoint operator $\mathcal{A}_c^*$ is bijective on $\mathscr{S}(\mathbb{R}^d)$ and so its adjoint operator $\mathcal{A}_c = (\mathcal{A}_c^*)^*$ is bijective on the dual space $\mathscr{S}'(\mathbb{R}^d)$. Thus, if (15.2) is valid, the equation $\mathcal{A}_c\, U = F$ is uniquely solvable for each tempered distribution F.

Especially for the differential operator $\Delta_\lambda := -\Delta + \lambda$ and $\lambda > 0$ we obtain

$$S_{\Delta_\lambda}(i\xi) = |\xi|^2 + \lambda \neq 0$$

for $\xi \in \mathbb{R}^d$, hence the solution u of $\Delta_\lambda u = f$ with $f \in \mathscr{S}(\mathbb{R}^d)$ is

$$u(x) \;=\; \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} e^{i\xi x} \cdot \frac{\hat{f}(\xi)}{|\xi|^2 + \lambda}\, d\xi, \quad (x \in \mathbb{R}^d).$$

Partial Fourier transformation is a method where the operator $\mathscr{F}$ is applied only to the spatial variables of the partial differential equation. The result is an ordinary differential equation with respect to the time variable t. Then solving this equation and applying inverse Fourier transformation leads to the solution of the original problem.

In the following we sketch how partial Fourier transformation is useful to solve the initial-value problem of the heat and wave equation in the entire space $\mathbb{R}^d$.

1. The initial-value problem of the heat equation is given by

$$\begin{cases} u_t - \Delta u = 0 & (t > 0) \\ u(0,x) \;=\; f(x) & (x \in \mathbb{R}). \end{cases}$$

Applying Fourier transformation to both sides of the differential equation with respect to the spatial variables $x_1, \ldots, x_d$, we obtain

$$\frac{\partial}{\partial t}\hat{u}(t,\xi) \;+\; \xi^2\,\hat{u}(t,\xi) \;=\; 0, \qquad\qquad \hat{u}(0,\xi) = \hat{f}(\xi), \tag{15.3}$$

where $\xi = (\xi_1, \ldots, \xi_d)$ are the variables in the *phase space.*

Then multiplying both sides of the first equation in (15.3) by $e^{|\xi|^2 t}$ leads to

$$\frac{\partial}{\partial t}\, e^{|\xi|^2 t}\,\hat{u}(t,\xi) \;=\; 0$$

with solution $\hat{u}(t,\xi) = \phi(\xi)\, e^{-|\xi|^2 t}$. The mapping $\xi \mapsto \phi(\xi)$ is the integration constant for each fixed point ξ. The second equation in (15.3) implies $\phi(\xi) = \hat{f}(\xi)$, hence the Fourier transformed solution $\hat{u}$ of the initial-value problem is nothing but

$$\hat{u}(t,\xi) = \hat{f}(\xi) \cdot e^{-|\xi|^2 t}. \tag{15.4}$$

The solution of the original equation is obtained by applying inverse Fourier transformation and the rule

$$\mathscr{F}^{-1}(f \cdot g) \;=\; \mathscr{F}^{-1}(f) * \mathscr{F}^{-1}(g)$$

to each side of (15.4):

$$\begin{aligned} u(t,x) \;&=\; \mathscr{F}^{-1}(\hat{f}(\xi) \cdot e^{-|\xi|^2 t}) \;=\; \mathscr{F}^{-1}(\hat{f}(\xi)) * \mathscr{F}^{-1}(e^{-|\xi|^2 t}) \\ &=\; f(x) * \mathscr{F}^{-1}(e^{-|\xi|^2 t}) \;=\; \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-|x-y|^2/(4t)}\, f(y)\, dy. \end{aligned}$$

The function

$$H_t(x,y) \;:=\; \frac{1}{\sqrt{4\pi t}}\, e^{-|x-y|^2/(4t)} \;=\; \Phi_{\partial_t - \Delta}(t, x-y)$$

is called a *heat kernel*, since it constitutes the kernel of the integral operator $\mathcal{H}_t : f \mapsto H_t * f$.

2. The method also enables to solve the initial-value problem for the wave equation

$$\begin{cases} u_{tt} - \Delta u = 0 & (t > 0), \\ u(0,x) = f(x) & (x \in \mathbb{R}^d), \\ u_t(0,x) = g(x) & (x \in \mathbb{R}^d). \end{cases}$$

Partial Fourier transformation let become this equation into

$$\hat{u}_{tt}(t,\xi) = |\xi|^2 \hat{u}(t,\xi), \qquad \hat{u}(0,\xi) = \hat{f}(\xi), \qquad \hat{u}_t(0,\xi) = \hat{g}(\xi),$$

and the general solution of the ordinary differential equation in phase space is

$$\hat{u}(t,\xi) = \hat{f}(\xi)\cos(t|\xi|) + \hat{g}(\xi) : \frac{\sin(t|\xi|)}{|\xi|}.$$

Unfortunately, performing inverse Fourier transformation to $\hat{u}$ for any $d > 1$ is not an easy task. But for $d = 1$ we obtain

$$u_1(x,t) = \mathscr{F}^{-1}\left(\hat{f}(\xi)\cdot\cos(|\xi|)\right) = \frac{1}{2}\left[f(x+t) + f(x-t)\right],$$

$$u_2(x,t) = \mathscr{F}^{-1}\left(\hat{g}(\xi)\cdot\frac{\sin(|\xi|)}{|\xi|}\right) = \frac{1}{2}\int_{-t}^{+t} g(x+s)\,ds,$$

thus the solution of the wave equation is given by $u = u_1 + u_2$, known as the D'ALEMBERT *solution*

$$u(t,x) = \frac{1}{2}\left[f(x+t) + f(x-t)\right] + \frac{1}{2t}\int_{-t}^{+t} g(x+s)\,ds.$$

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